GreenHub RVO Pareto analysis

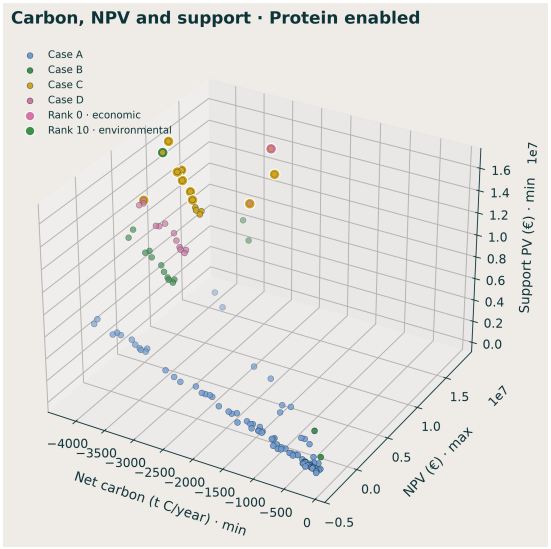
Current no-PEF BioSTEAM launch

Data assembled 2026-11-02T18:24:18Z UTC | Seed 20261102 | 128 sampled designs per protein group

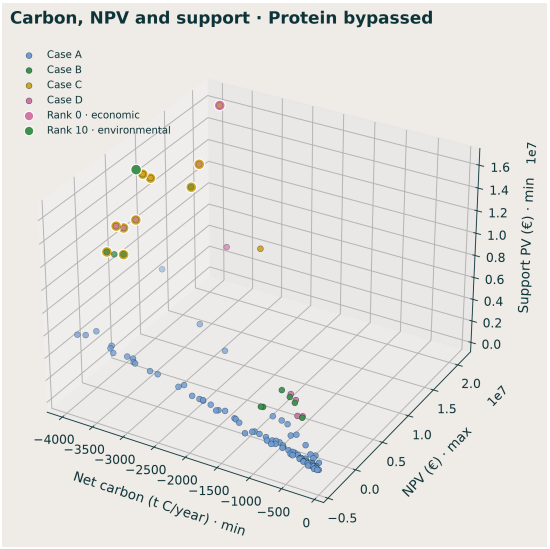
Separate protein-enabled and protein-bypassed process frontiers compare net carbon, screening NPV, and assumed public-support present value. The report includes selected candidates, baseline comparisons, indexed figures, model documentation, source batches, and selected model-run evidence.

RVO support amounts are user-defined what-if assumptions. They are not verified awards, eligibility determinations, or official SDE++ tariff calculations.

Three-objective front - protein enabled



Three-objective front - protein bypassed



Analysis summary

Each of 128 sampled process configurations in each protein group was simulated once, then paired with four financial support cases. Four explicit baseline references per group were also evaluated. PEF is disabled throughout (F1). Negative profits and dominated designs remain in the saved batches.

Protein group	Evaluated rows	Carbon / NPV	Support / NPV	Carbon / NPV / support
Protein enabled	516	3	19	129
Protein bypassed	516	4	26	106

Support cases

Case	Programme / named authority	CAPEX grant	Operating premium	Term
A	A · RVO DEI+ 30% + RVO SDE++ €20/t feed proxy	30%	EUR 20/t/y	15 y
B	B · RVO DEI+ 45% + RVO SDE++ €40/t feed proxy	45%	EUR 40/t/y	15 y
C	C · RVO DEI+ 60% + RVO SDE++ €60/t feed proxy	60%	EUR 60/t/y	15 y
D	D · RVO SDE++ €50/t feed proxy (OPEX only)	0%	EUR 50/t/y	15 y

Assumptions and scope

NPV is a pre-tax, 20-year screening metric at an 8% discount rate. It uses high direct uninstalled equipment CAPEX, partial operating margin, upfront CAPEX grants, and a 15-year operating-support proxy. Installation, financing, escalation, replacements, working capital, residual value, and taxes are excluded.

Net carbon is operational gate-to-gate climate load converted to carbon equivalent minus assumed carbon retained in press-cake furniture and nutrient-rich solids. This mixed screening indicator is not a complete LCA or durable carbon-removal claim. Full methods, exclusions, and official-programme caveats are copied in the documentation folder.

Selected candidates - Protein enabled

Ranks 0 to 10 are selected from the three-objective front. Rank 0 maximizes NPV; rank 10 is the minimum-net-carbon endpoint among selected candidates. Switches use Protein-Biogas-N-P-K; 1 means pathway present and 0 means bypassed.

Rank	P-B-N-P-K	Routes	Case	Feed t/y	Net C t C/y	NPV EUR	Support PV EUR
0	P1-B1-N1-P1-K1	PR3/BG4/NR7	C	24,204	-3,208.0	18,039,491	13,798,787
1	P1-B1-N1-P1-K1	PR2/BG4/NR7	C	22,224	-2,946.8	16,181,913	12,675,052
2	P1-B1-N1-P1-K1	PR3/BG4/NR7	D	24,204	-3,208.0	14,599,369	10,358,665
3	P1-B0-N0-P0-K0	PR1/BG0/NR0	C	26,896	-3,772.0	9,242,464	14,152,223
4	P1-B1-N0-P1-K0	PR1/BG3/NR2	C	27,114	-3,786.6	9,296,779	15,027,774
5	P1-B1-N1-P0-K0	PR2/BG3/NR3	C	29,783	-4,150.8	10,491,516	16,558,923
6	P1-B1-N0-P0-K0	PR1/BG3/NR0	C	27,345	-3,818.2	8,934,227	14,992,753
7	P1-B0-N0-P0-K0	PR1/BG0/NR0	C	25,672	-3,600.4	8,959,951	13,519,177
8	P1-B0-N0-P0-K0	PR3/BG0/NR0	C	25,039	-3,500.2	8,396,748	13,191,755
9	P1-B0-N0-P0-K0	PR3/BG0/NR0	D	29,975	-4,190.2	7,026,312	12,828,571
10	P1-B0-N0-P0-K0	PR3/BG0/NR0	C	29,975	-4,190.2	9,942,938	15,745,197

Selected candidates - Protein bypassed

Ranks 0 to 10 are selected from the three-objective front. Rank 0 maximizes NPV; rank 10 is the minimum-net-carbon endpoint among selected candidates. Switches use Protein-Biogas-N-P-K; 1 means pathway present and 0 means bypassed.

Rank	P-B-N-P-K	Routes	Case	Feed t/y	Net C t C/y	NPV EUR	Support PV EUR
0	P0-B1-N1-P1-K1	PR0/BG2/NR7	C	29,075	-3,879.8	20,883,904	16,279,667
1	P0-B1-N1-P1-K1	PR0/BG2/NR7	D	29,075	-3,879.8	17,047,456	12,443,219
2	P0-B1-N1-P1-K1	PR0/BG2/NR7	B	29,075	-3,879.8	15,569,666	10,965,428
3	P0-B0-N0-P0-K0	PR0/BG0/NR0	C	27,648	-3,887.9	8,712,470	14,460,167
4	P0-B1-N0-P0-K0	PR0/BG2/NR1	C	27,824	-3,894.3	7,479,761	15,269,912
5	P0-B0-N0-P0-K0	PR0/BG0/NR0	D	27,648	-3,887.9	6,084,795	11,832,492
6	P0-B1-N0-P0-K0	PR0/BG2/NR1	D	27,824	-3,894.3	4,117,756	11,907,907
7	P0-B1-N0-P1-K0	PR0/BG3/NR2	D	28,376	-3,972.2	3,643,226	12,144,318
8	P0-B0-N0-P0-K0	PR0/BG0/NR0	B	27,648	-3,887.9	3,914,179	9,661,876
9	P0-B1-N0-P1-K0	PR0/BG3/NR2	B	28,376	-3,972.2	2,037,927	10,539,019
10	P0-B1-N0-P1-K0	PR0/BG3/NR2	C	28,376	-3,972.2	7,170,175	15,671,267

Baseline references

Exact current F1/no-PEF baseline references are simulated separately for each support case. The plotted baseline figures compare protein-enabled and protein-bypassed routes; detailed values and run IDs remain in the source batches.

Group	Case	P-B-N-P-K	Feed (t/y)	Net carbon (t C/y)	NPV (EUR)	Support PV (EUR)
Protein enabled	A	P1-B1-N1-P1-K0	8,760	-1,220.0	-3,606,926	1,836,087
Protein enabled	B	P1-B1-N1-P1-K0	8,760	-1,220.0	-1,939,072	3,503,940
Protein enabled	C	P1-B1-N1-P1-K0	8,760	-1,220.0	-271,219	5,171,794
Protein enabled	D	P1-B1-N1-P1-K0	8,760	-1,220.0	-1,693,961	3,749,052
Protein bypassed	A	P0-B1-N1-P1-K0	8,760	-1,225.6	-3,942,265	1,821,670
Protein bypassed	B	P0-B1-N1-P1-K0	8,760	-1,225.6	-2,281,620	3,482,316
Protein bypassed	C	P0-B1-N1-P1-K0	8,760	-1,225.6	-620,974	5,142,961
Protein bypassed	D	P0-B1-N1-P1-K0	8,760	-1,225.6	-2,014,884	3,749,052

Complete model warnings, limitations, and sample configurations are preserved in `evidence/current_run/`. The source PNGs are embedded below without edits to their visual styling.

pareto net carbon vs npv protein enabled

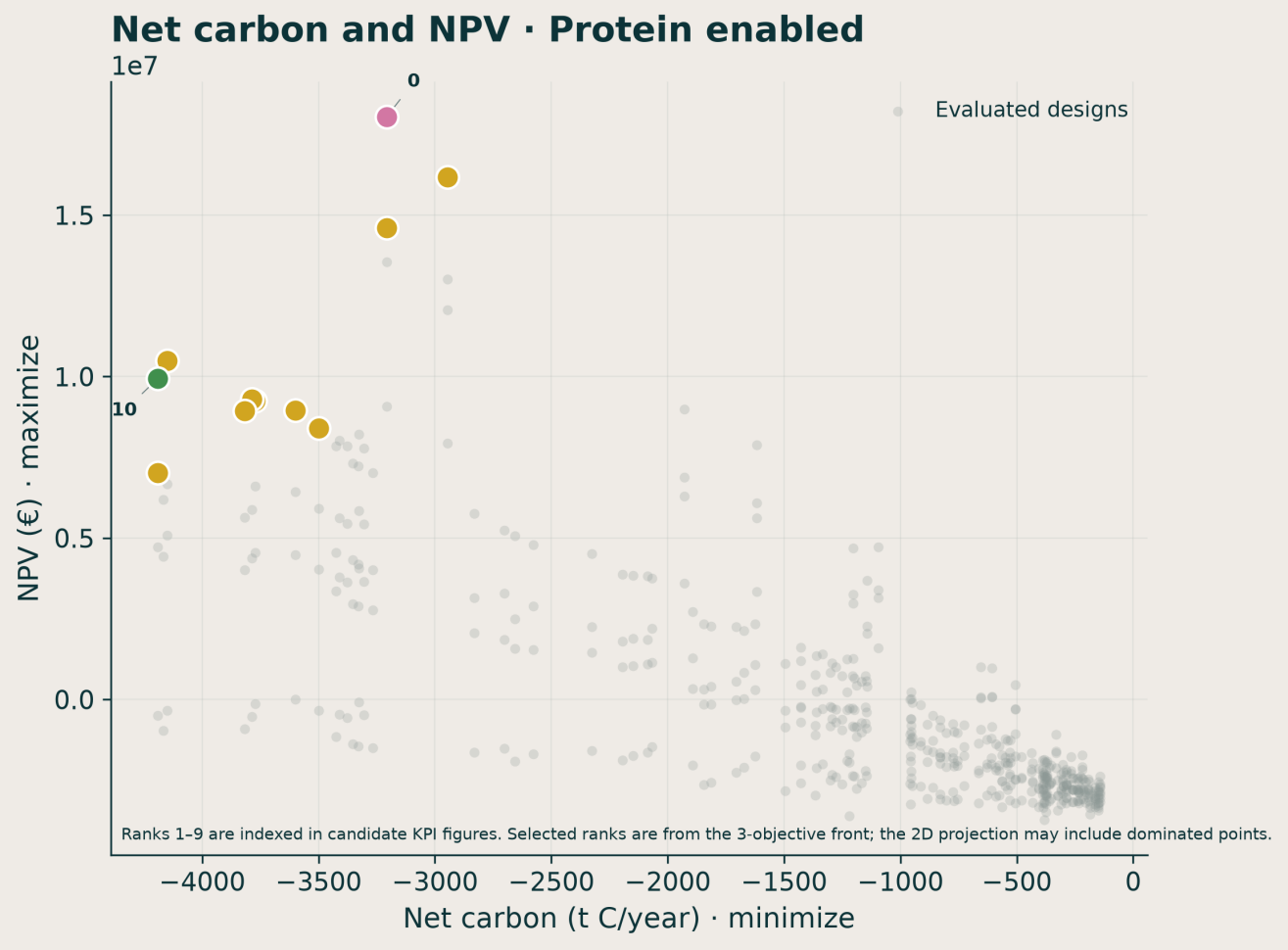


Figure 01. Net carbon and NPV · Protein enabled

protein_enabled / all support cases | Net carbon: t C/year; NPV: € | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

pareto subsidy granted vs npv protein enabled

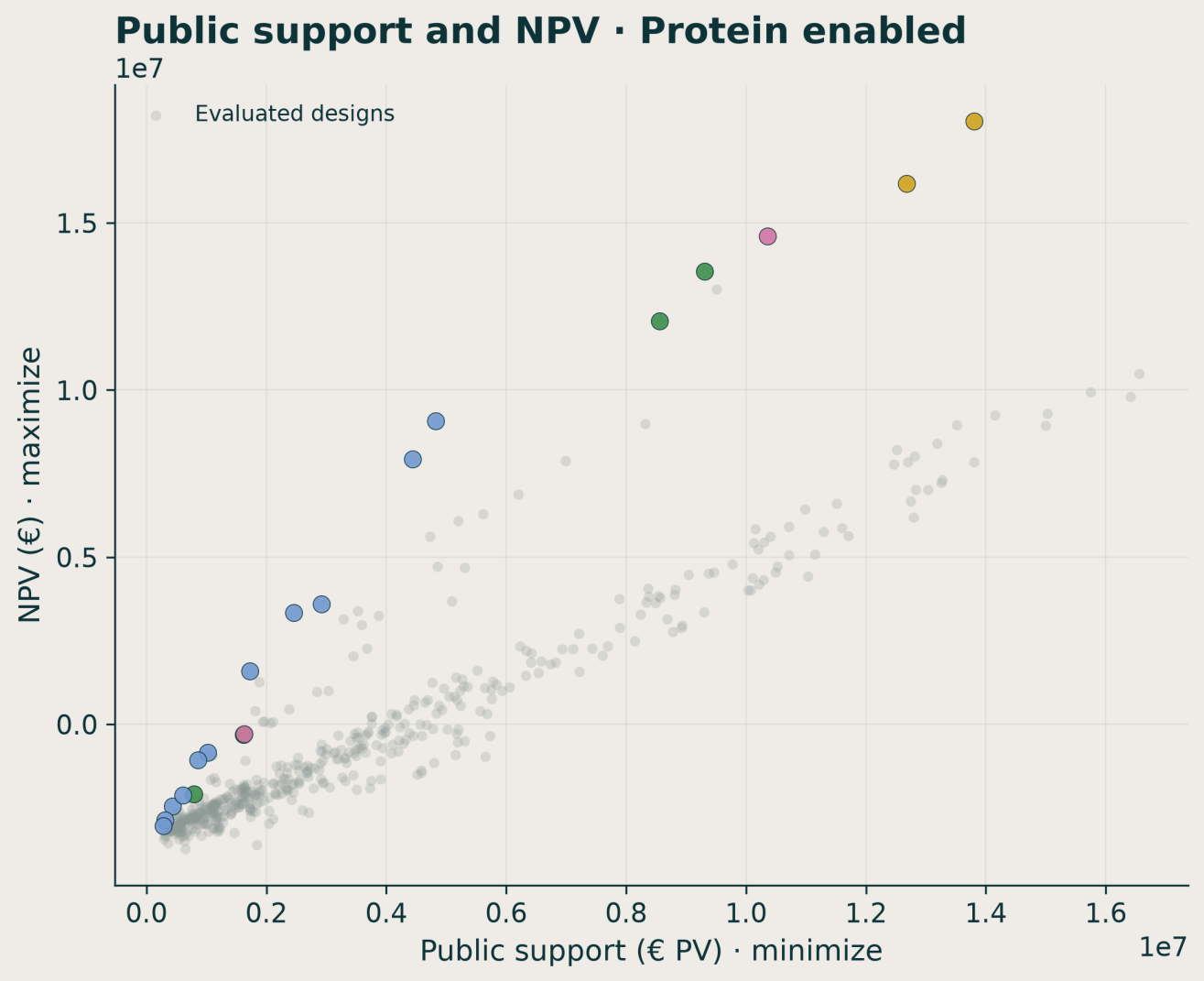


Figure 02. Public support and NPV · Protein enabled

protein_enabled / all support cases | Public support: € PV; NPV: € | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

pareto net carbon vs npv protein enabled case A

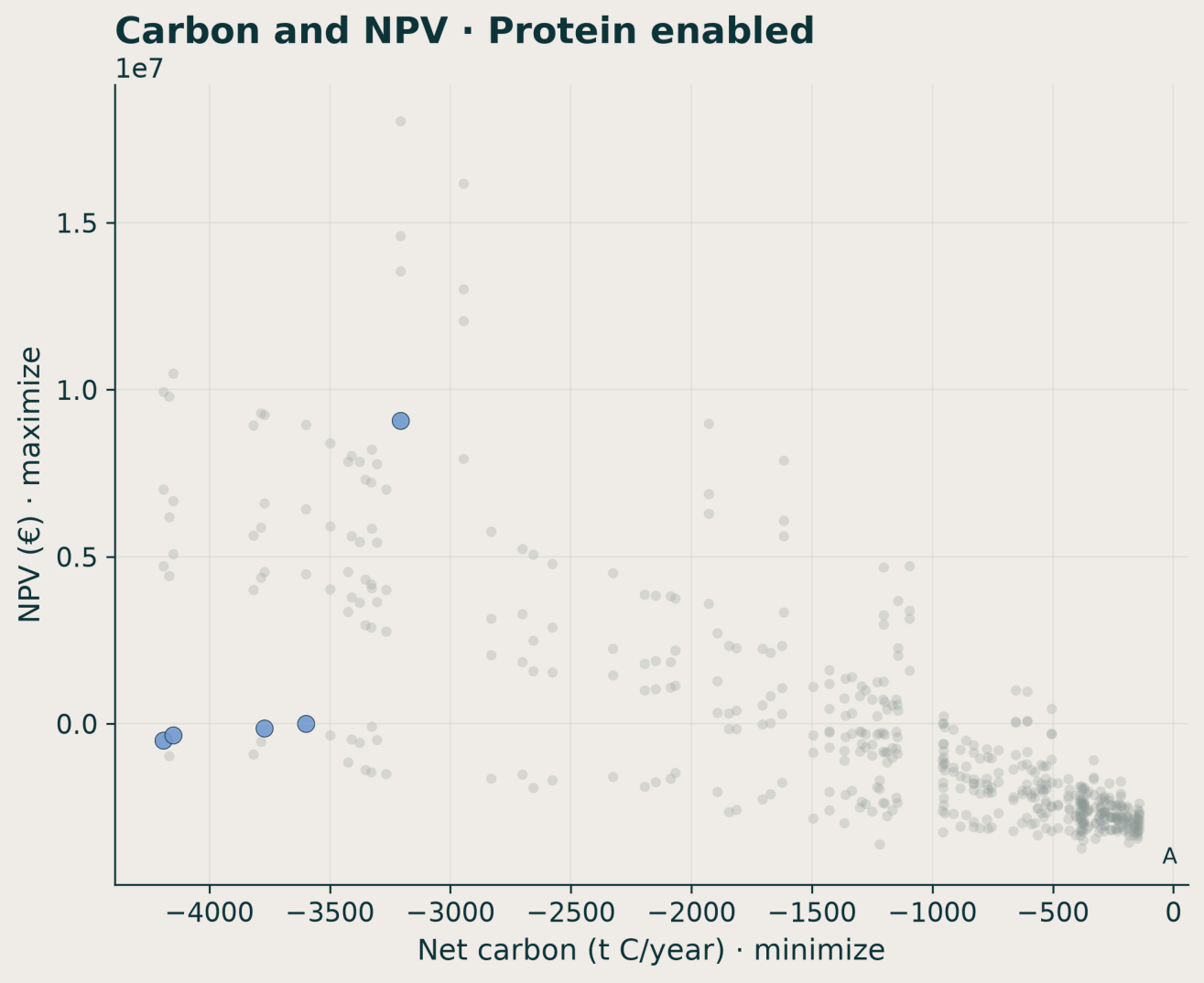


Figure 03. Carbon and NPV · Protein enabled

protein_enabled / case A | Net carbon: t C/year; NPV: € | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

pareto net carbon vs npv protein enabled case B

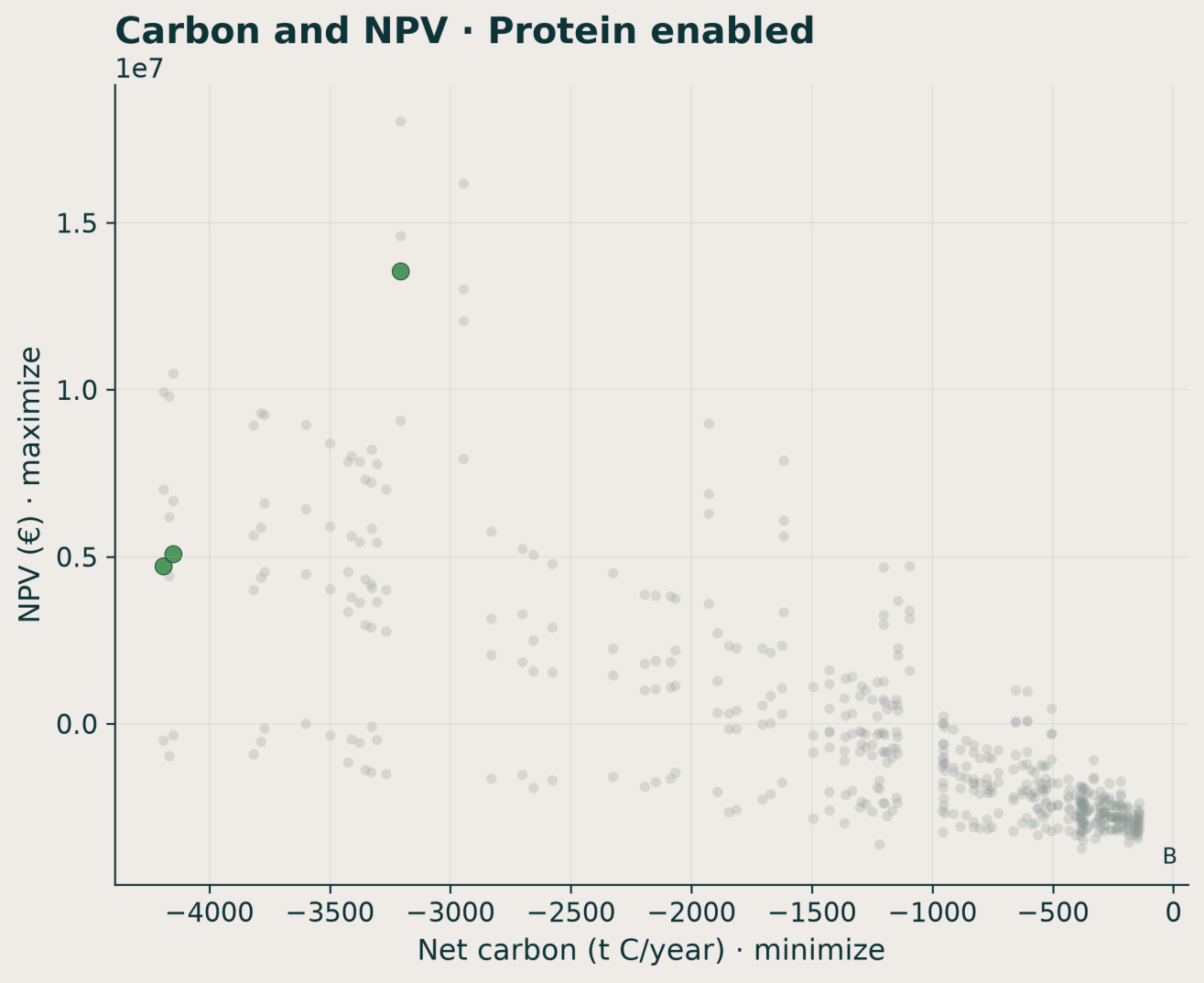


Figure 04. Carbon and NPV · Protein enabled

protein_enabled / case B | Net carbon: t C/year; NPV: € | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

pareto net carbon vs npv protein enabled case C

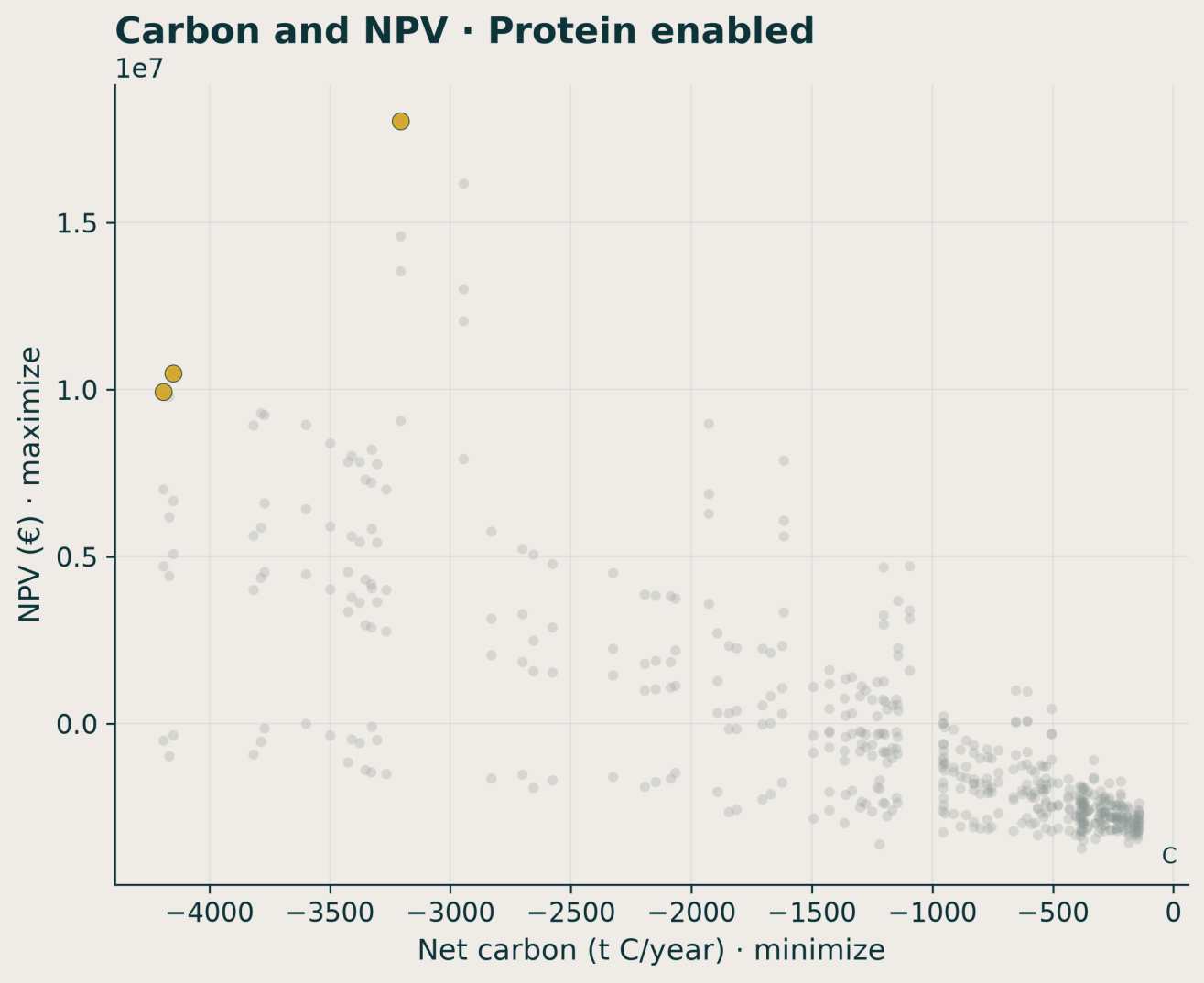


Figure 05. Carbon and NPV · Protein enabled

protein_enabled / case C | Net carbon: t C/year; NPV: € | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

pareto net carbon vs npv protein enabled case D

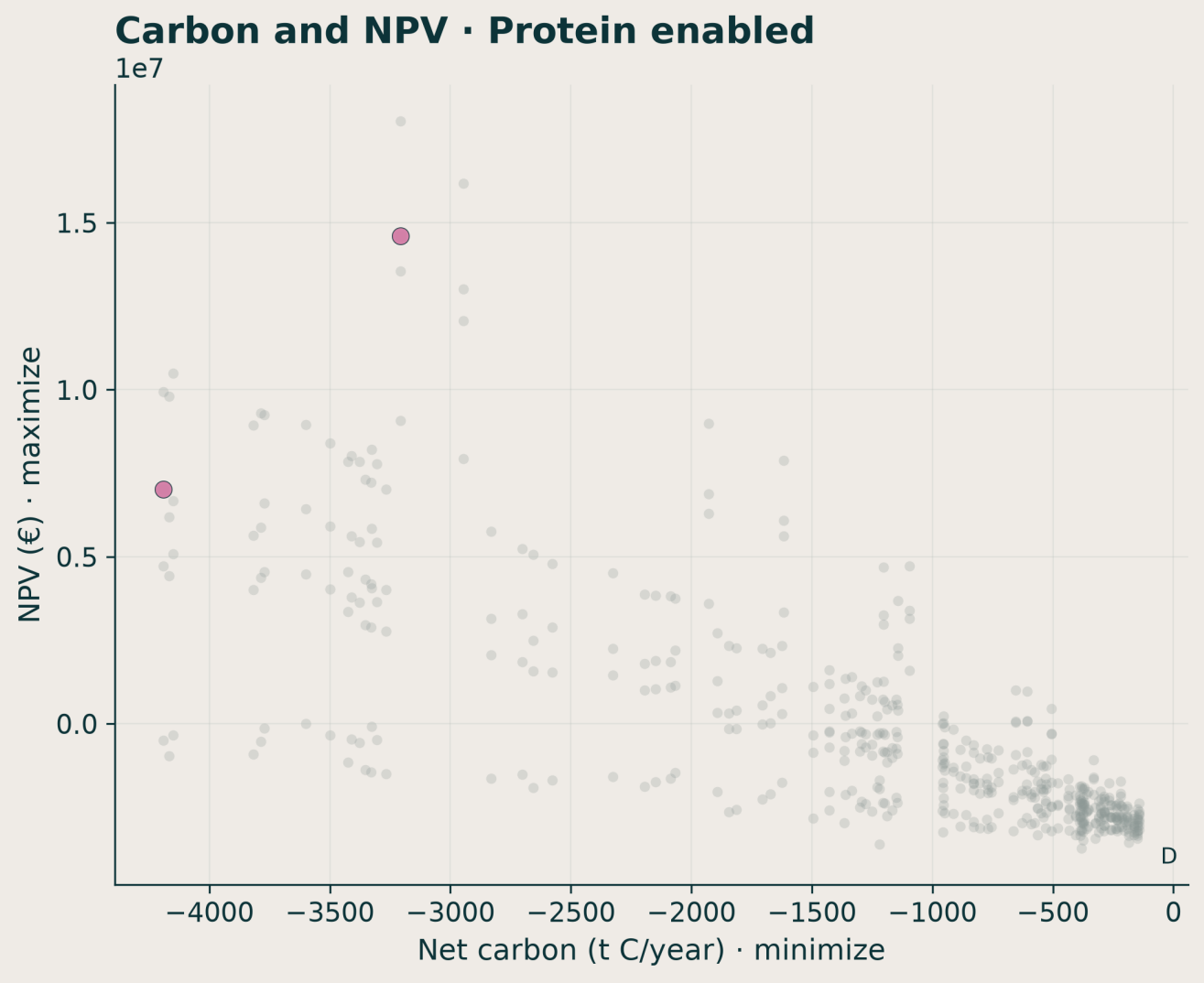


Figure 06. Carbon and NPV · Protein enabled

protein_enabled / case D | Net carbon: t C/year; NPV: € | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

pareto 3d carbon npv support protein enabled

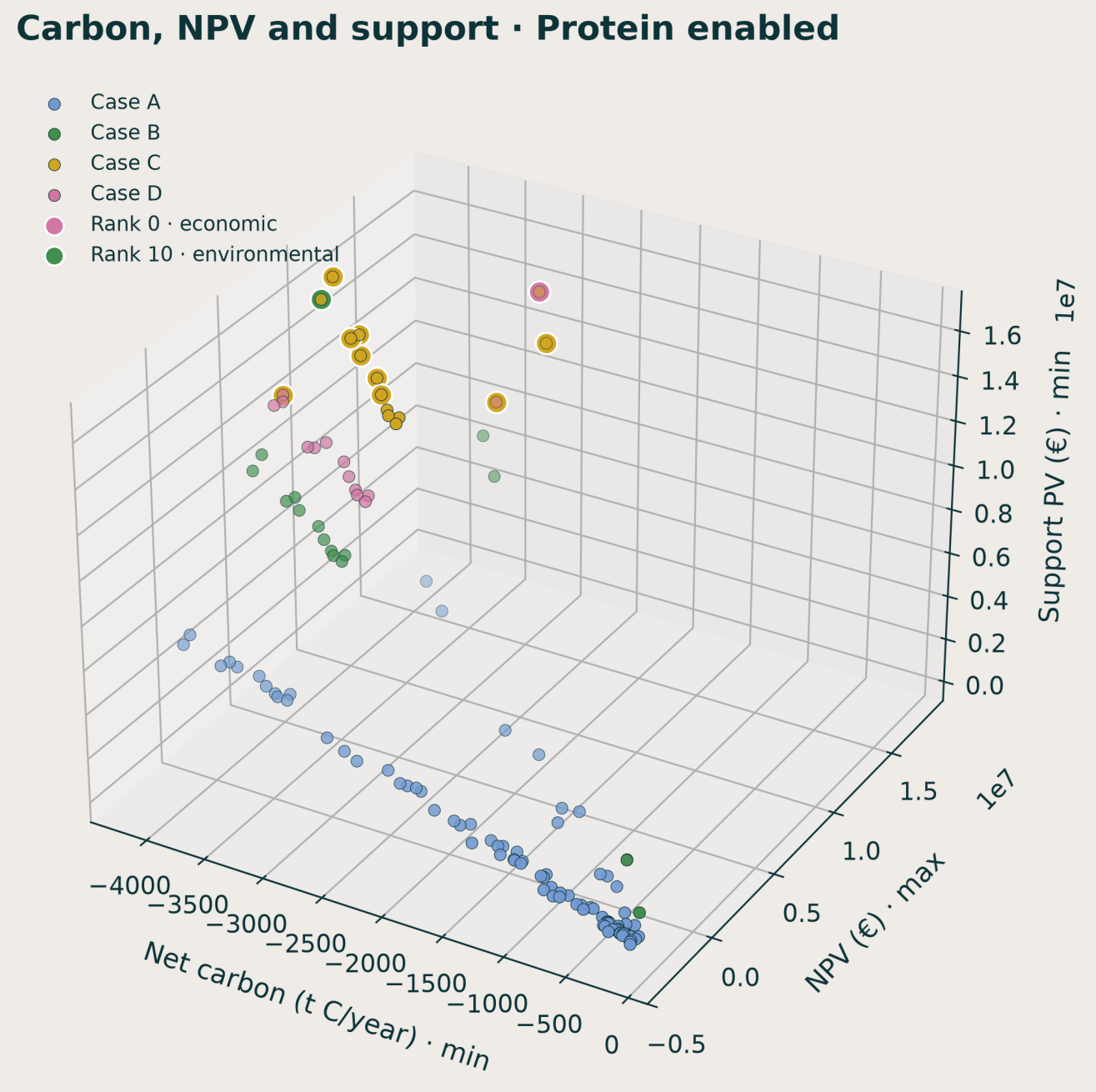


Figure 07. Three-objective Pareto front

Protein enabled | t C/year; €; € present value | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate kpi heatmap protein enabled

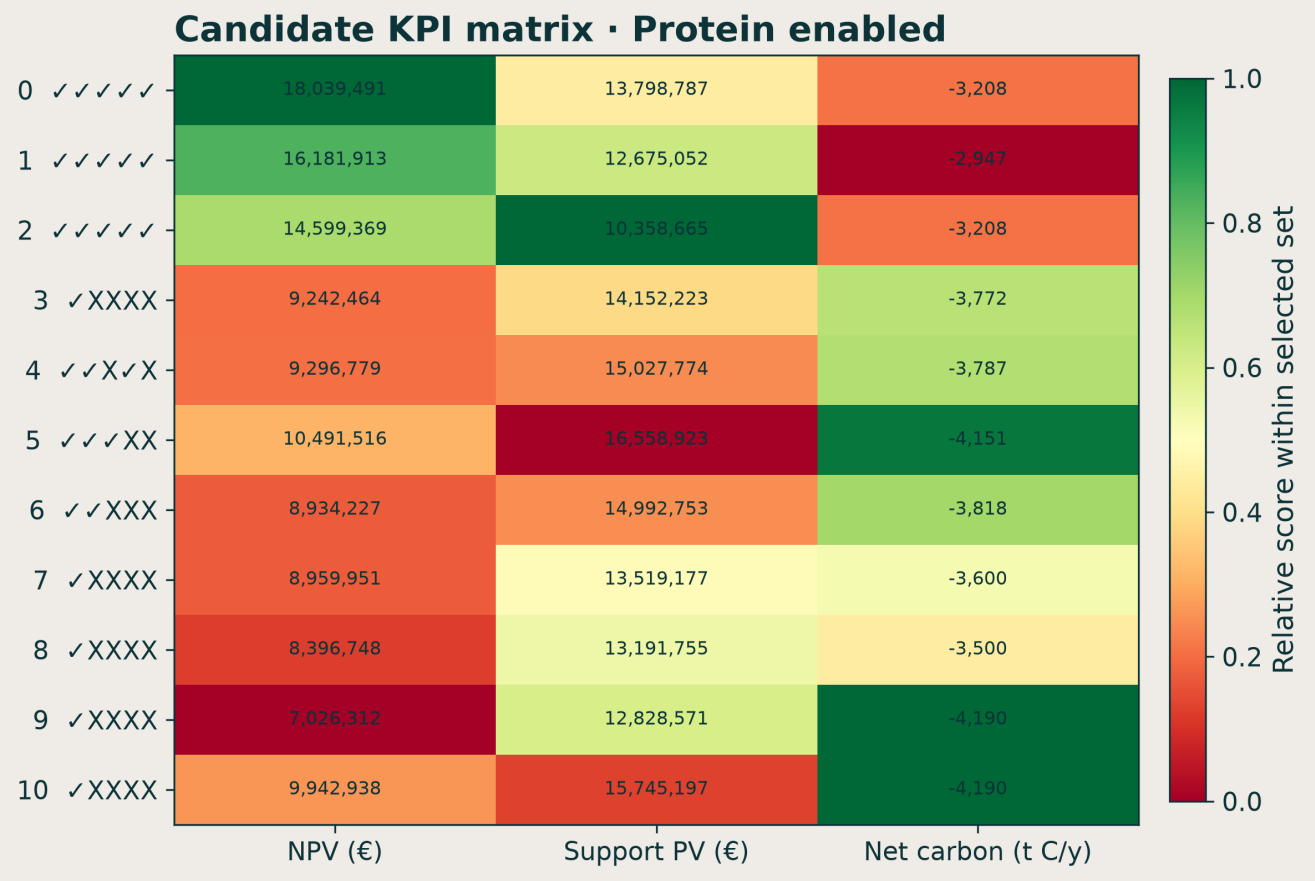


Figure 08. Selected candidate KPI comparison; color normalized by objective direction

Protein enabled | €, € present value, t C/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate economics protein enabled

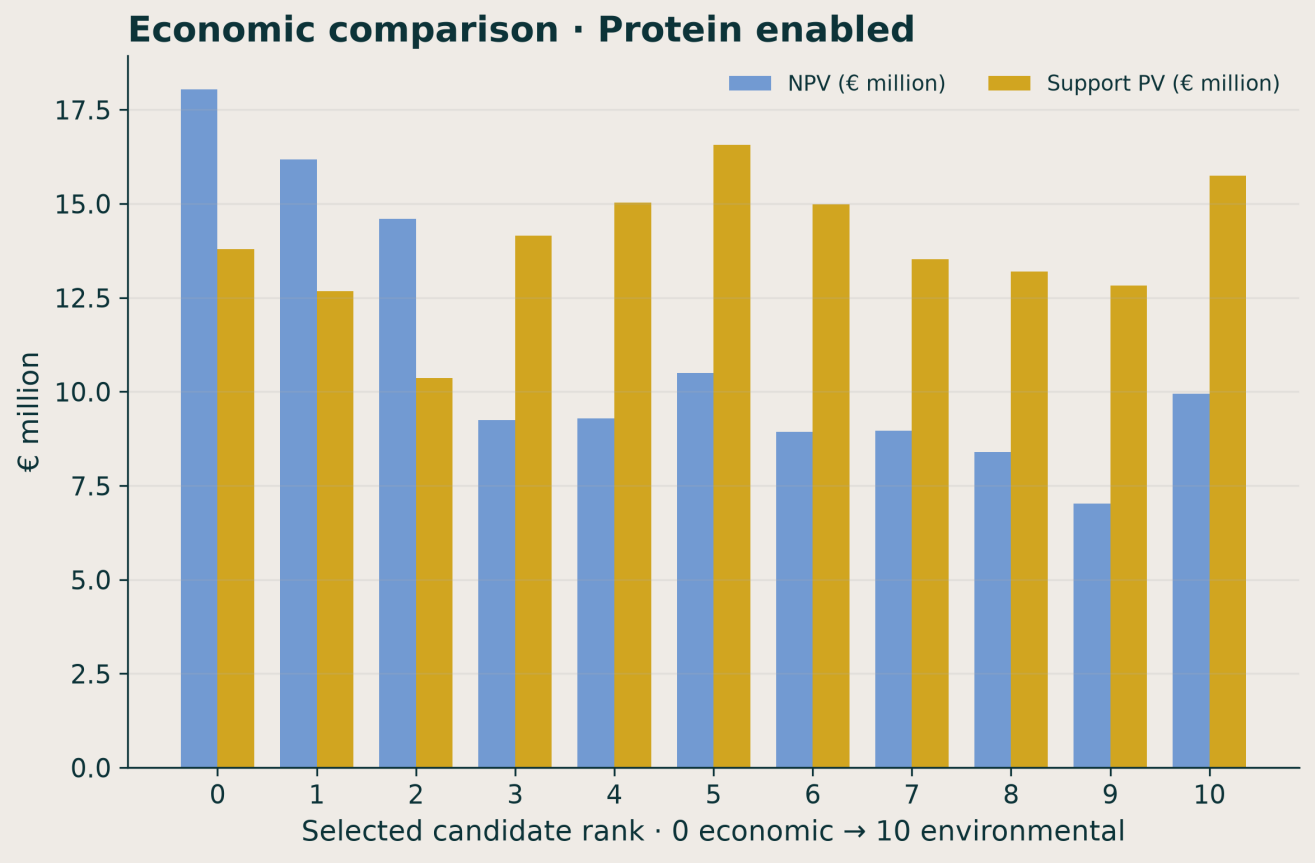


Figure 09. NPV and assumed public support PV for selected Pareto candidates

Protein enabled | € million | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate environment protein enabled

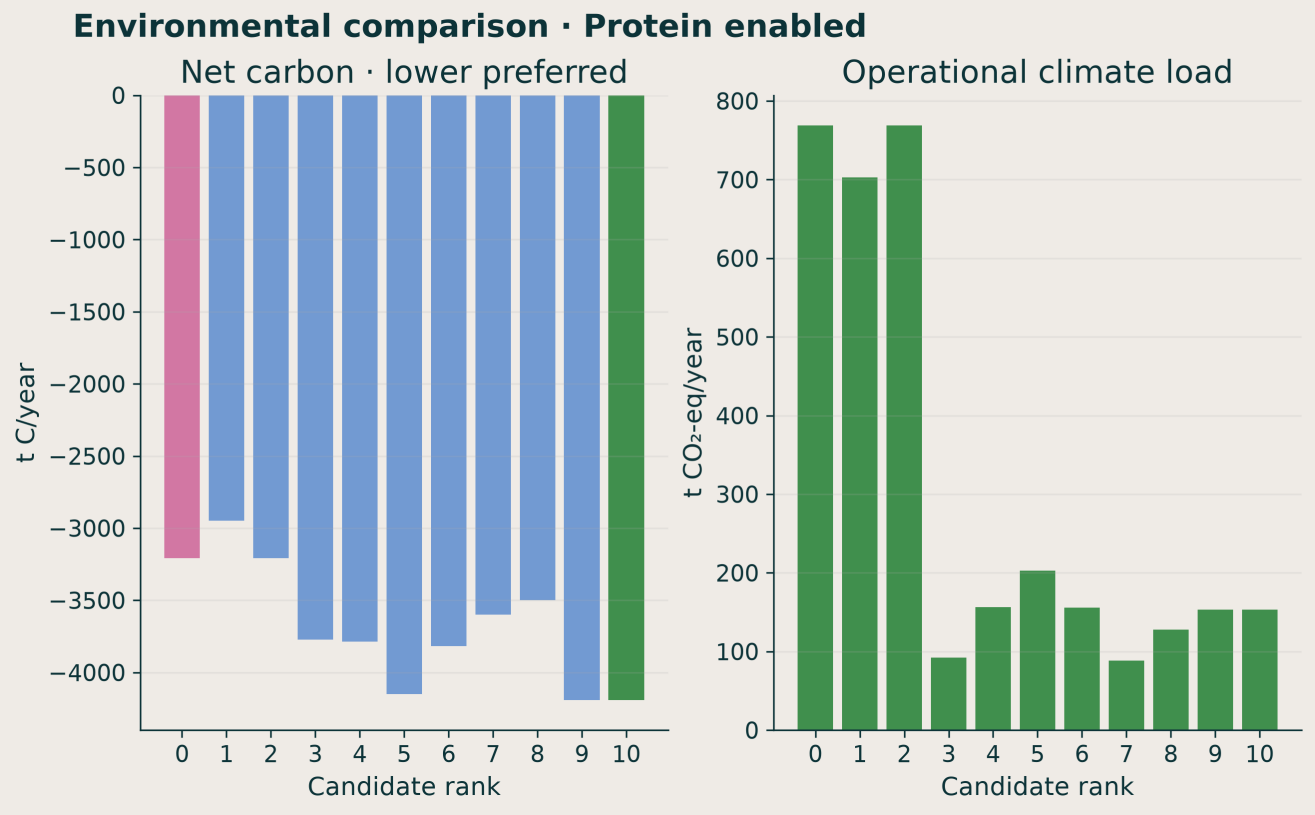


Figure 10. Net stored carbon objective and attributional operational climate load

Protein enabled | t C/year; t CO□-eq/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

carbon load vs stored carbon protein enabled

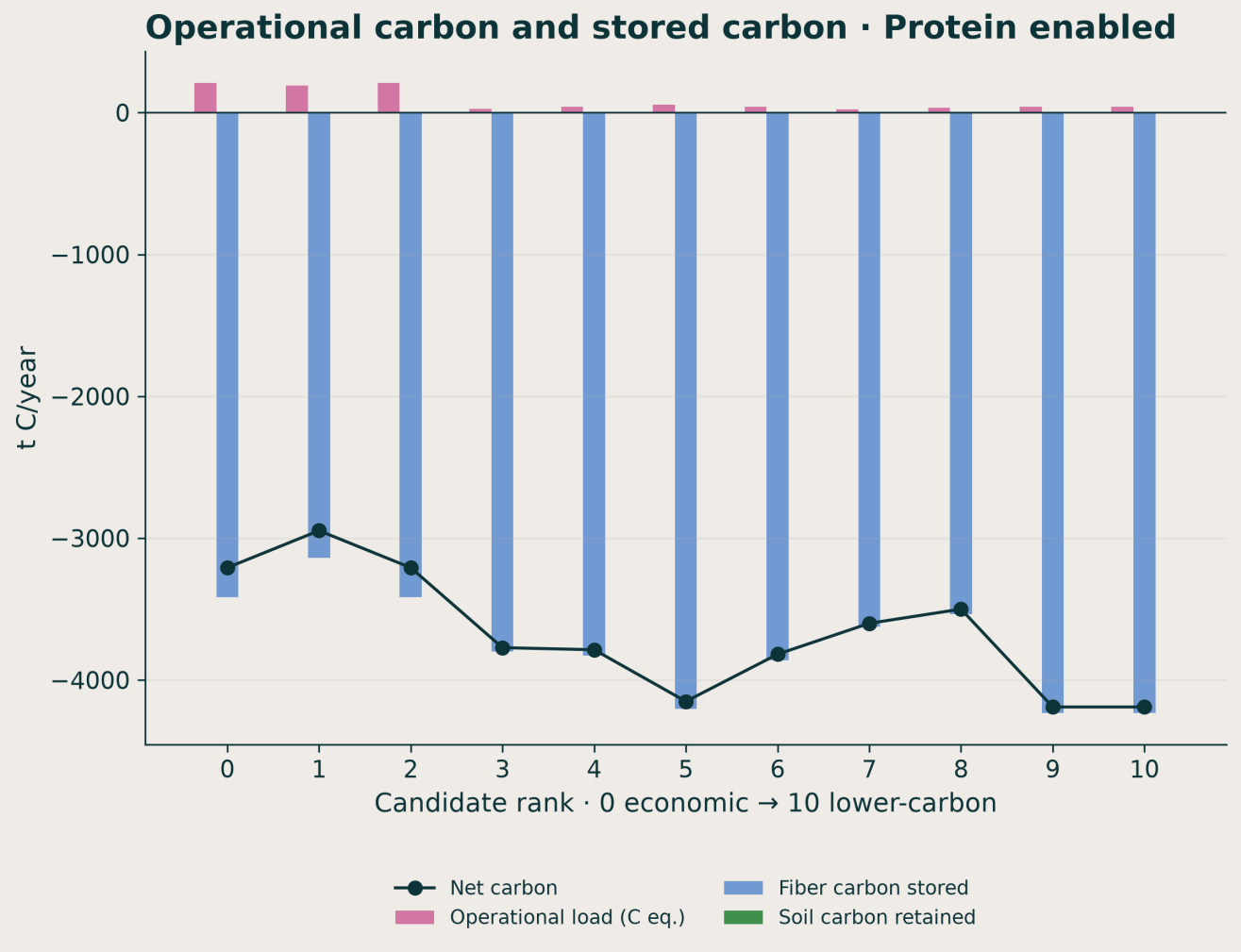


Figure 11. Attributional operational carbon-equivalent load minus fiber and soil retained product carbon

Protein enabled | t C/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate products

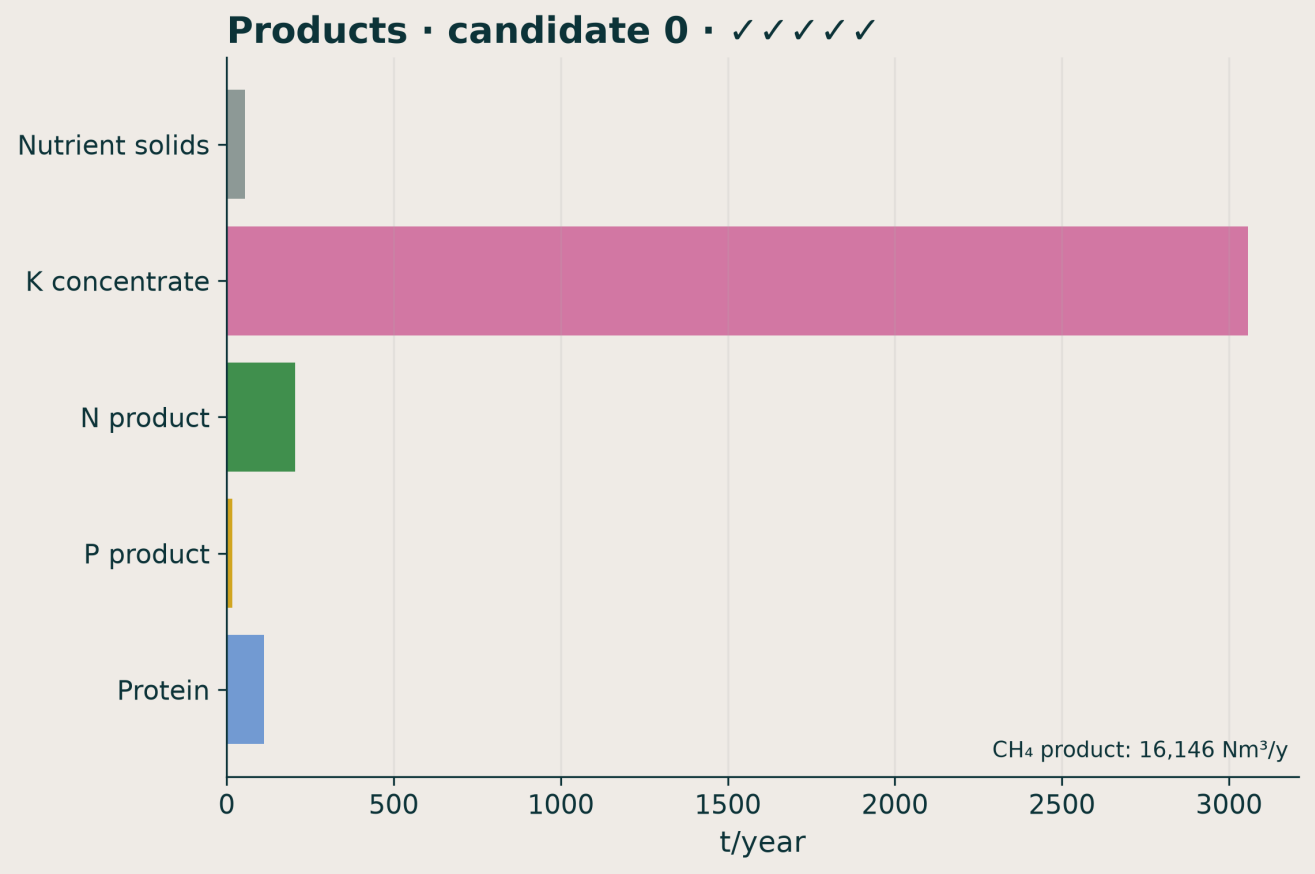


Figure 12. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_enabled; rank 0; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD1', 'biogas_upgrading': 'BG4', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate utilities

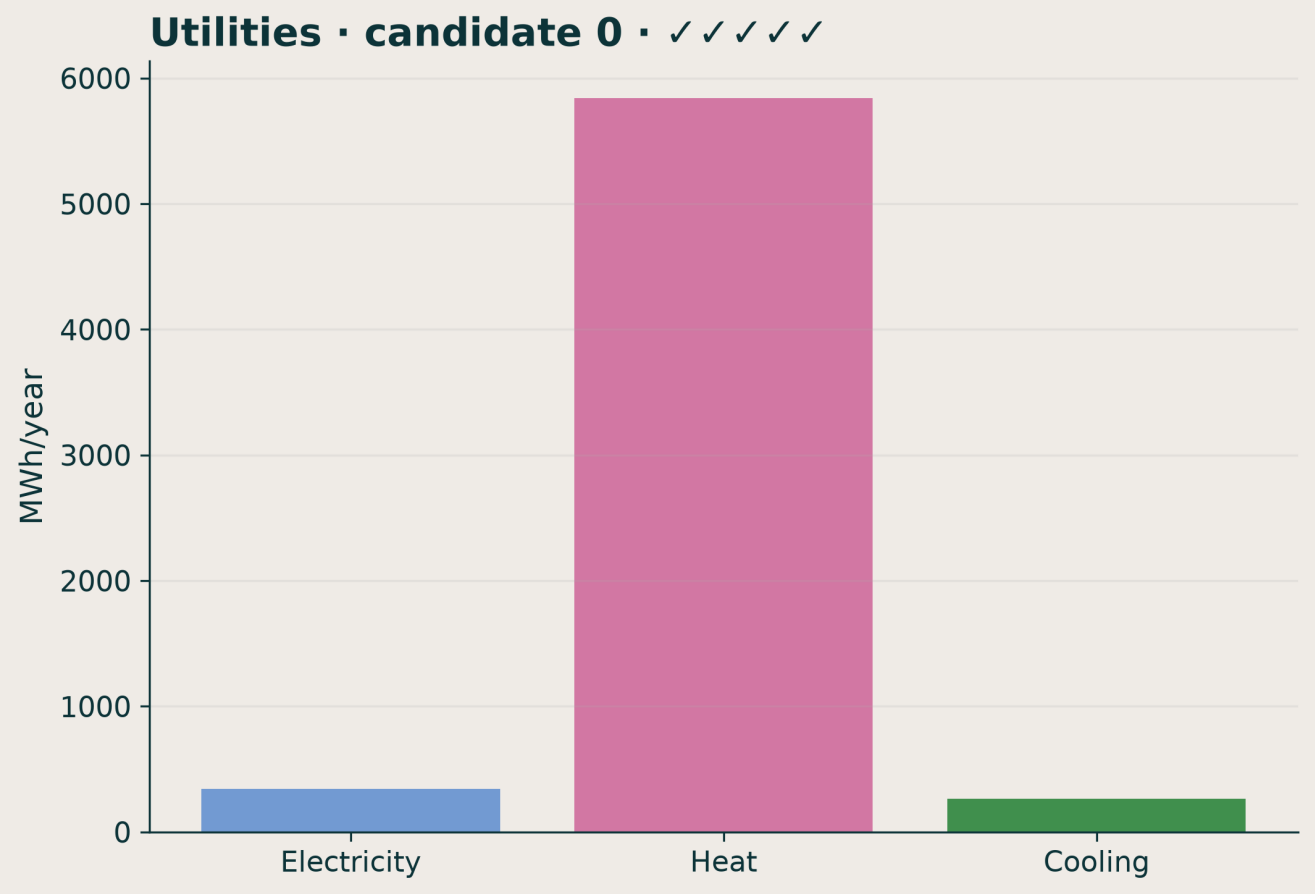


Figure 13. Annual electricity, heating and cooling duty

protein_enabled; rank 0; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD1', 'biogas_upgrading': 'BG4', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate kpis

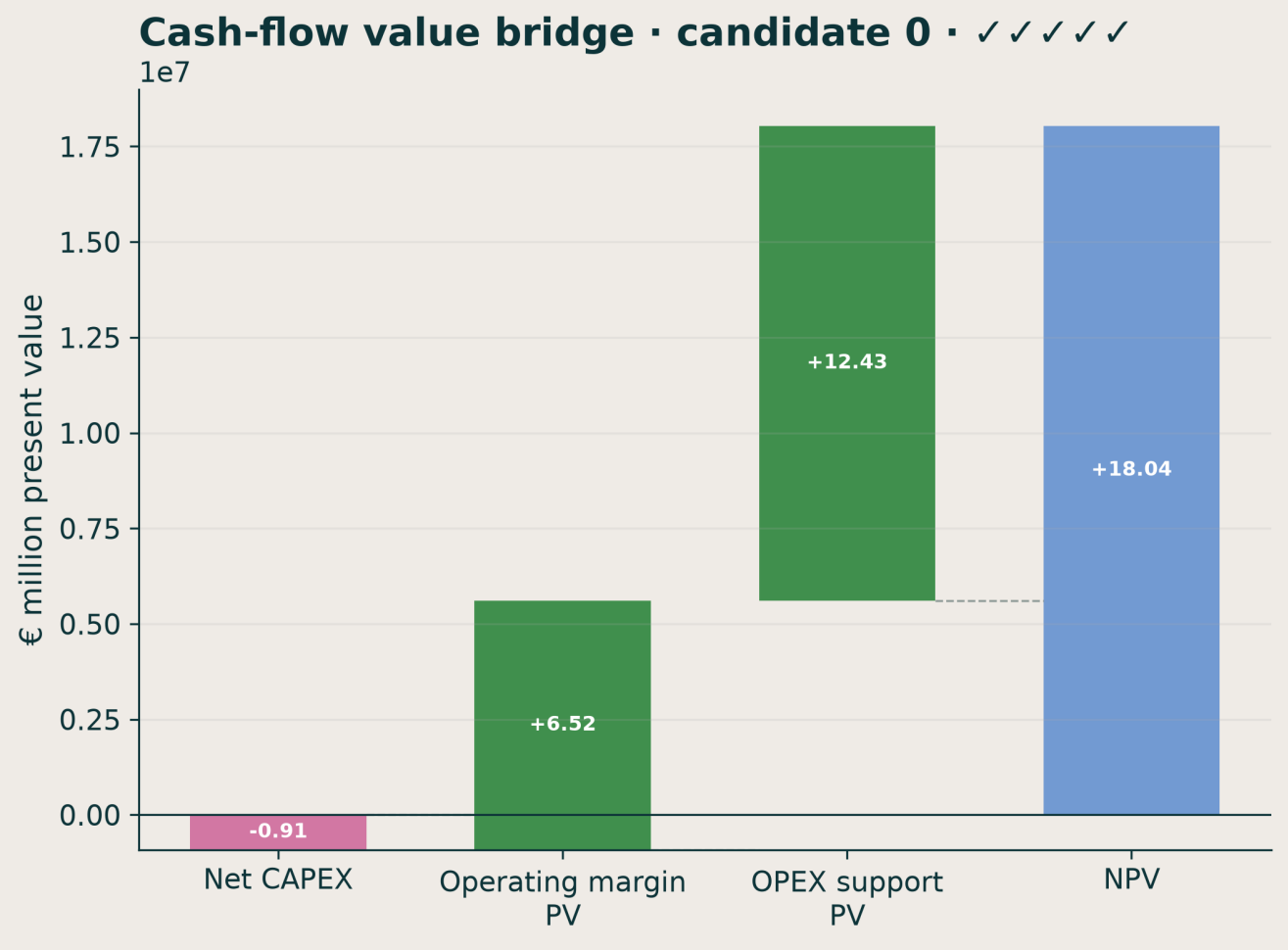


Figure 14. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 0; support C | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate products

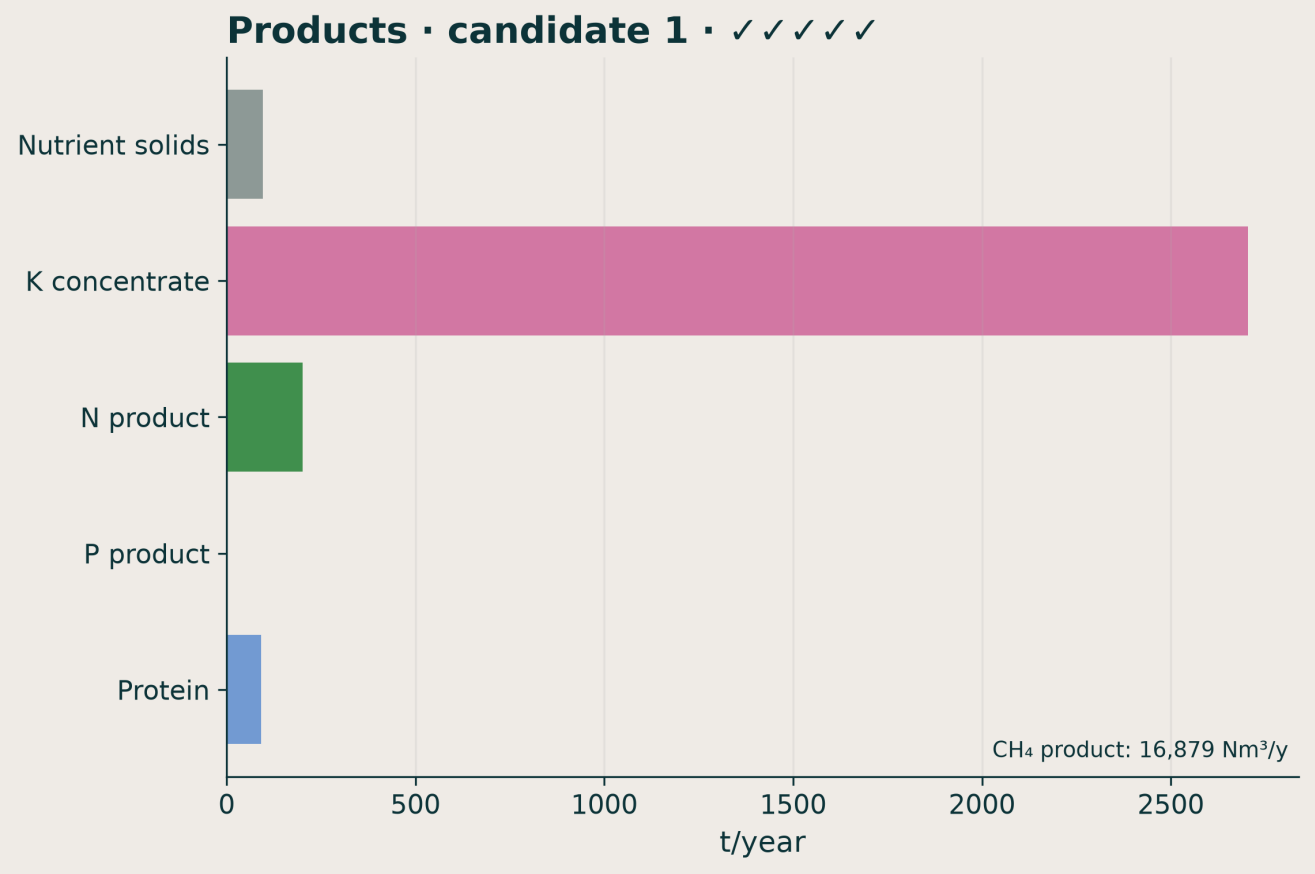


Figure 15. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_enabled; rank 1; {'feed_preparation': 'F1', 'protein_recovery': 'PR2', 'anaerobic_digestion': 'AD1', 'biogas_upgrading': 'BG4', 'nutrient_recovery': 'NR7', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate utilities

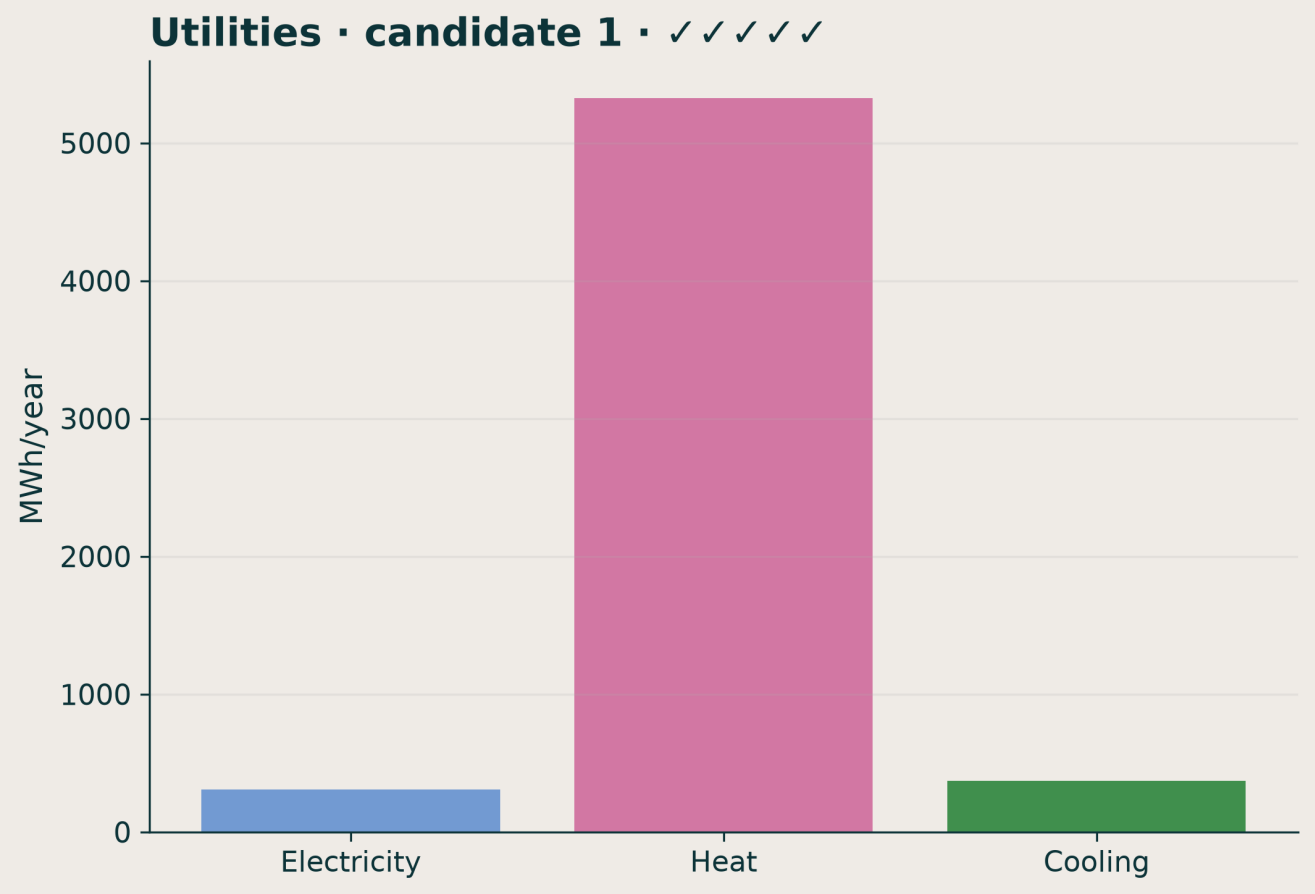


Figure 16. Annual electricity, heating and cooling duty

protein_enabled; rank 1; {'feed_preparation': 'F1', 'protein_recovery': 'PR2', 'anaerobic_digestion': 'AD1', 'biogas_upgrading': 'BG4', 'nutrient_recovery': 'NR7', 'process_order': 'A'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate kpis

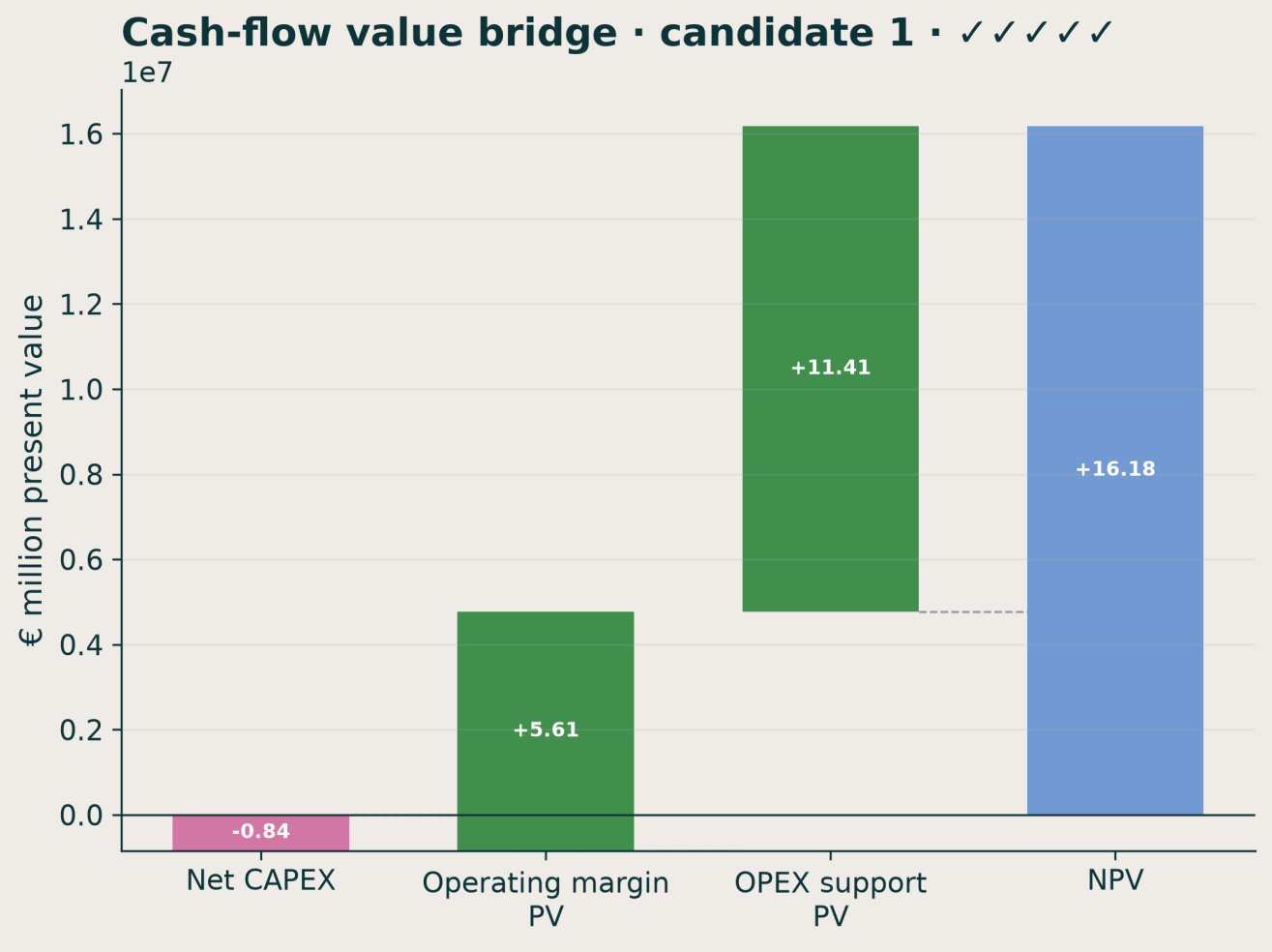


Figure 17. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 1; support C | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate products

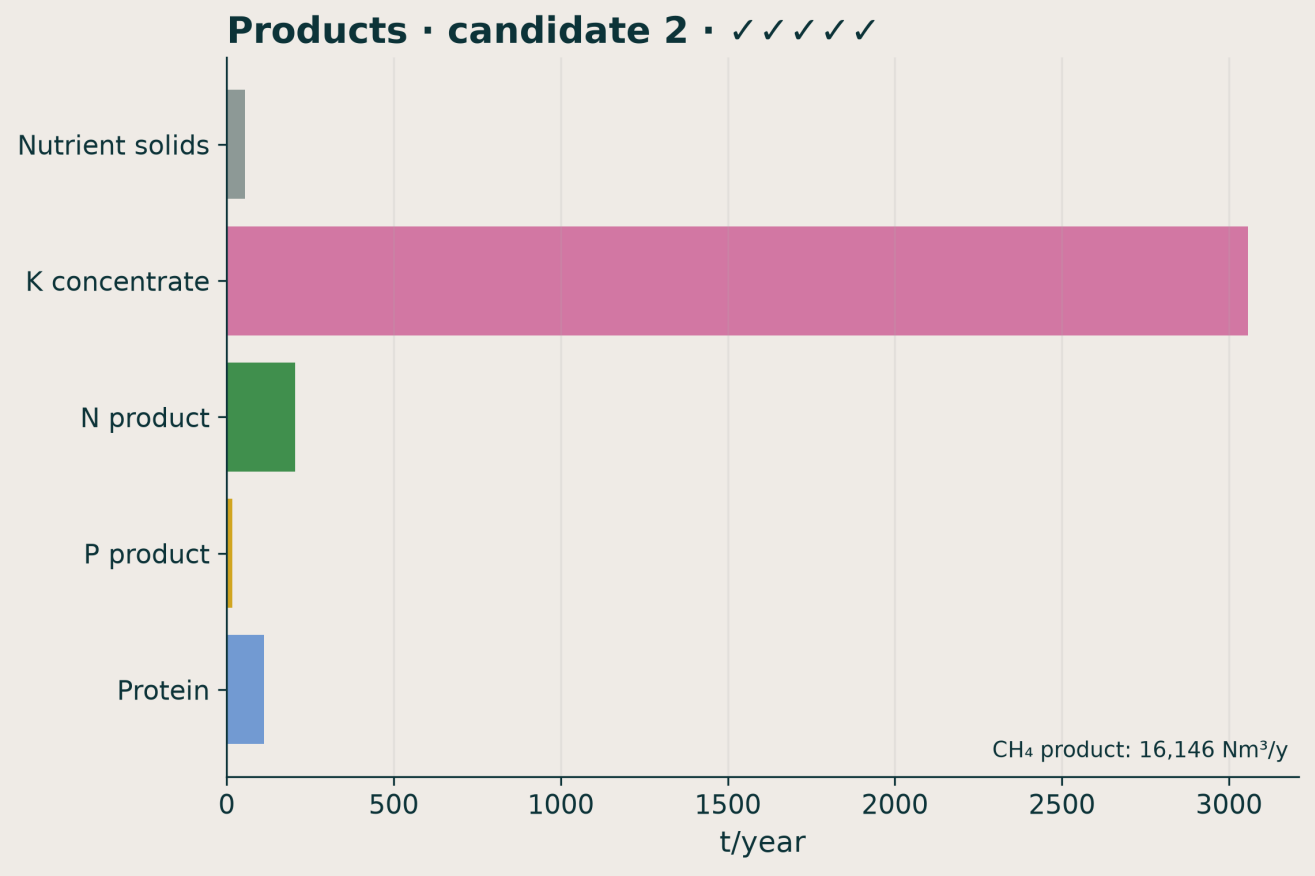


Figure 18. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_enabled; rank 2; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD1', 'biogas_upgrading': 'BG4', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate utilities

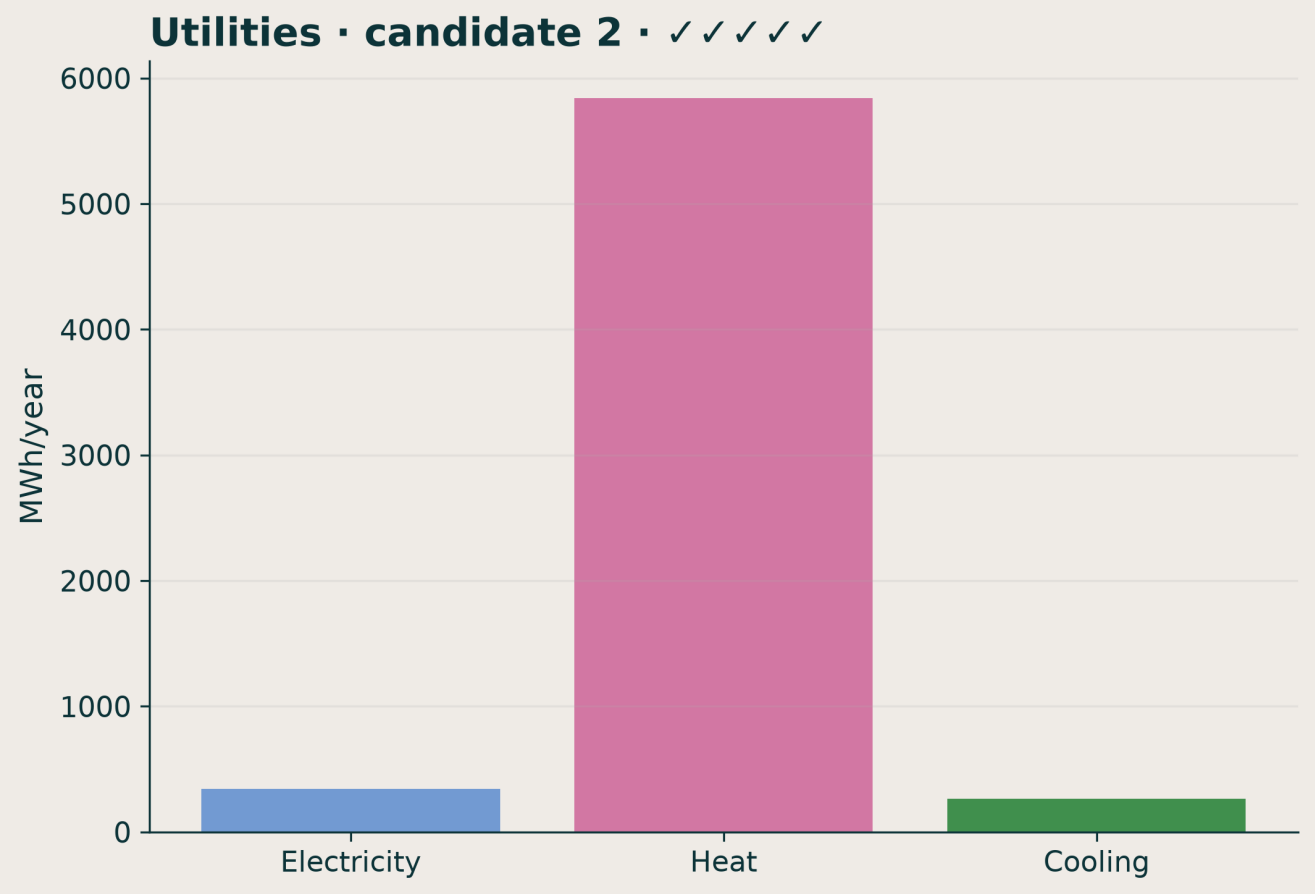


Figure 19. Annual electricity, heating and cooling duty

protein_enabled; rank 2; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD1', 'biogas_upgrading': 'BG4', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate kpis

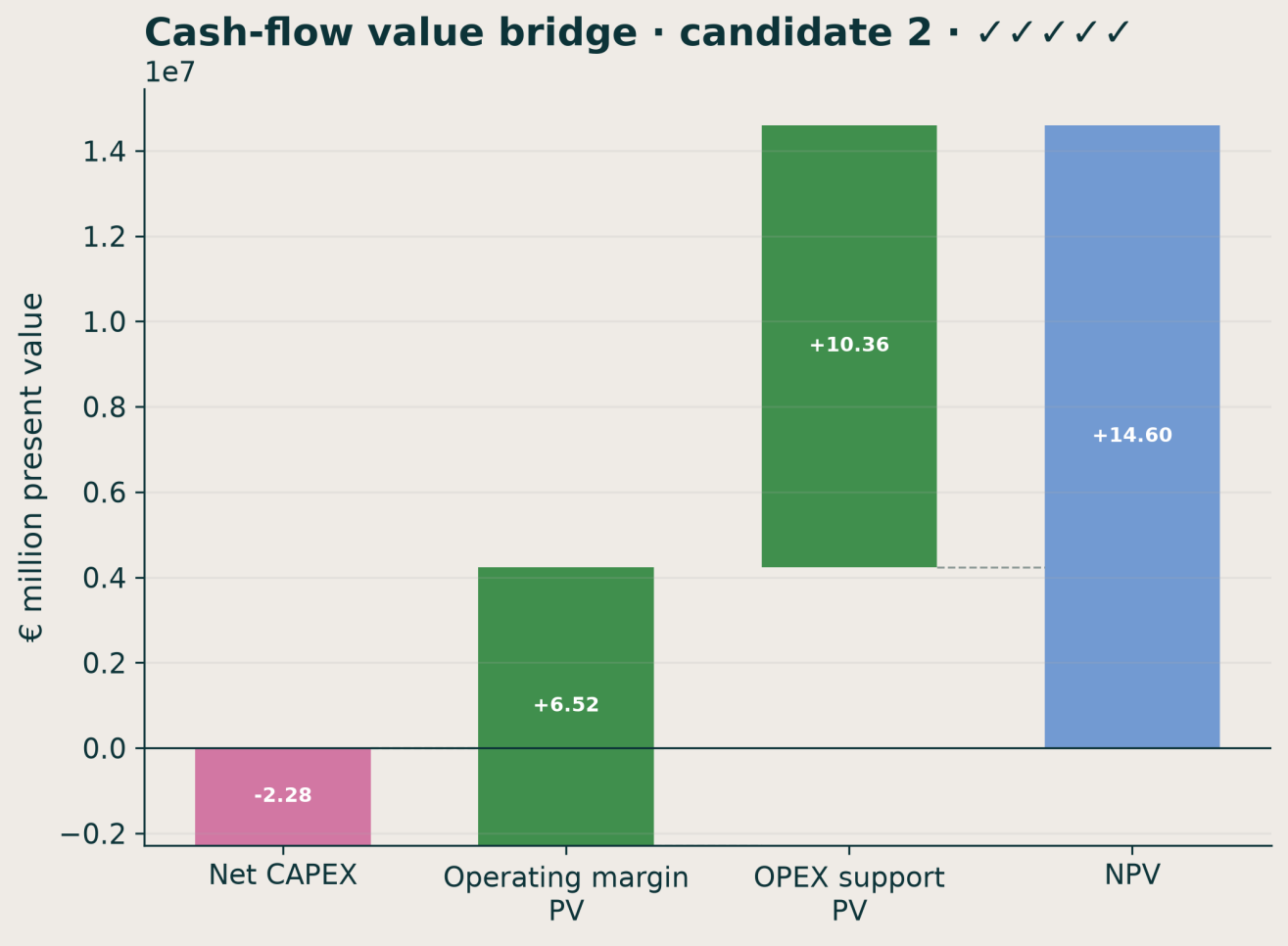


Figure 20. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 2; support D | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate products

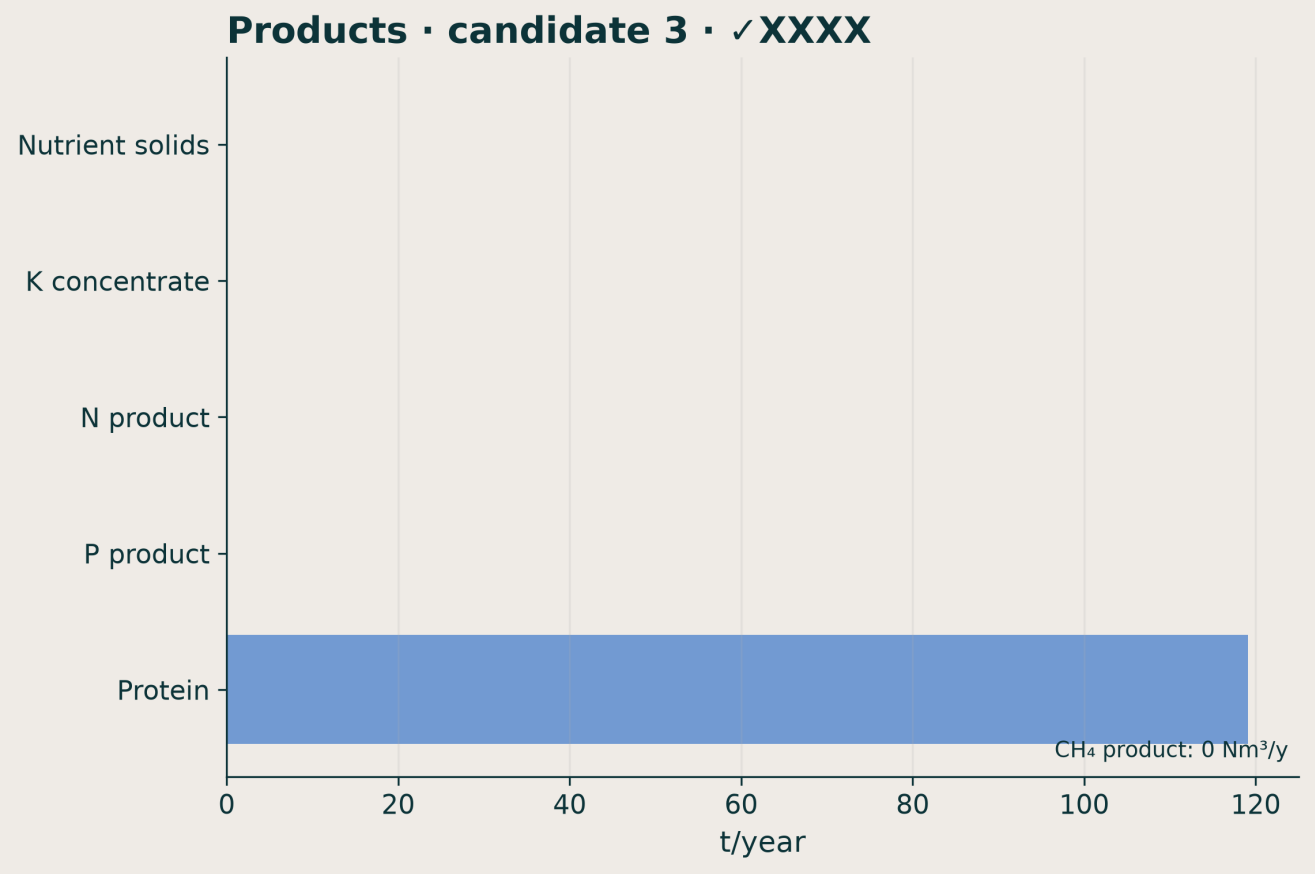


Figure 21. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_enabled; rank 3; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate utilities

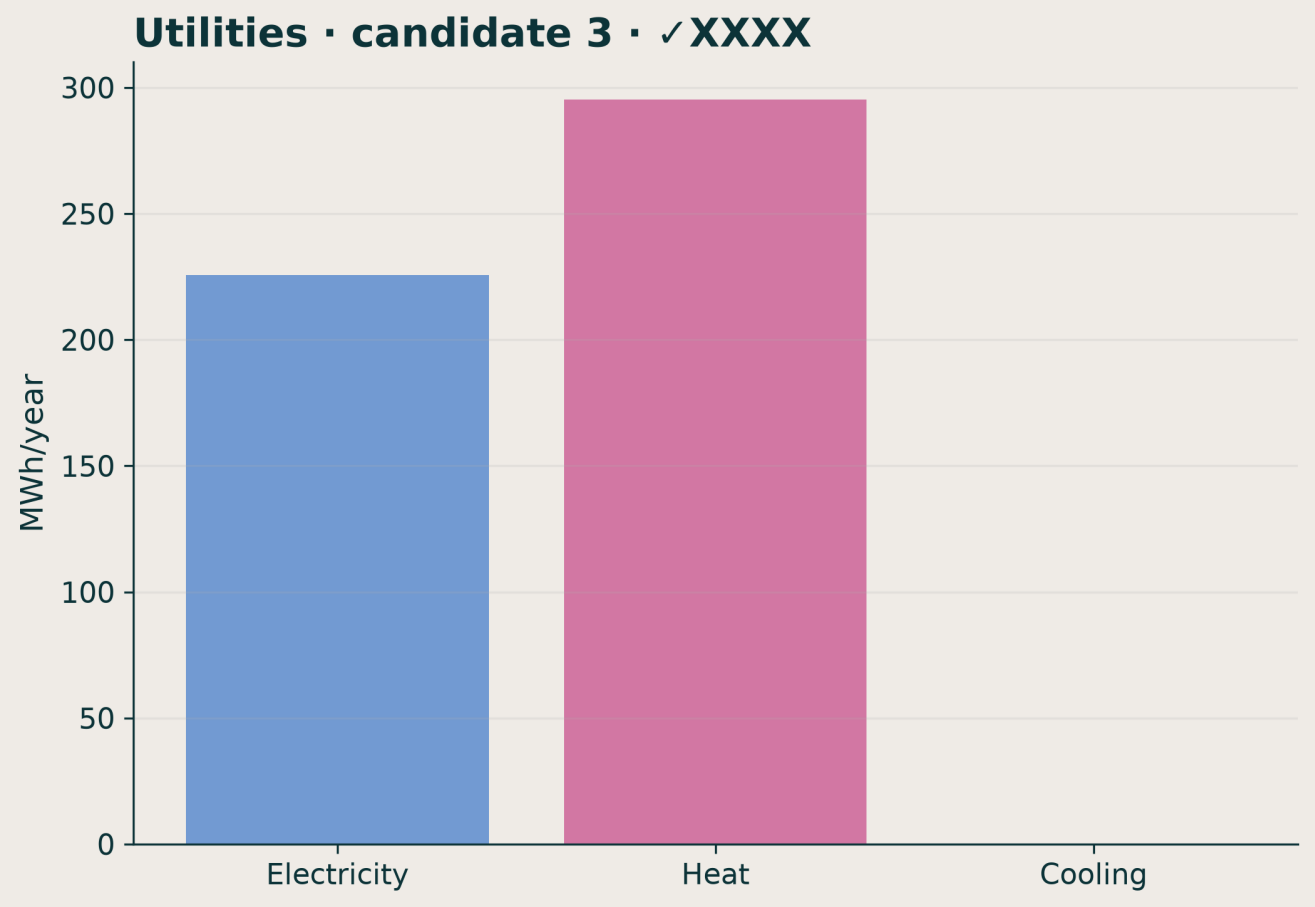


Figure 22. Annual electricity, heating and cooling duty

protein_enabled; rank 3; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate kpis

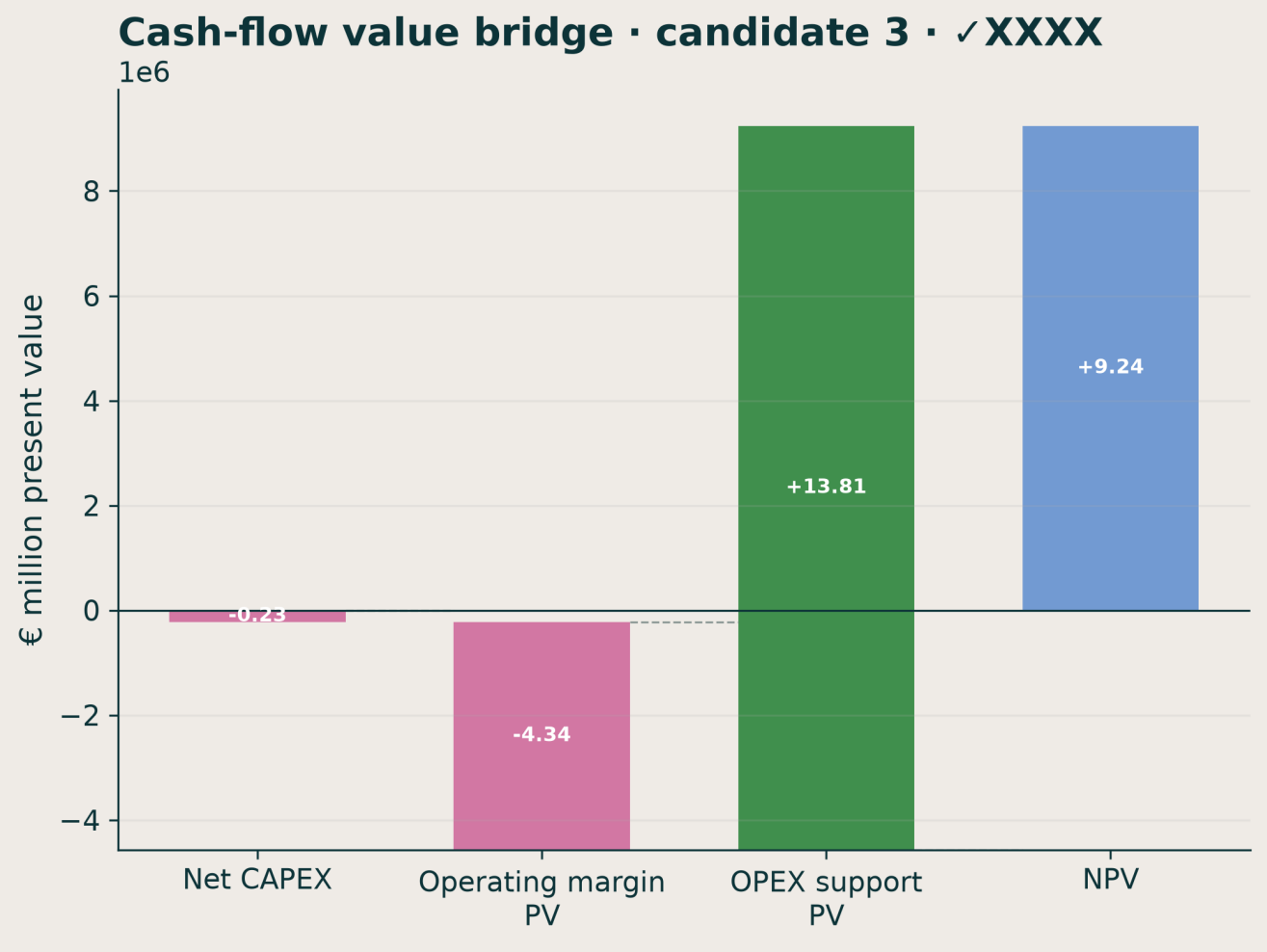


Figure 23. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 3; support C | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate products

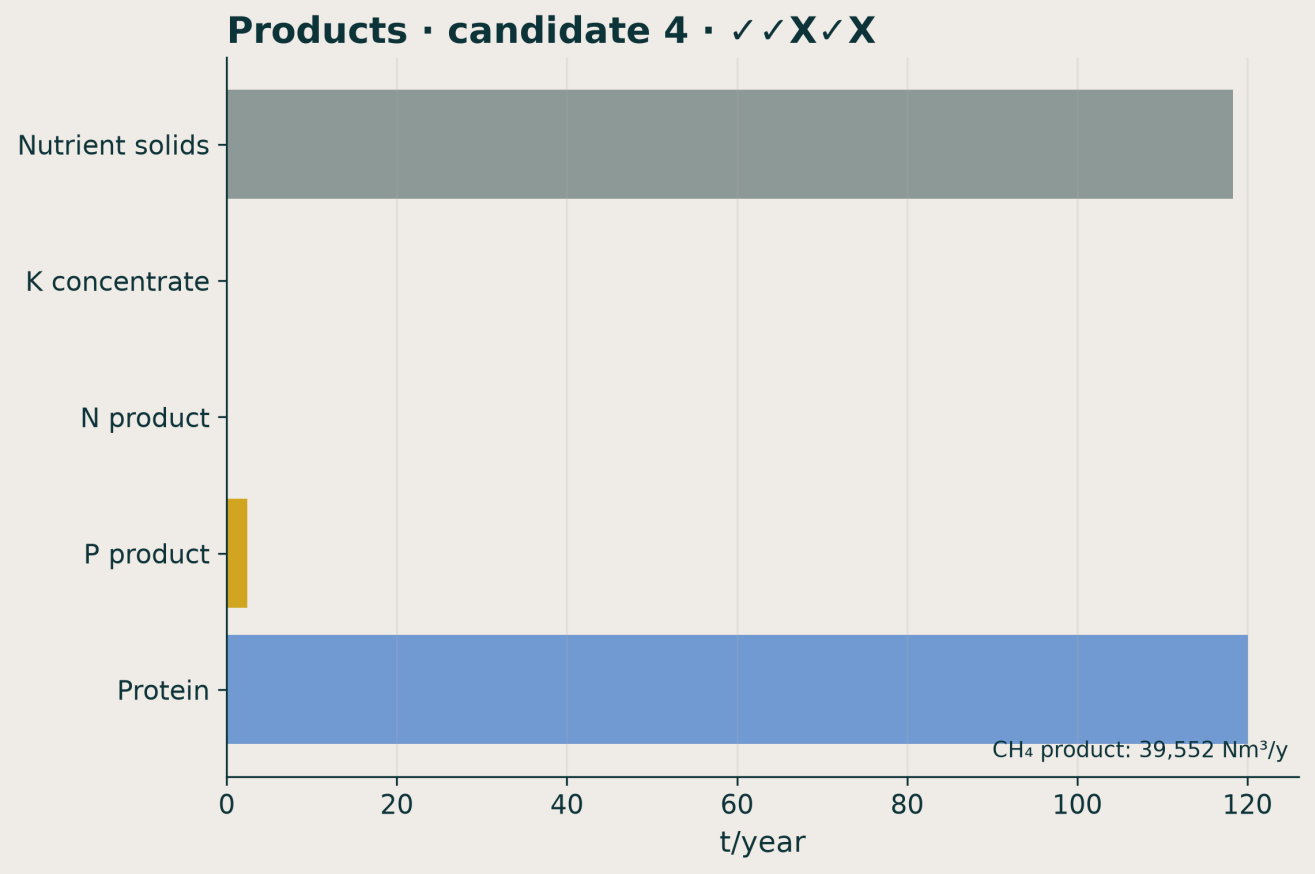


Figure 24. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_enabled; rank 4; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR2', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate utilities

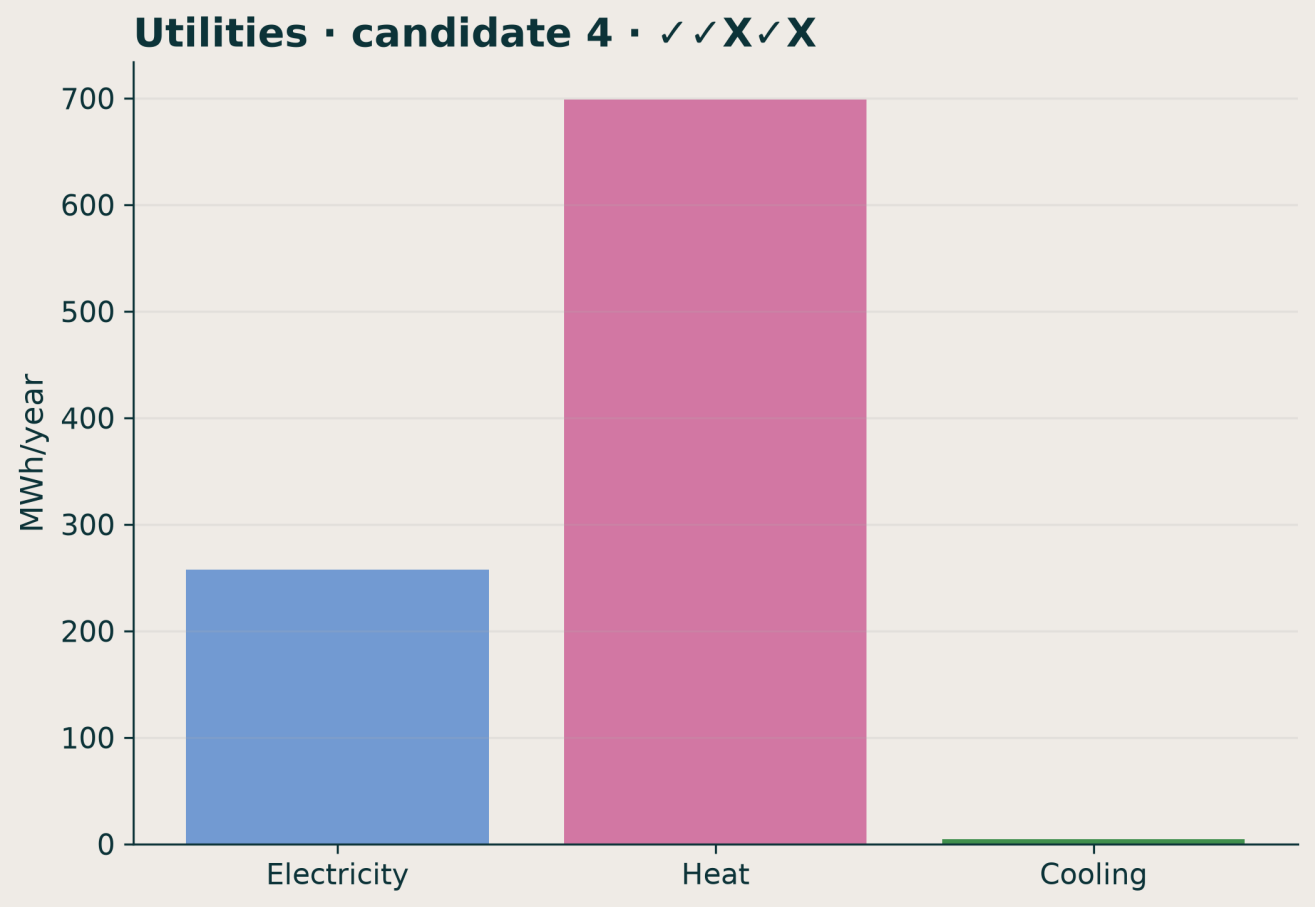


Figure 25. Annual electricity, heating and cooling duty

protein_enabled; rank 4; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR2', 'process_order': 'A'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate kpis

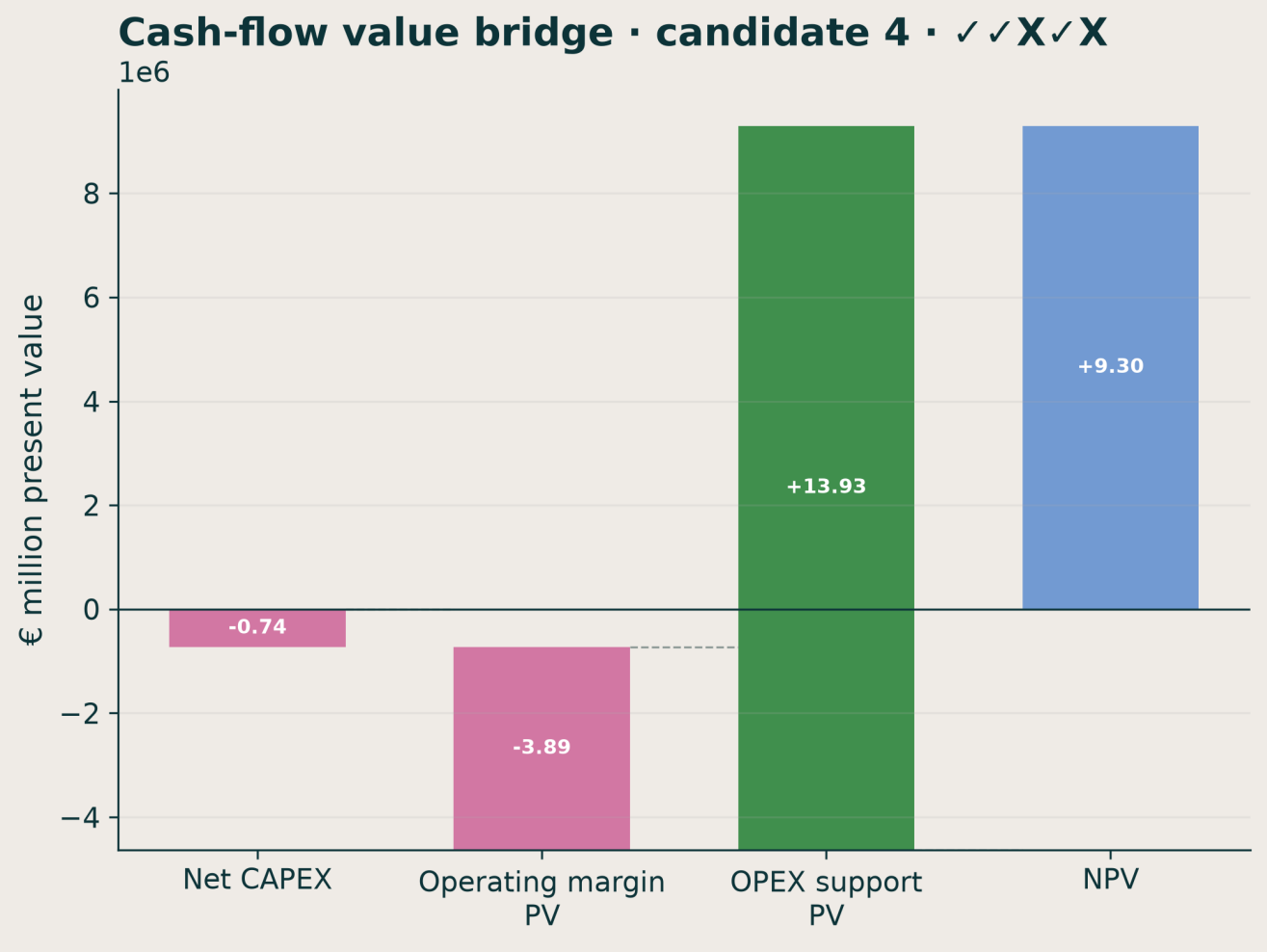


Figure 26. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 4; support C | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate products

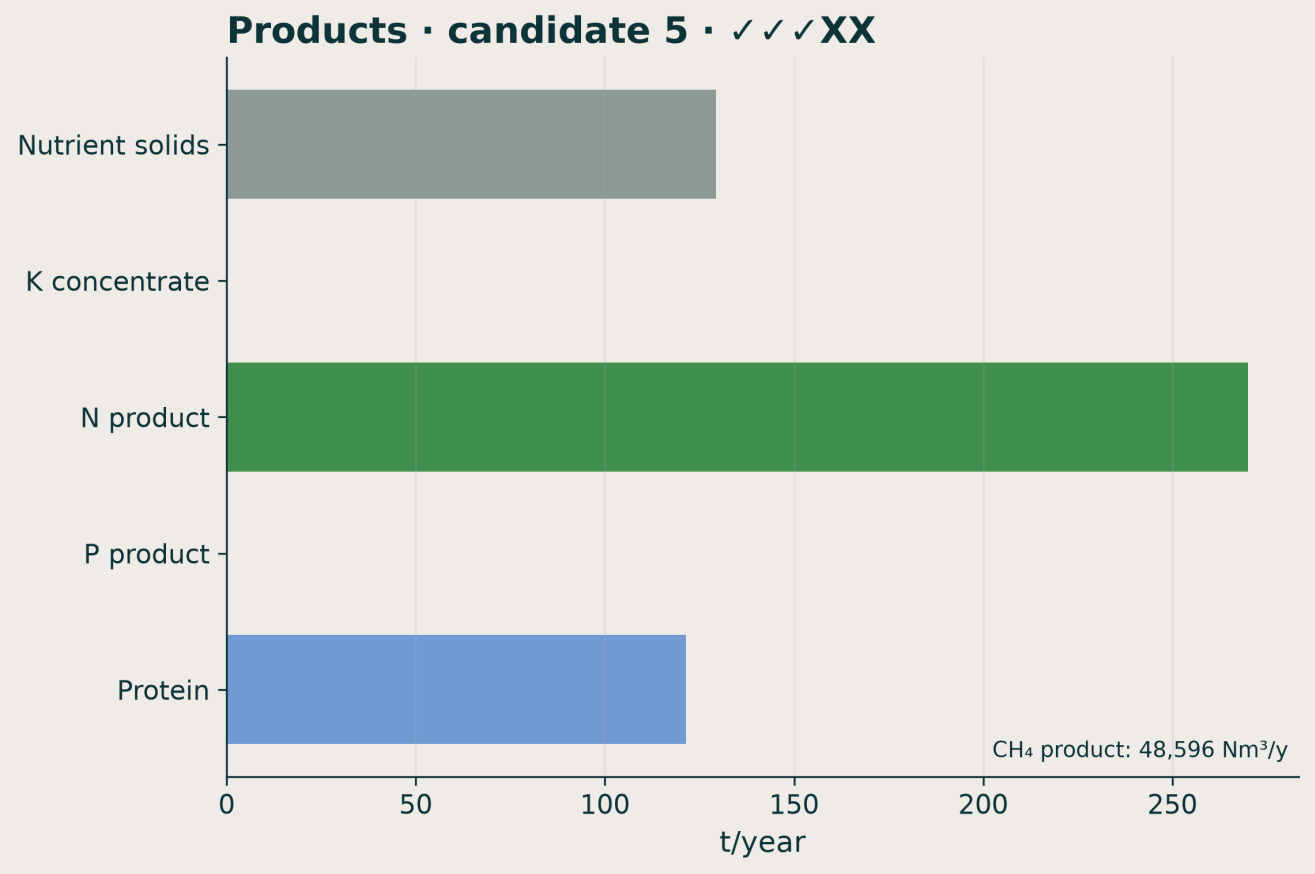


Figure 27. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_enabled; rank 5; {'feed_preparation': 'F1', 'protein_recovery': 'PR2', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR3', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate utilities

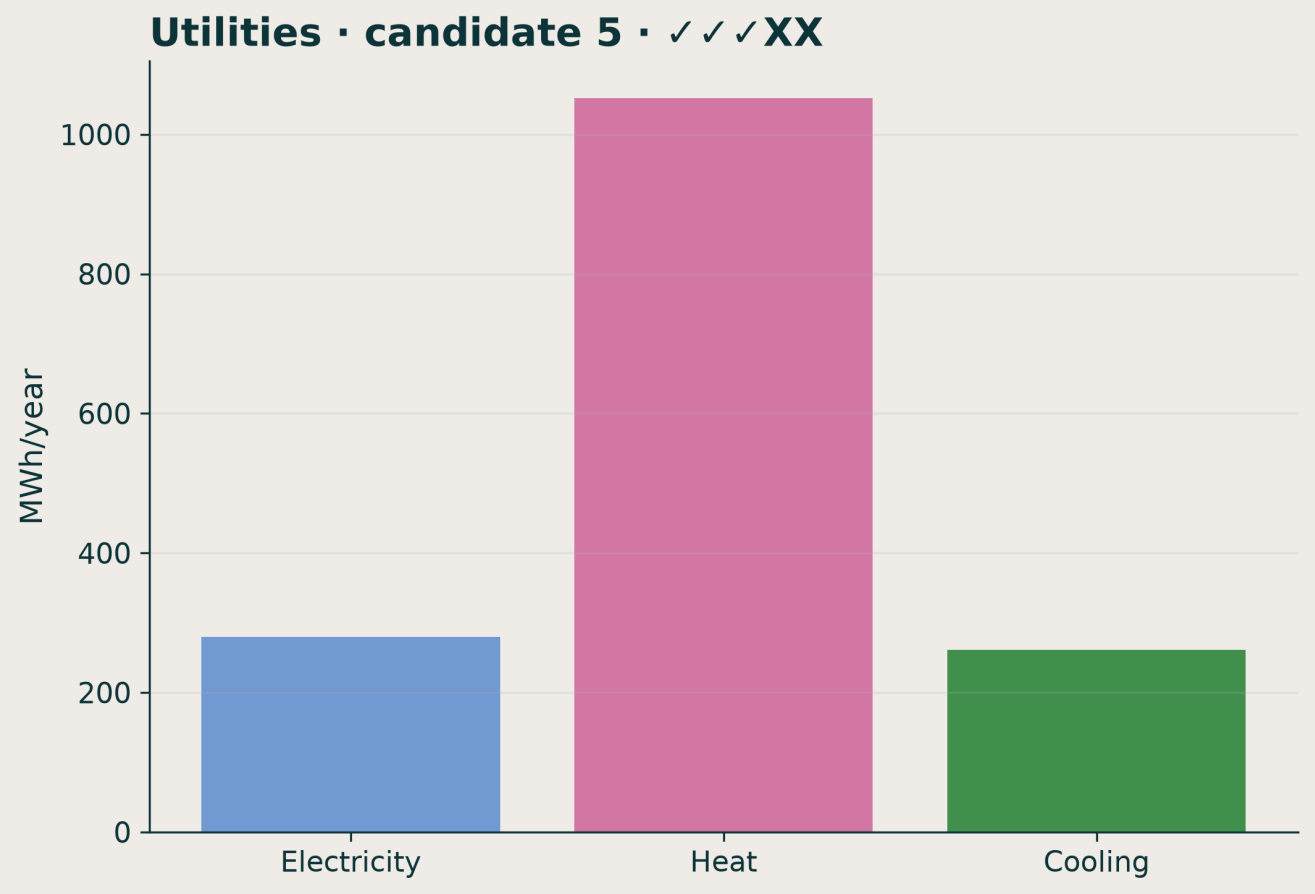


Figure 28. Annual electricity, heating and cooling duty

protein_enabled; rank 5; {'feed_preparation': 'F1', 'protein_recovery': 'PR2', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR3', 'process_order': 'A'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate kpis

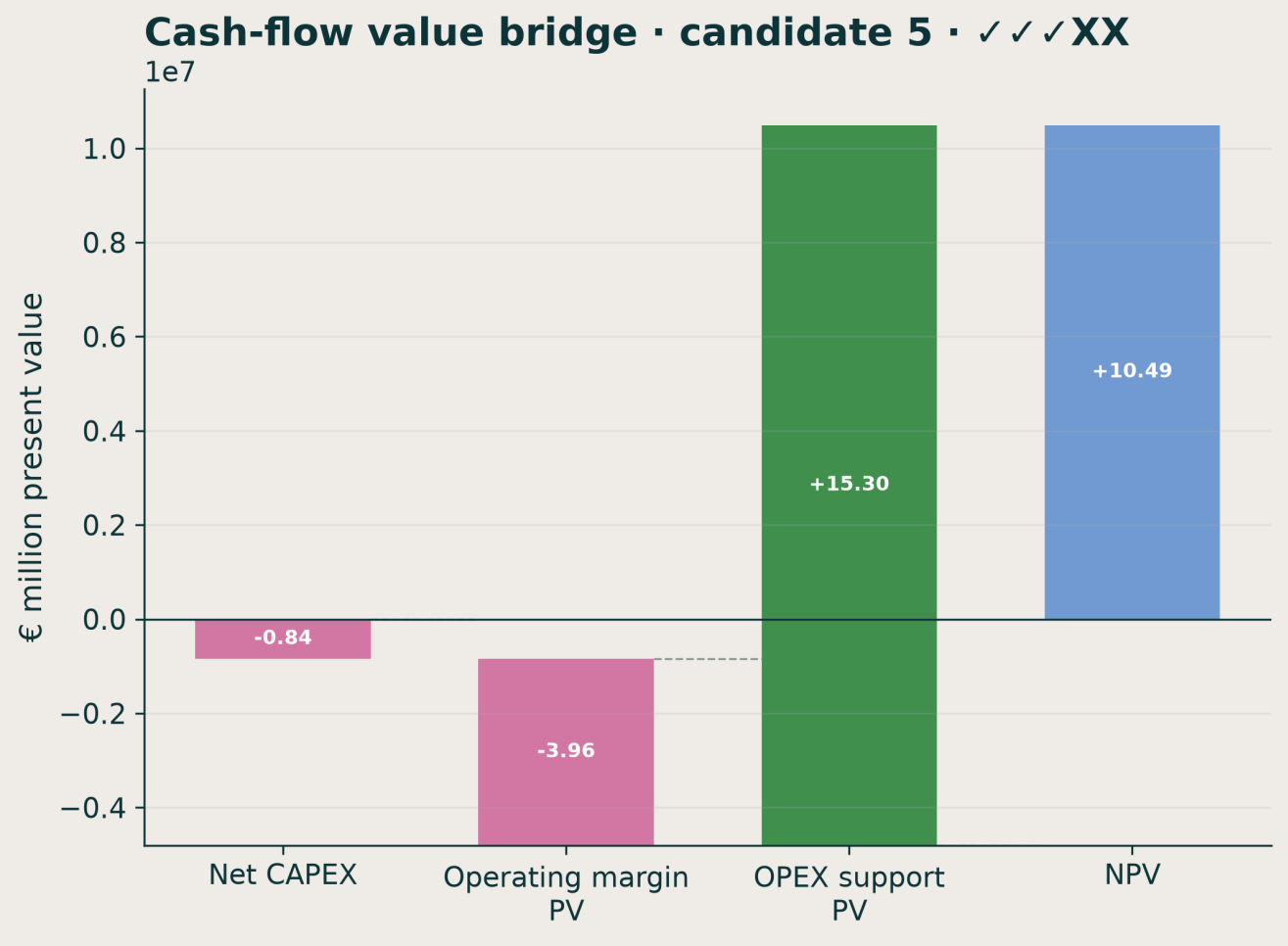


Figure 29. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 5; support C | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate products

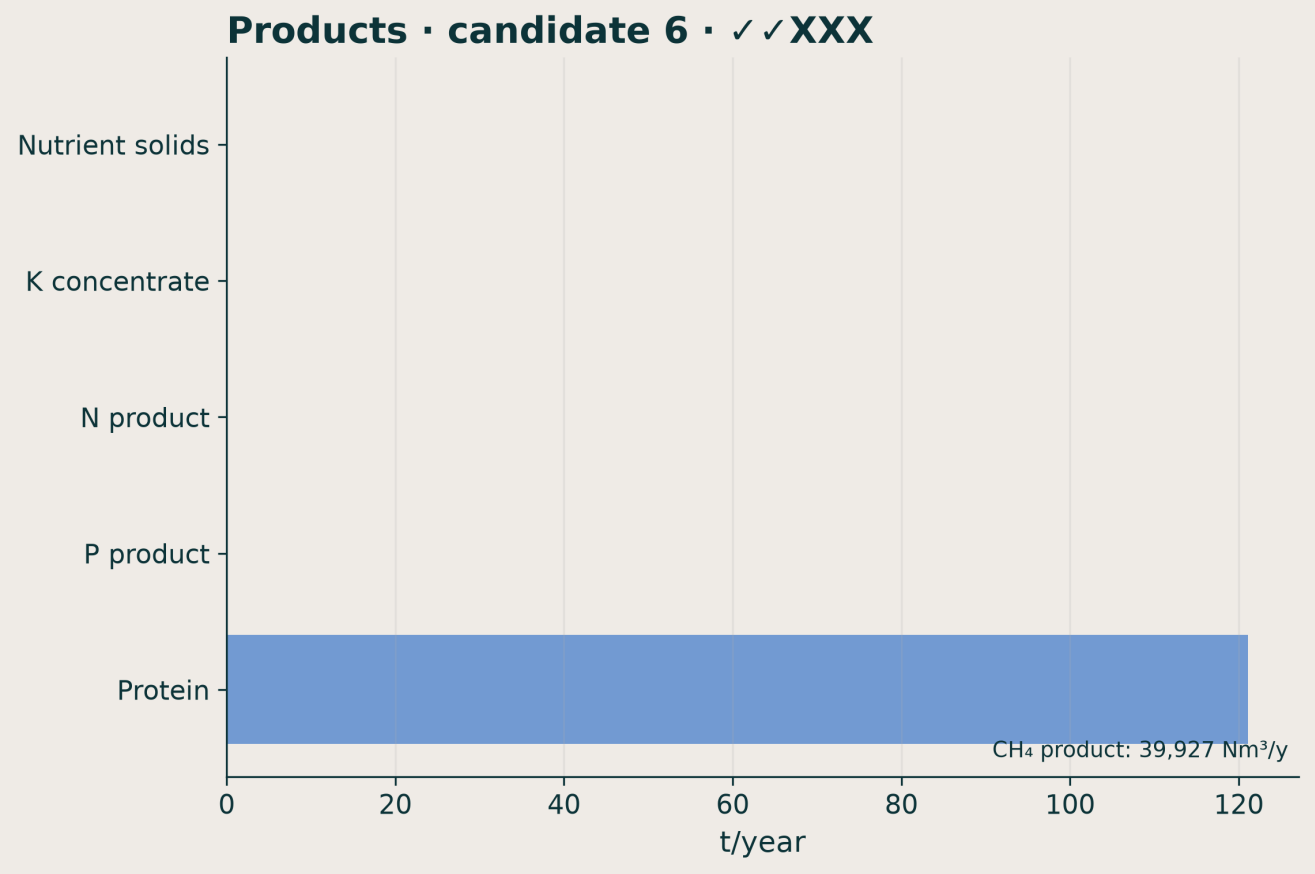


Figure 30. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_enabled; rank 6; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate utilities

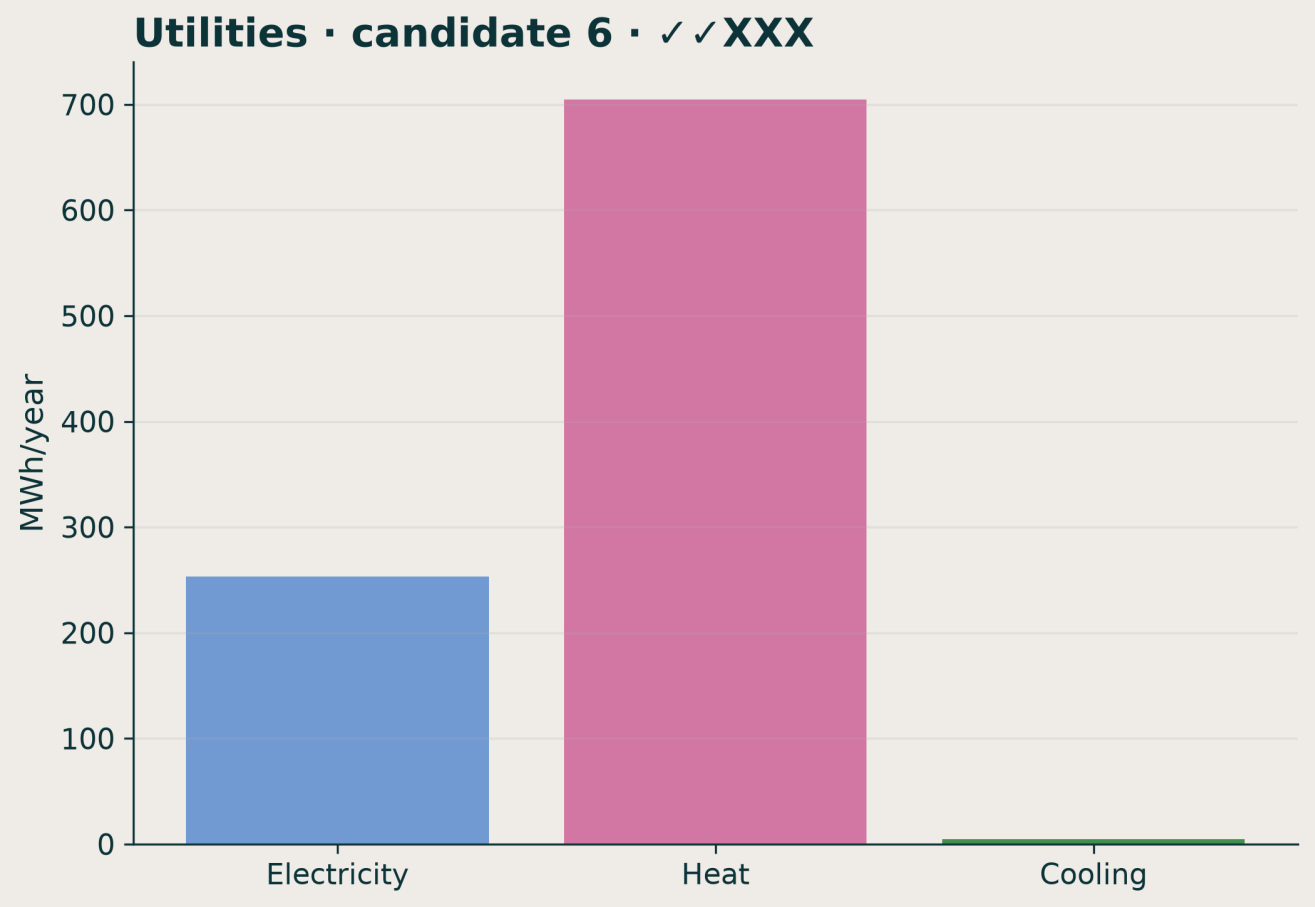


Figure 31. Annual electricity, heating and cooling duty

protein_enabled; rank 6; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate kpis

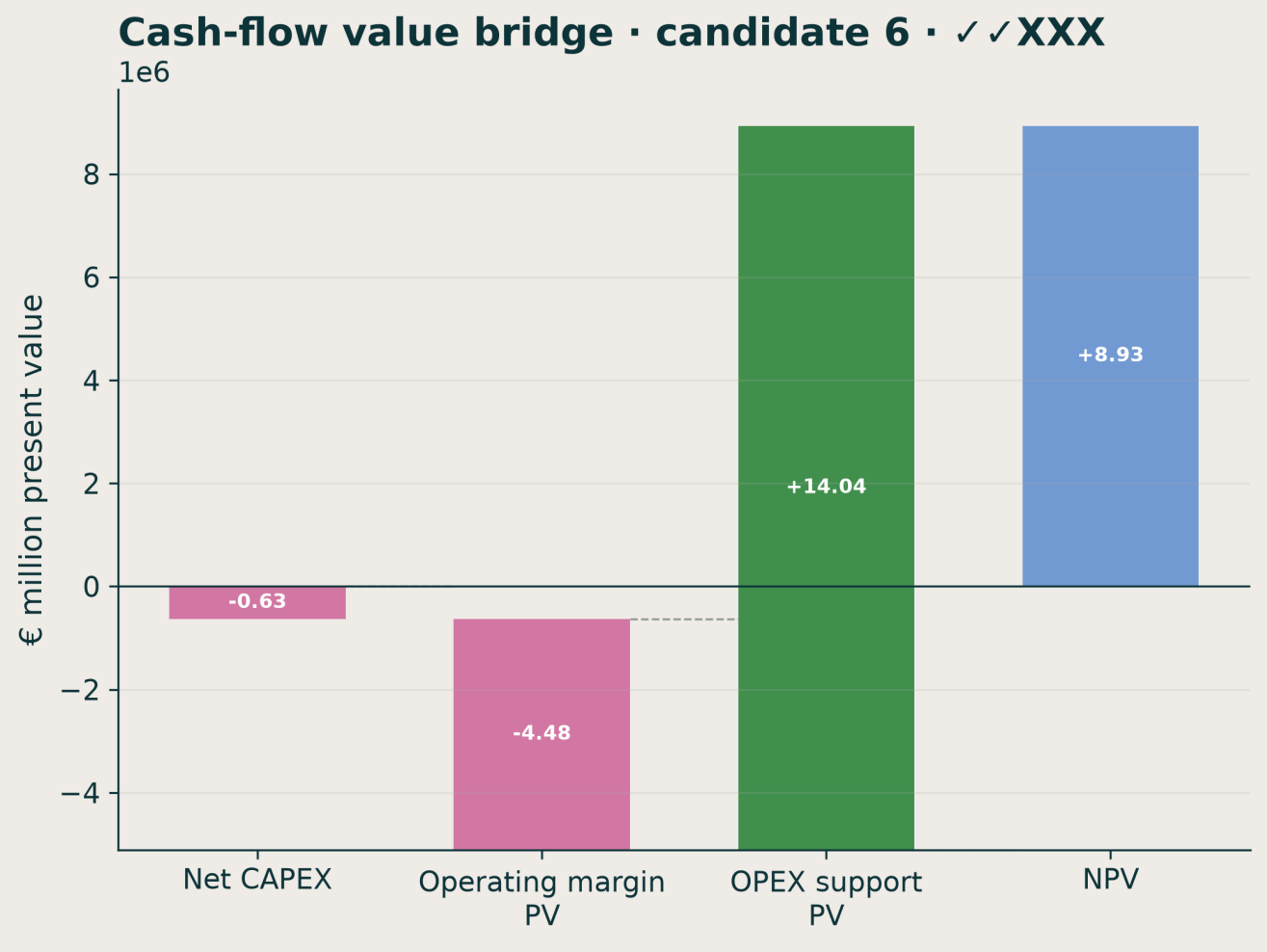


Figure 32. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 6; support C | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate products

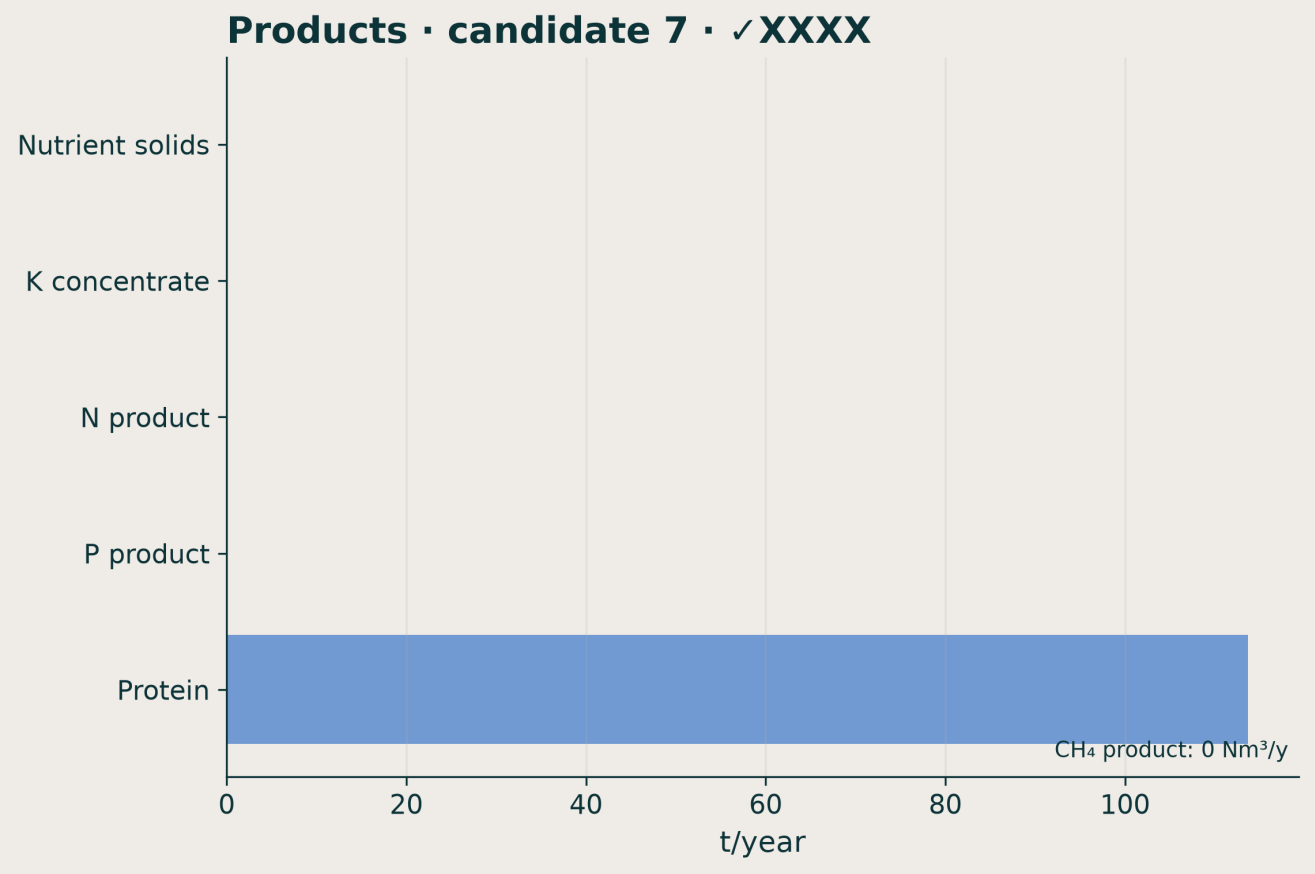


Figure 33. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_enabled; rank 7; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate utilities

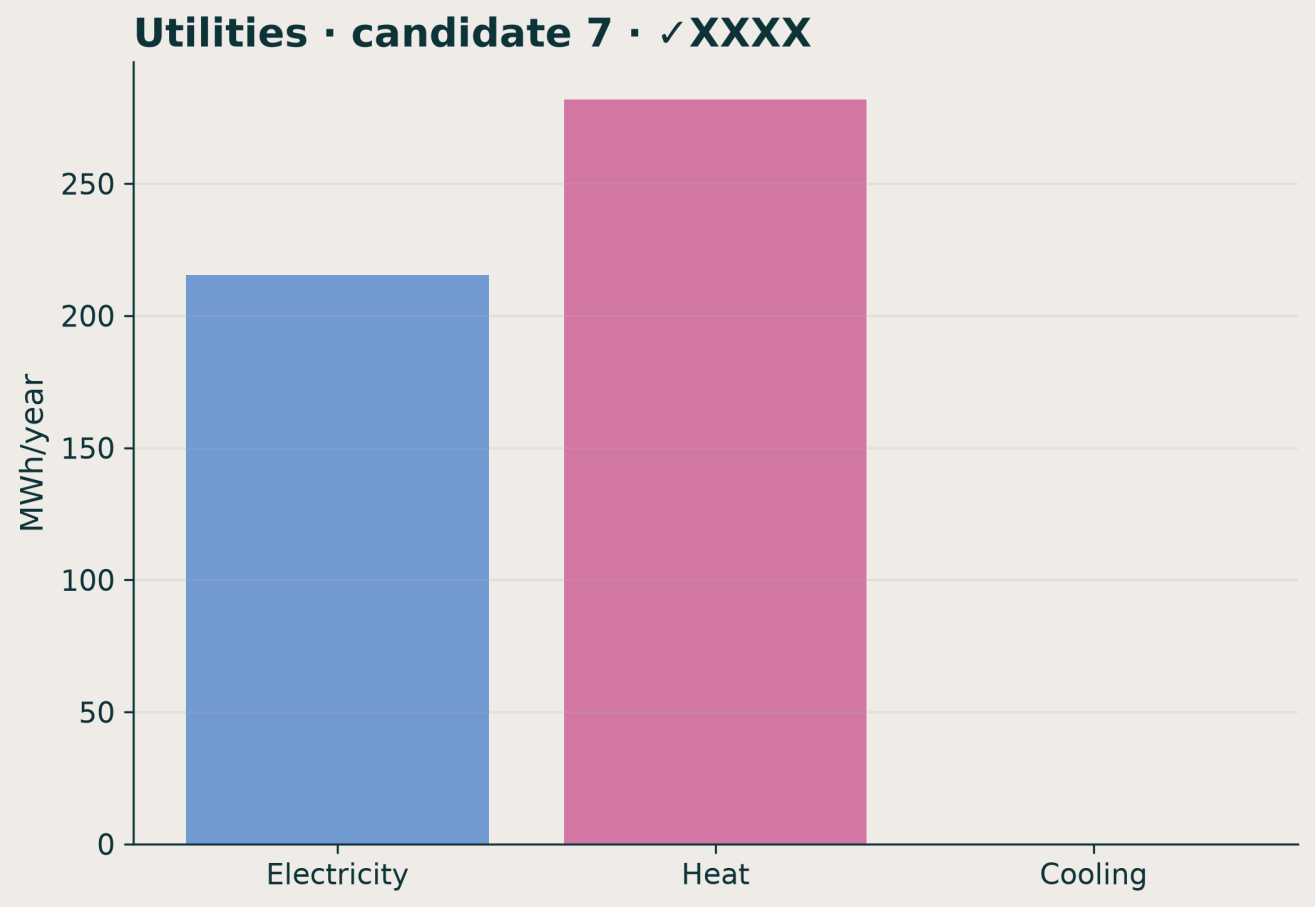


Figure 34. Annual electricity, heating and cooling duty

protein_enabled; rank 7; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate kpis

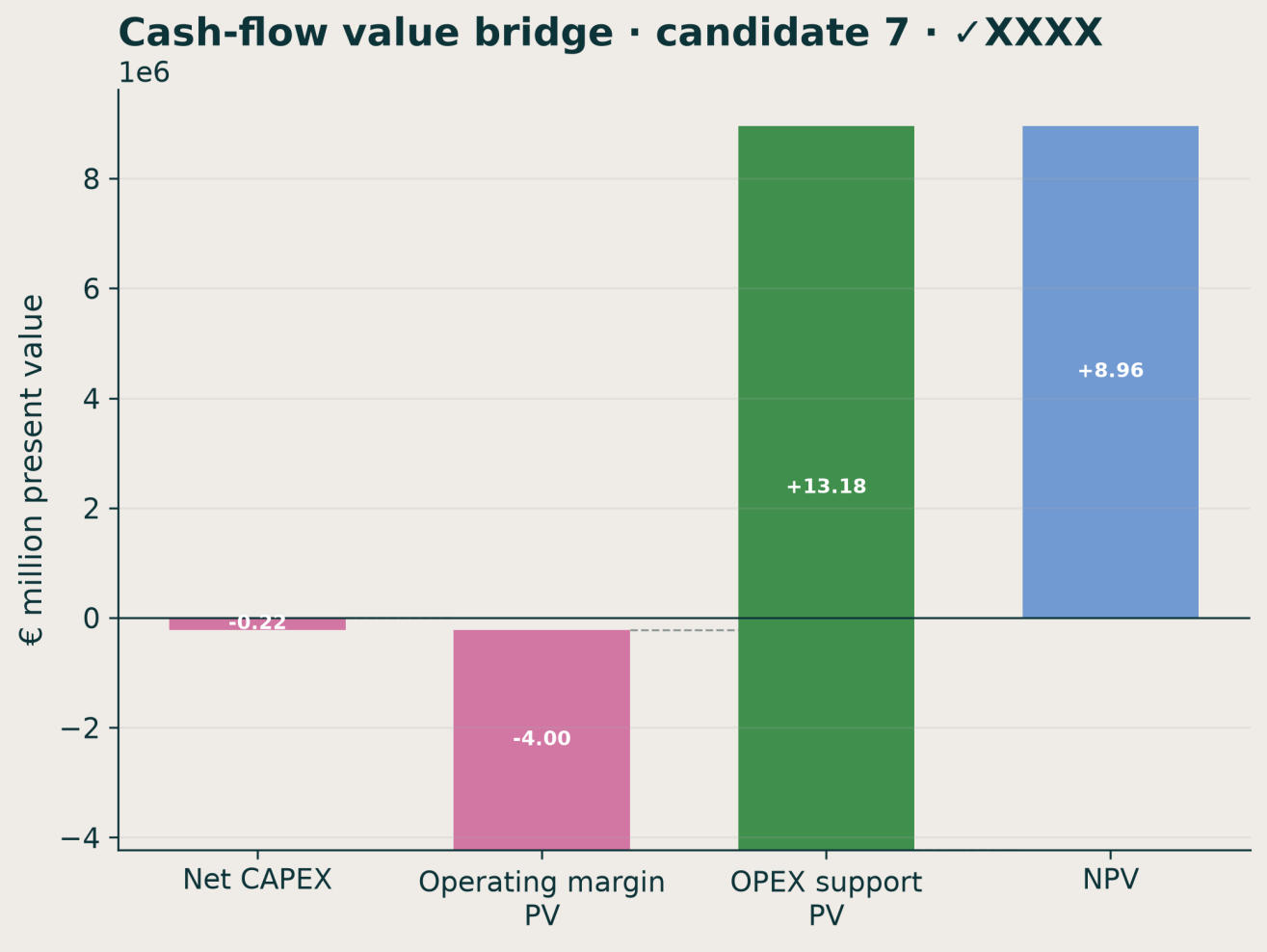


Figure 35. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 7; support C | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate products

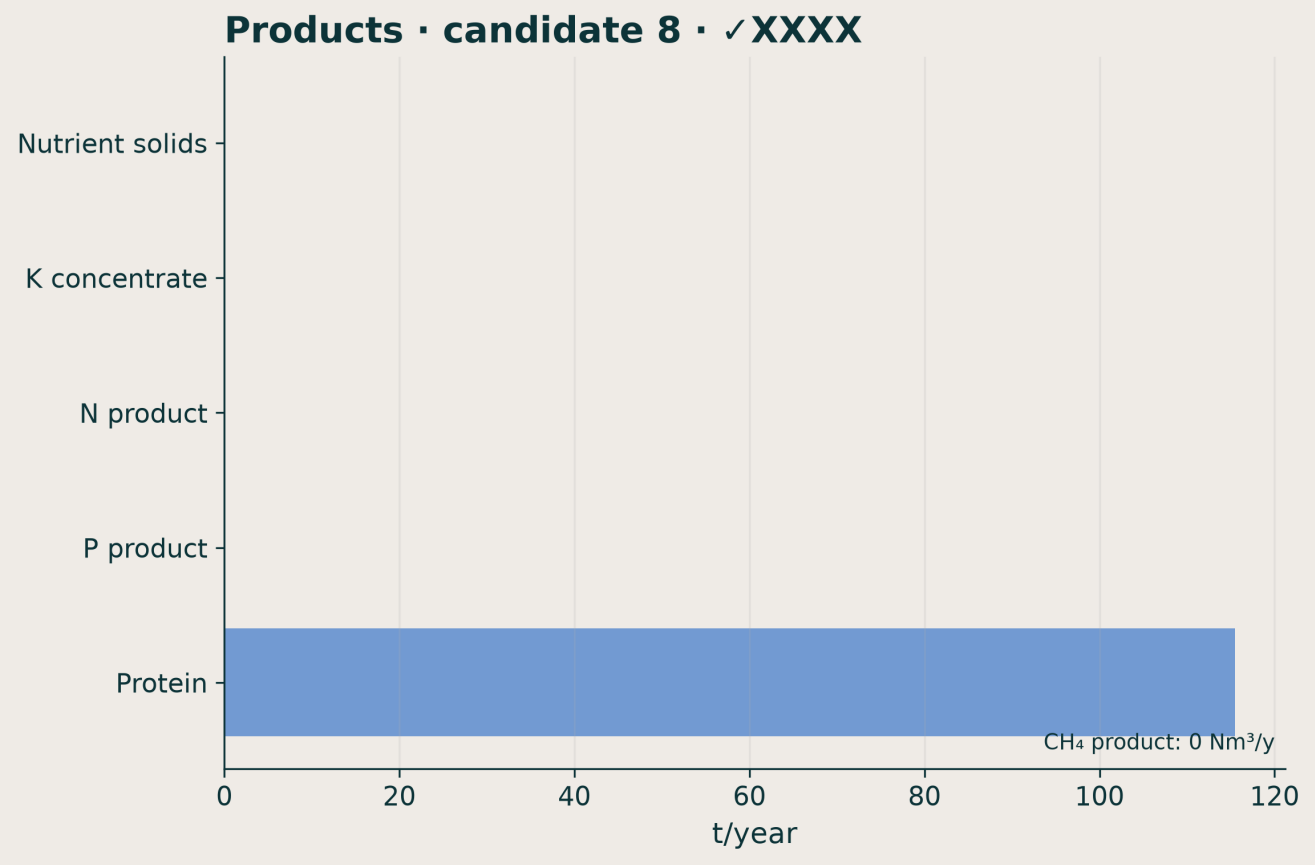


Figure 36. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_enabled; rank 8; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate utilities

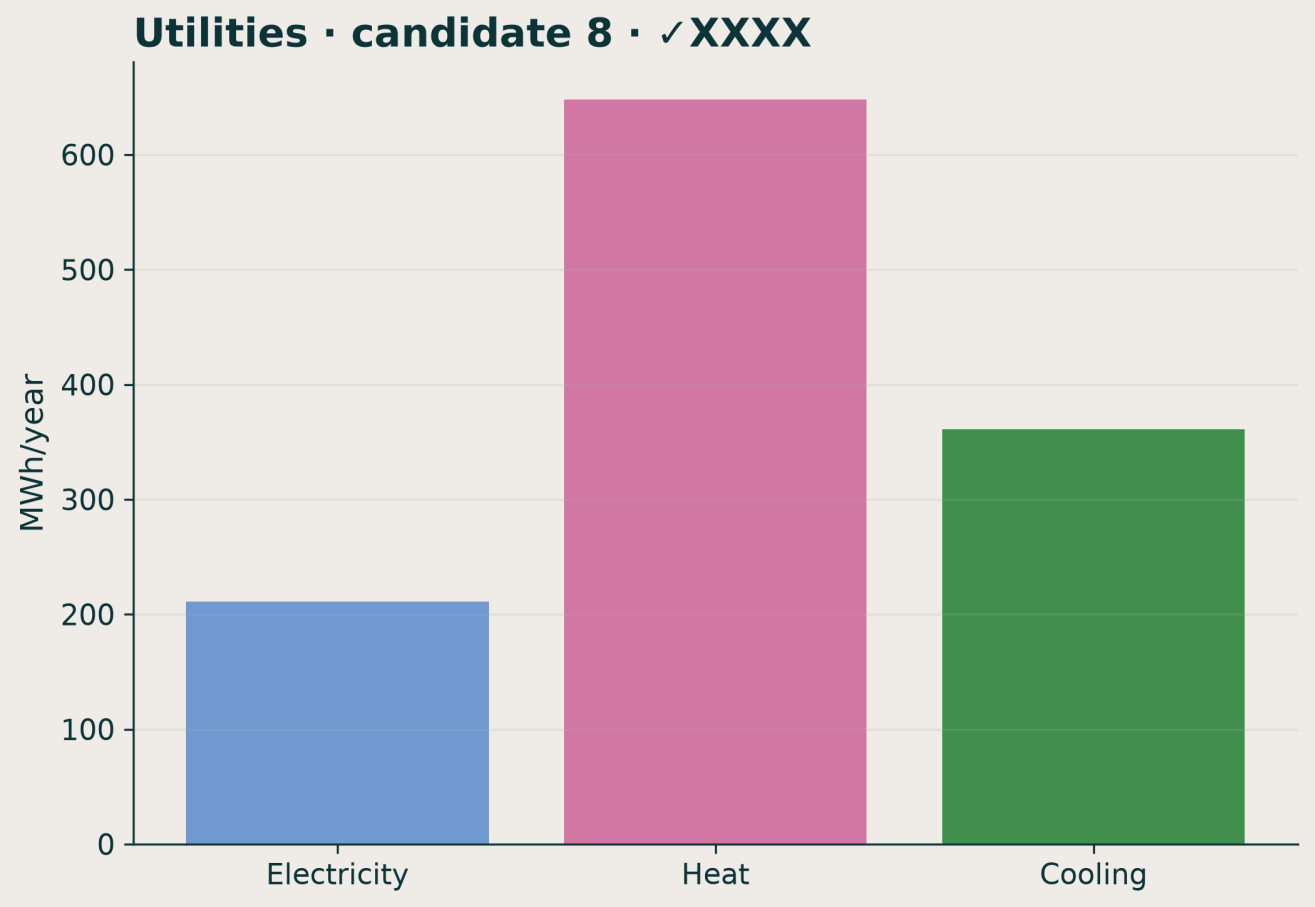


Figure 37. Annual electricity, heating and cooling duty

protein_enabled; rank 8; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate kpis

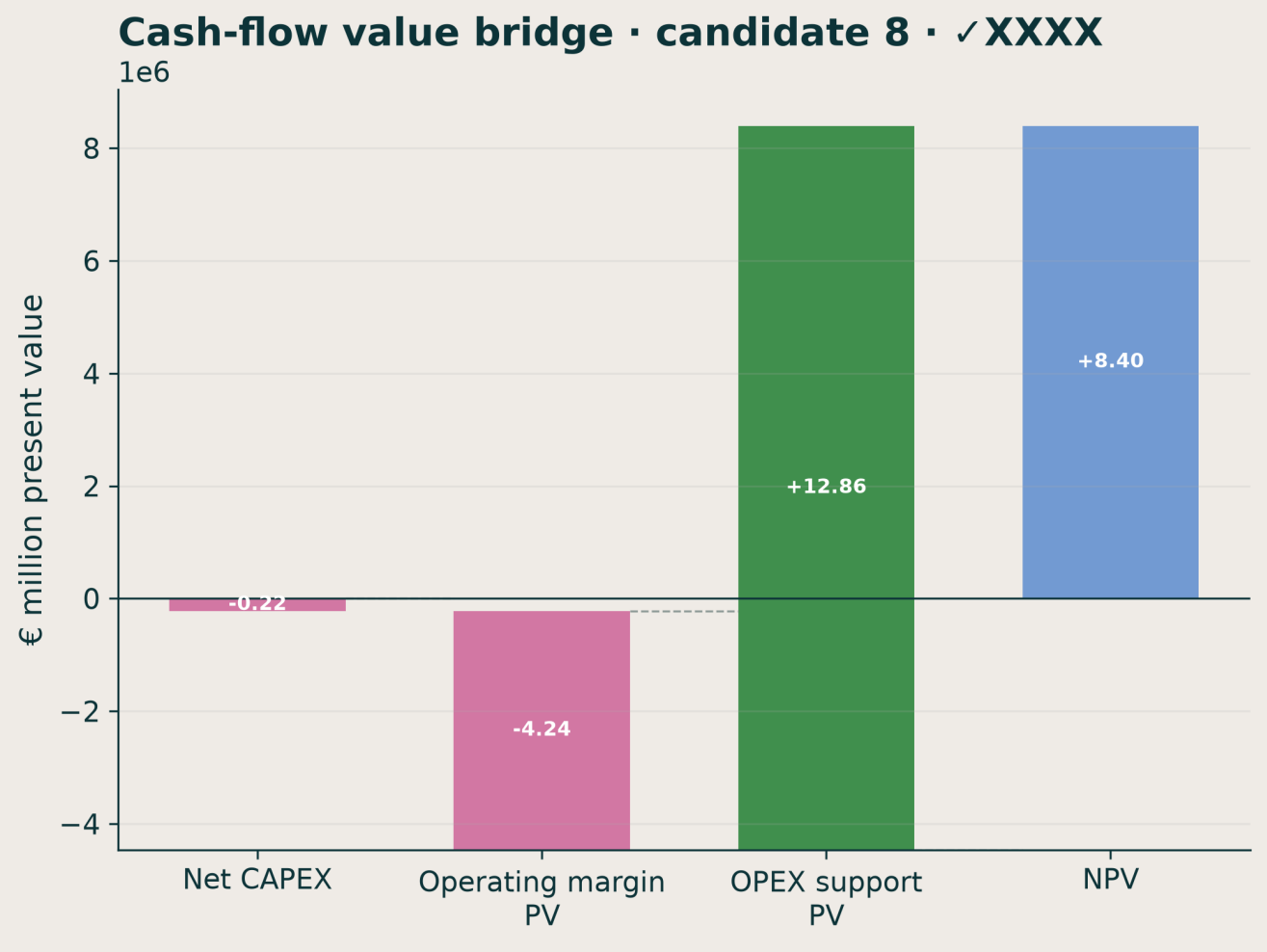


Figure 38. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 8; support C | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate products

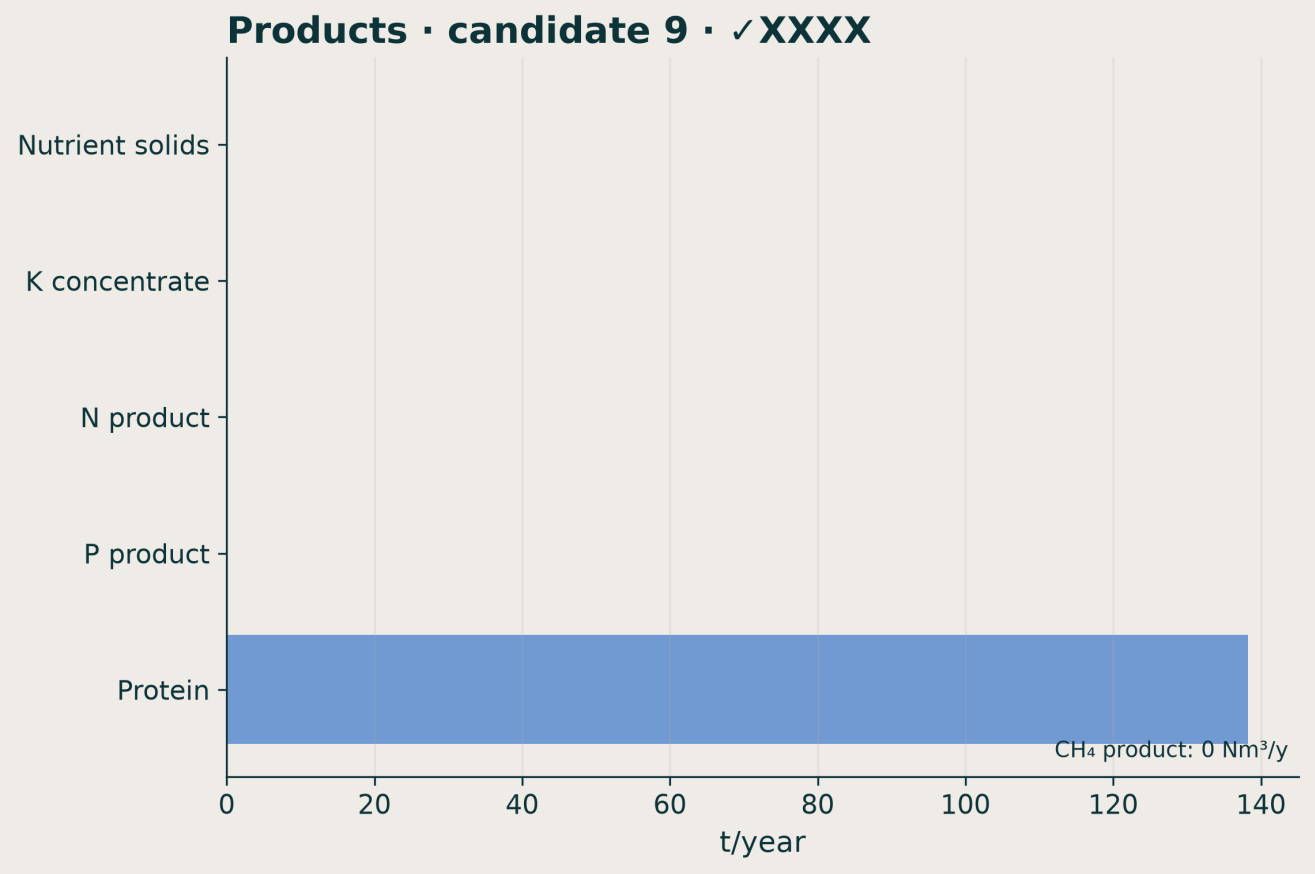


Figure 39. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_enabled; rank 9; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate utilities

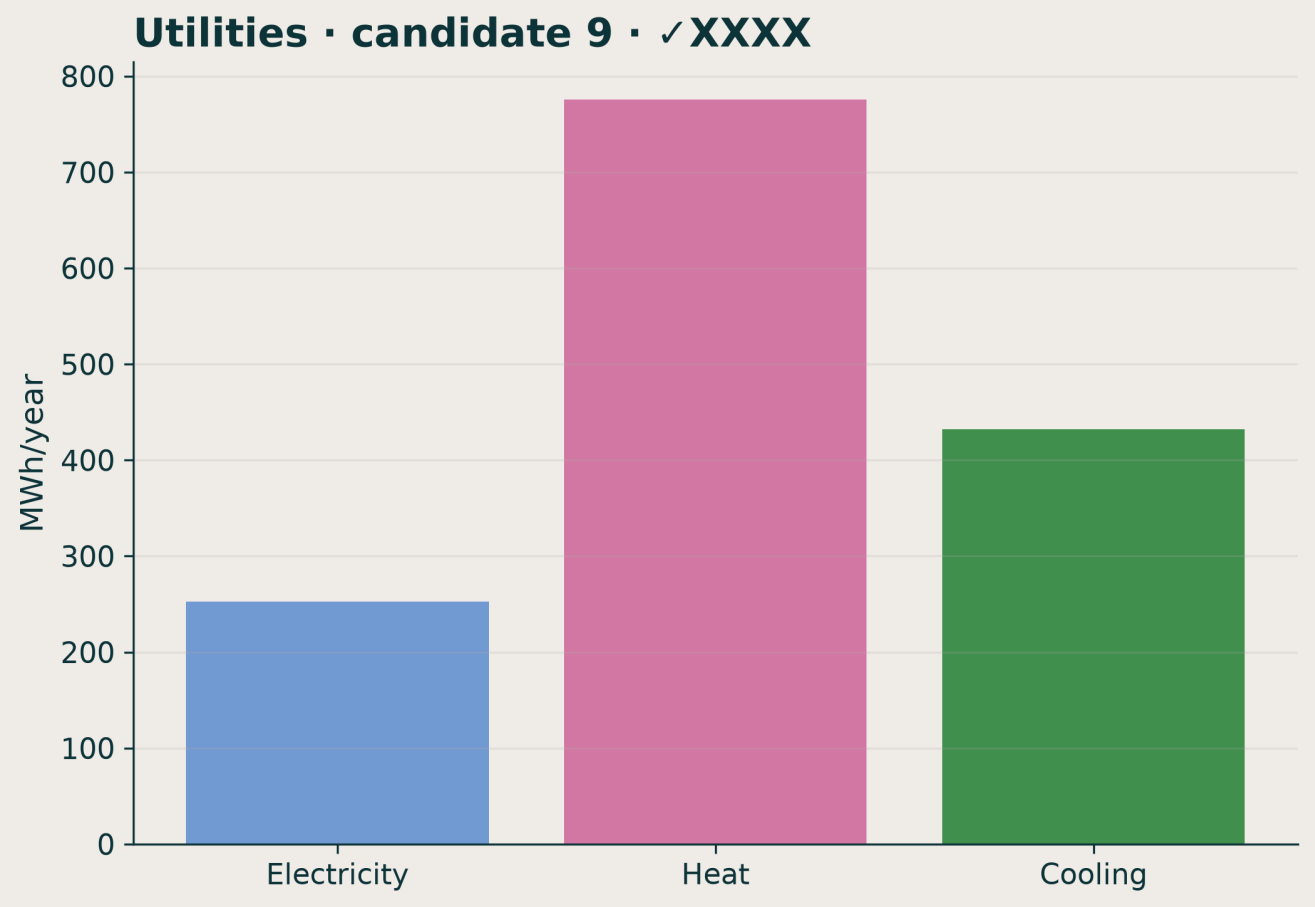


Figure 40. Annual electricity, heating and cooling duty

protein_enabled; rank 9; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate kpis

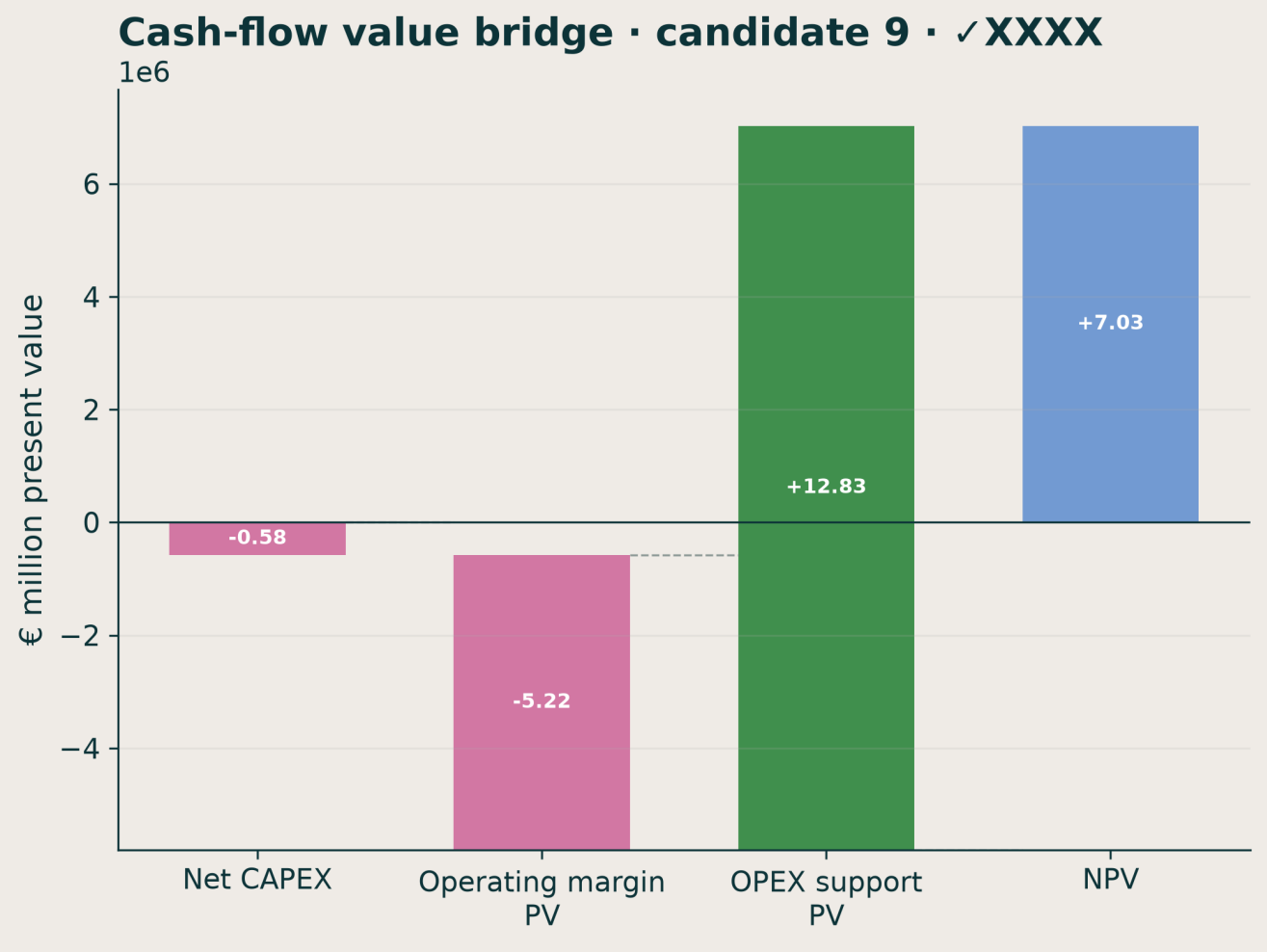


Figure 41. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 9; support D | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate products

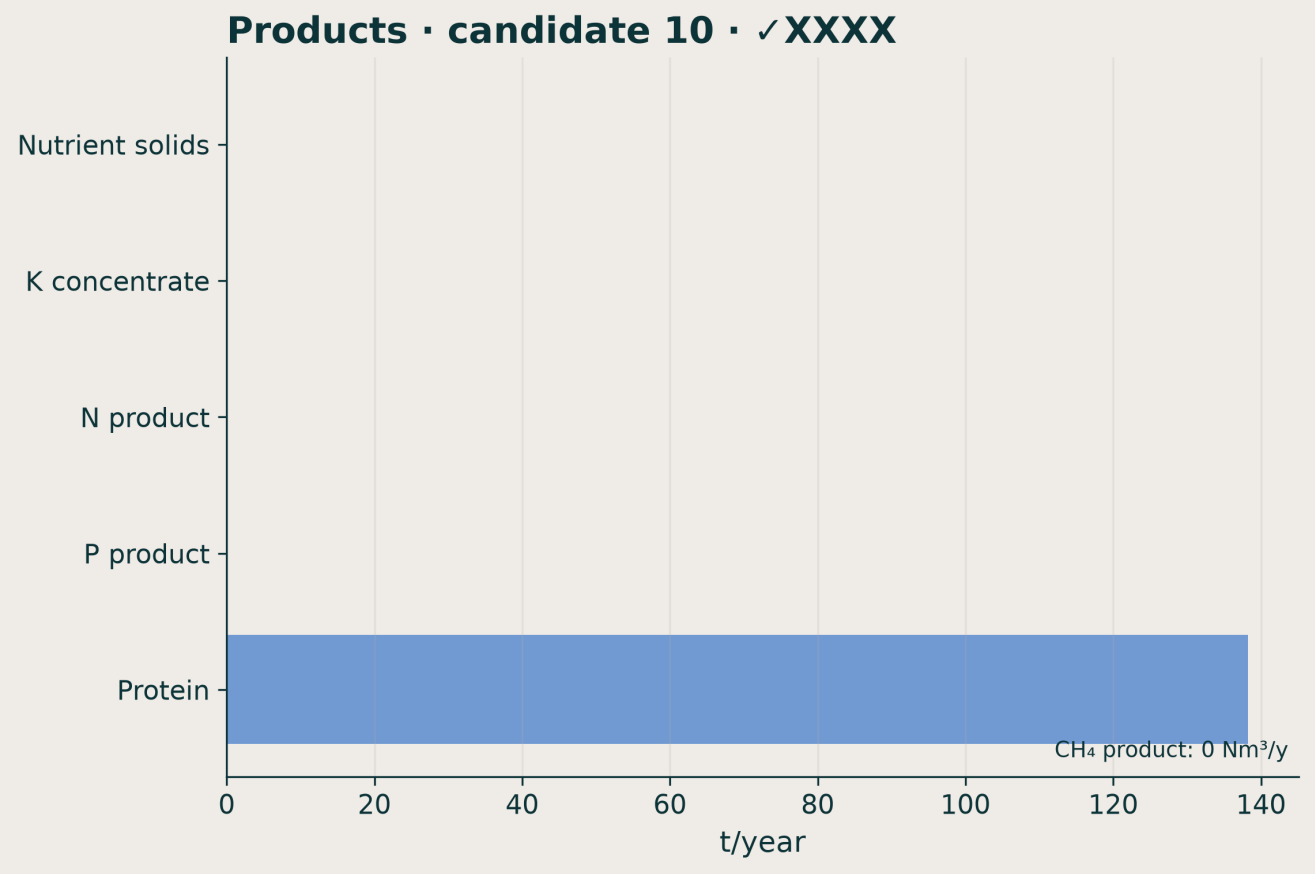


Figure 42. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_enabled; rank 10; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate utilities

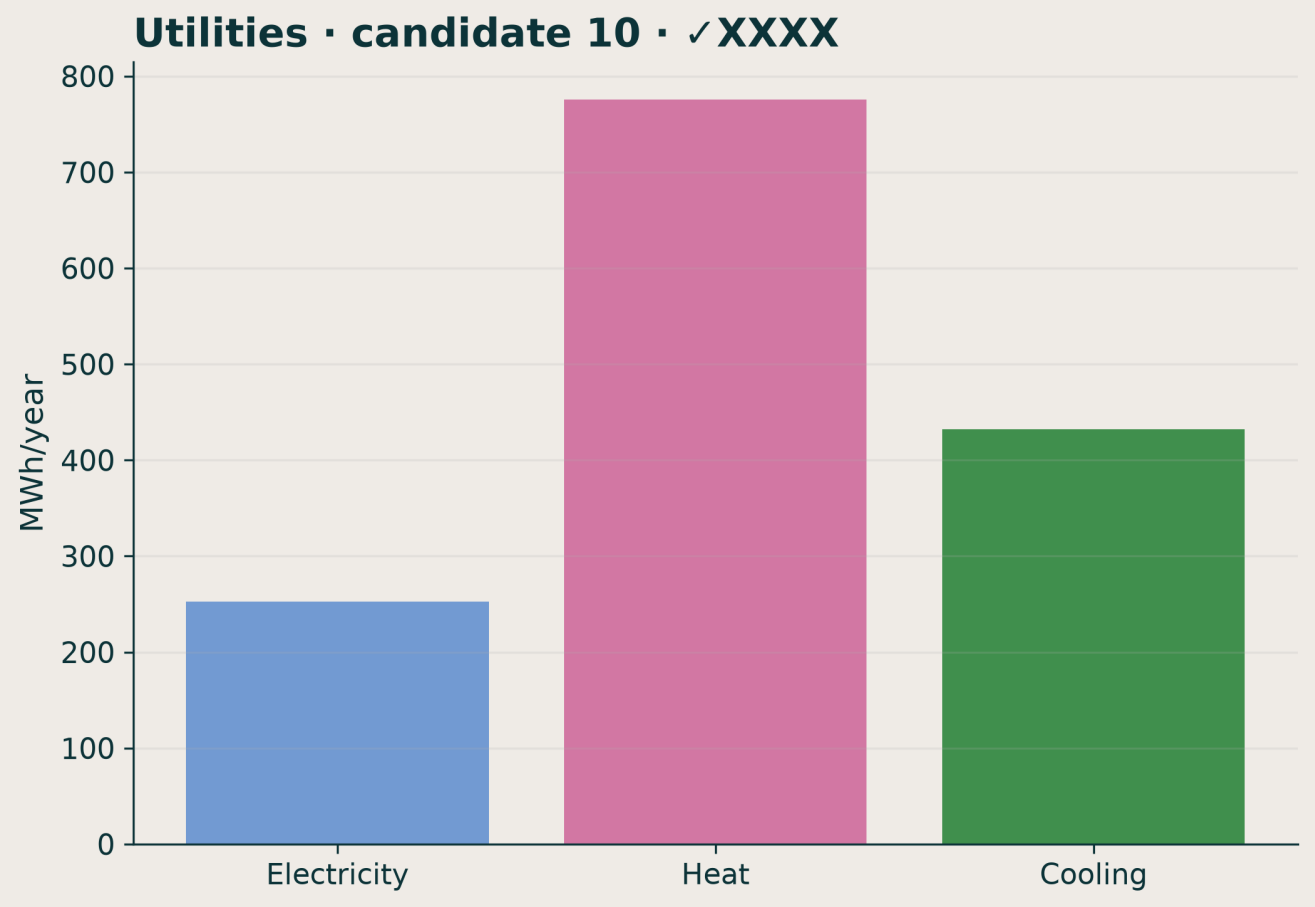


Figure 43. Annual electricity, heating and cooling duty

protein_enabled; rank 10; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

candidate kpis

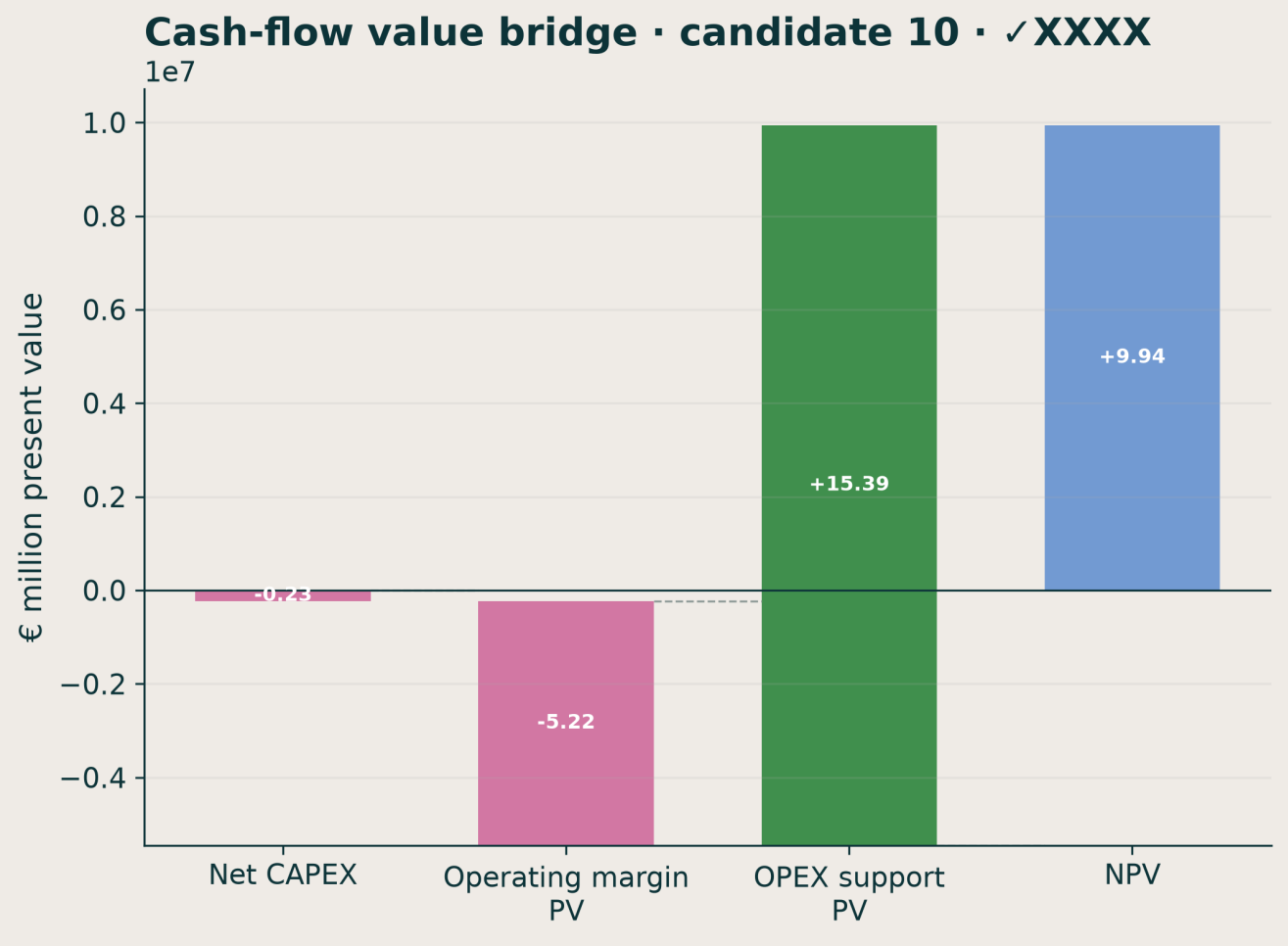


Figure 44. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 10; support C | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_enabled_batch.json

pareto net carbon vs npv protein bypass

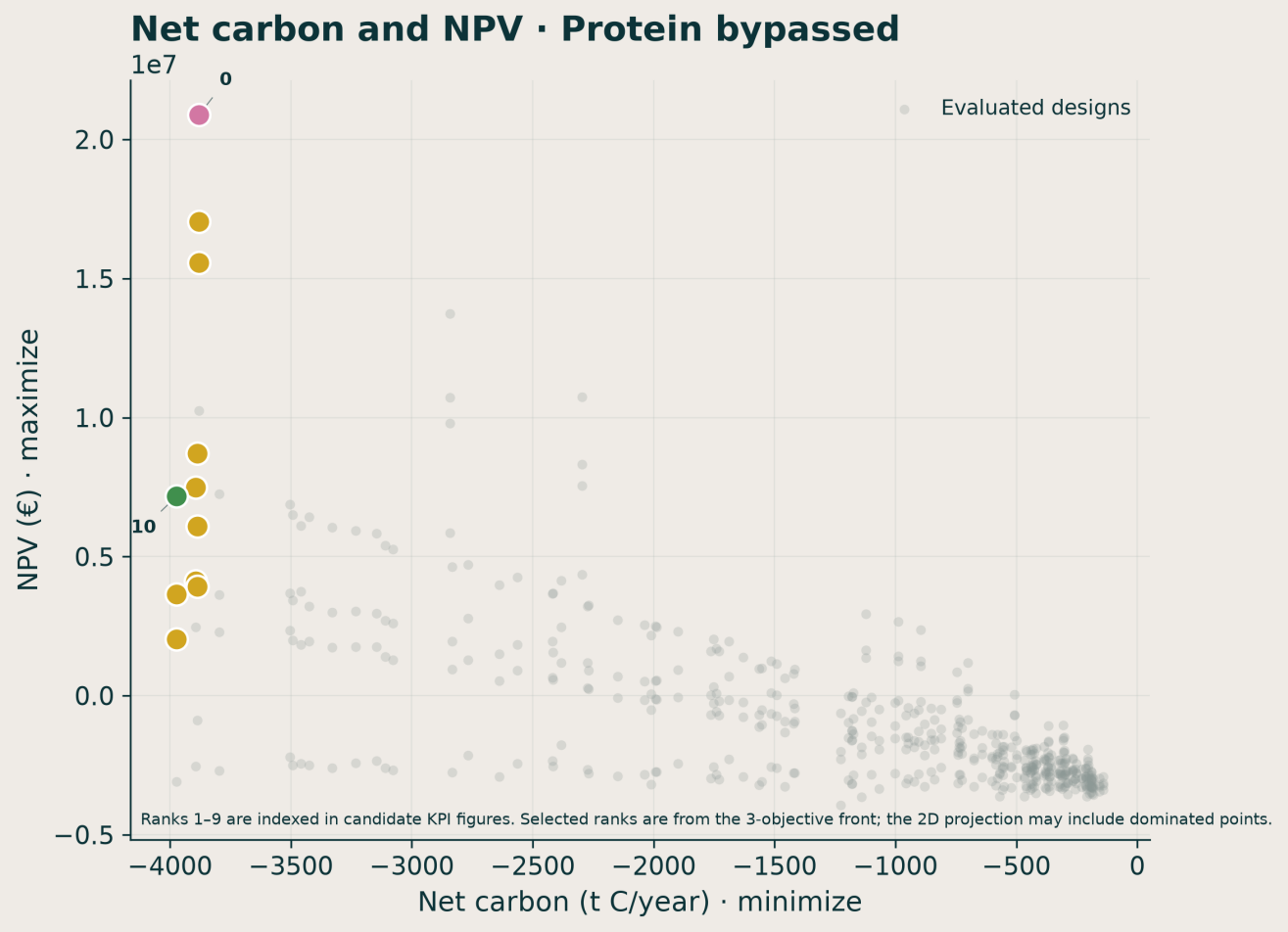


Figure 45. Net carbon and NPV · Protein bypassed

protein_bypass / all support cases | Net carbon: t C/year; NPV: € | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

pareto subsidy granted vs npv protein bypass

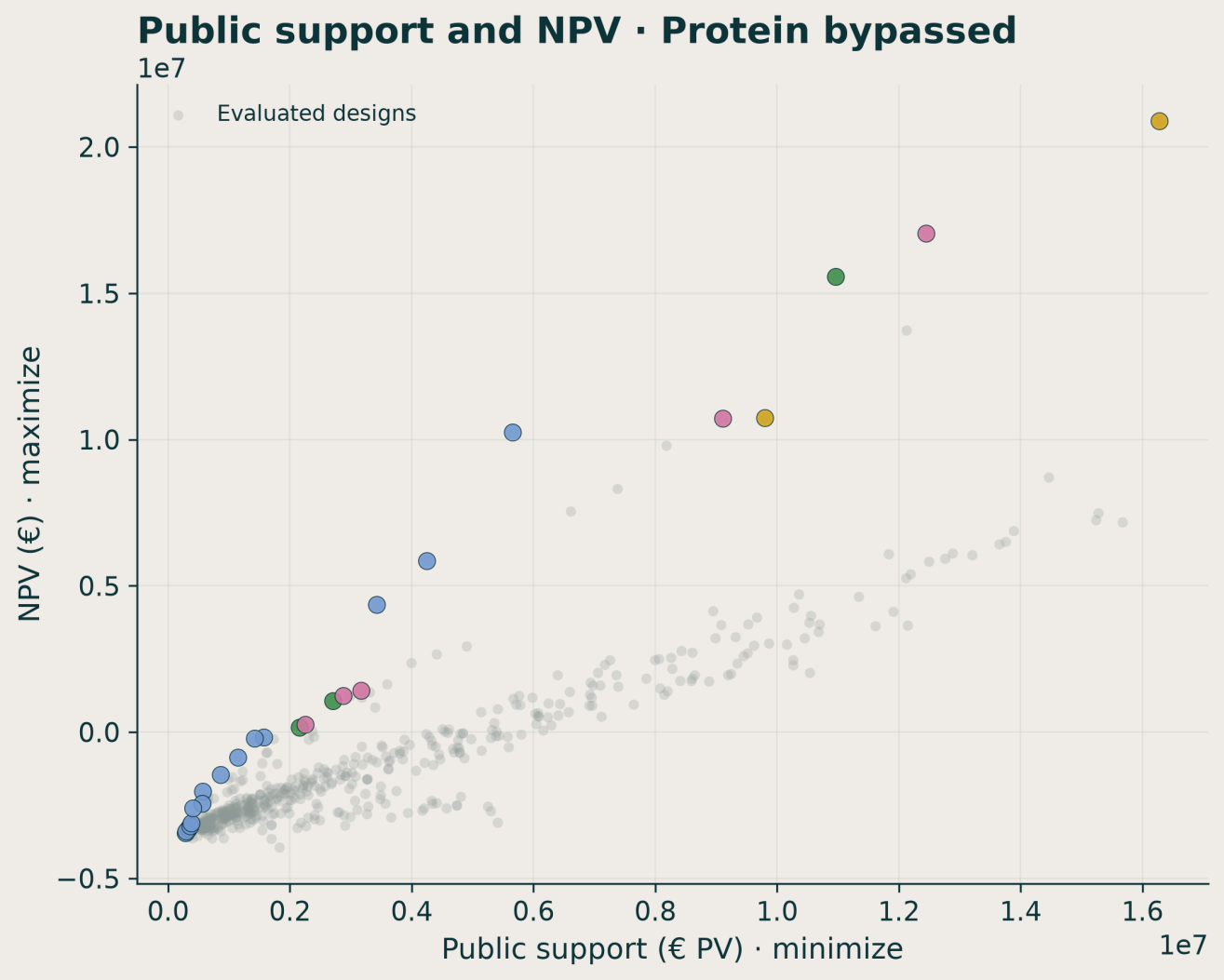


Figure 46. Public support and NPV · Protein bypassed

protein_bypass / all support cases | Public support: € PV; NPV: € | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

pareto net carbon vs npv protein bypass case A

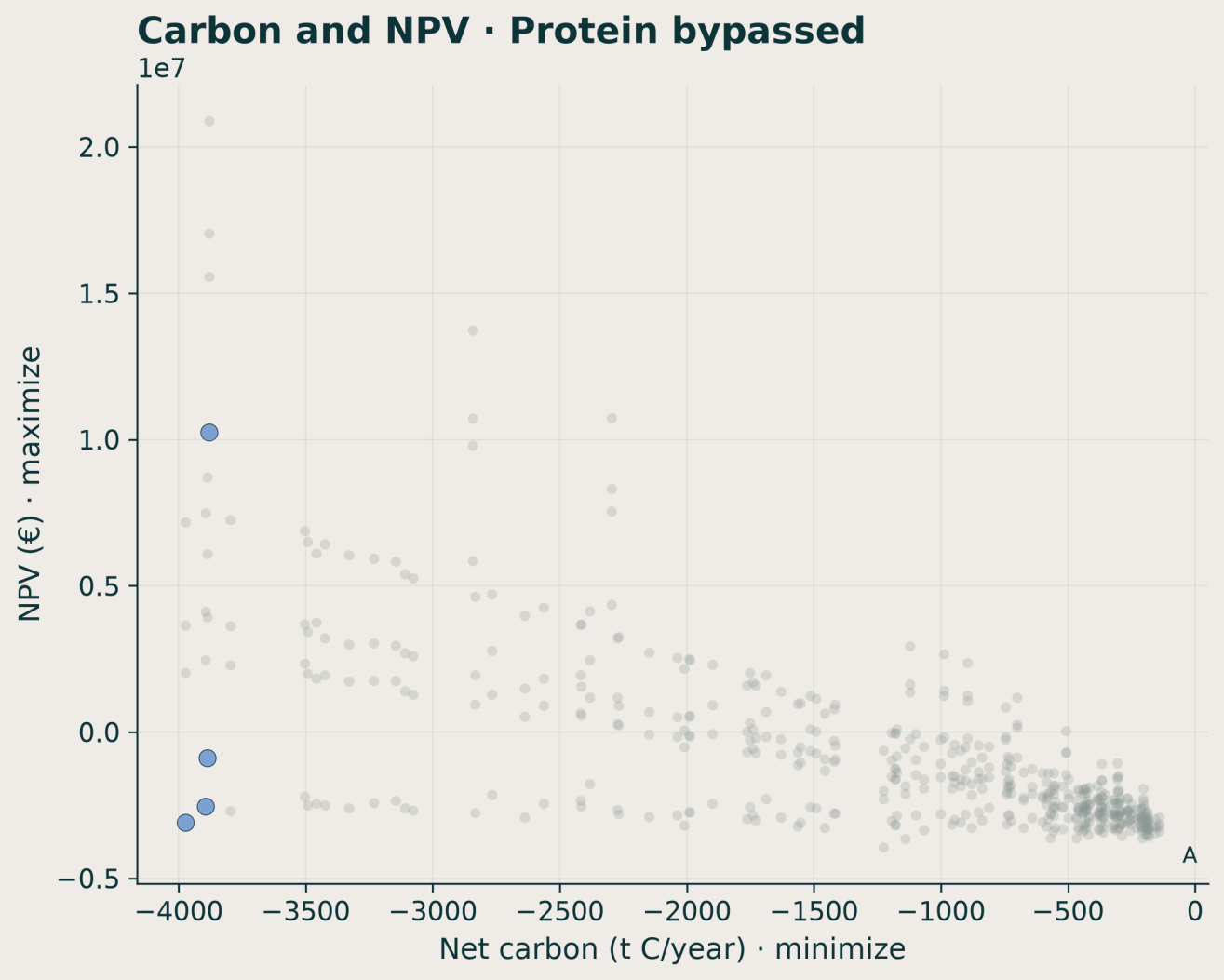


Figure 47. Carbon and NPV · Protein bypassed

protein_bypass / case A | Net carbon: t C/year; NPV: € | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

pareto net carbon vs npv protein bypass case B

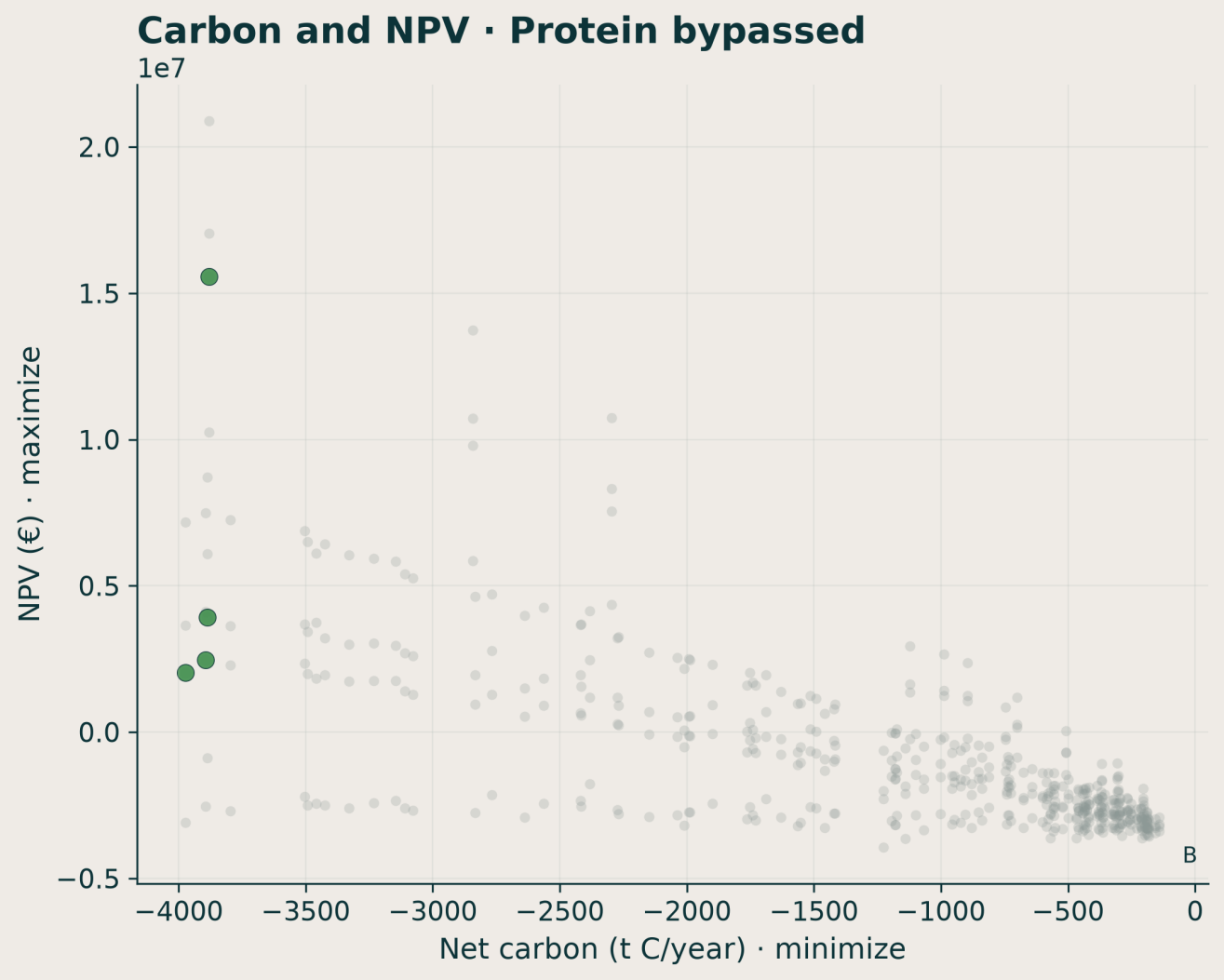


Figure 48. Carbon and NPV · Protein bypassed

protein_bypass / case B | Net carbon: t C/year; NPV: € | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

pareto net carbon vs npv protein bypass case C

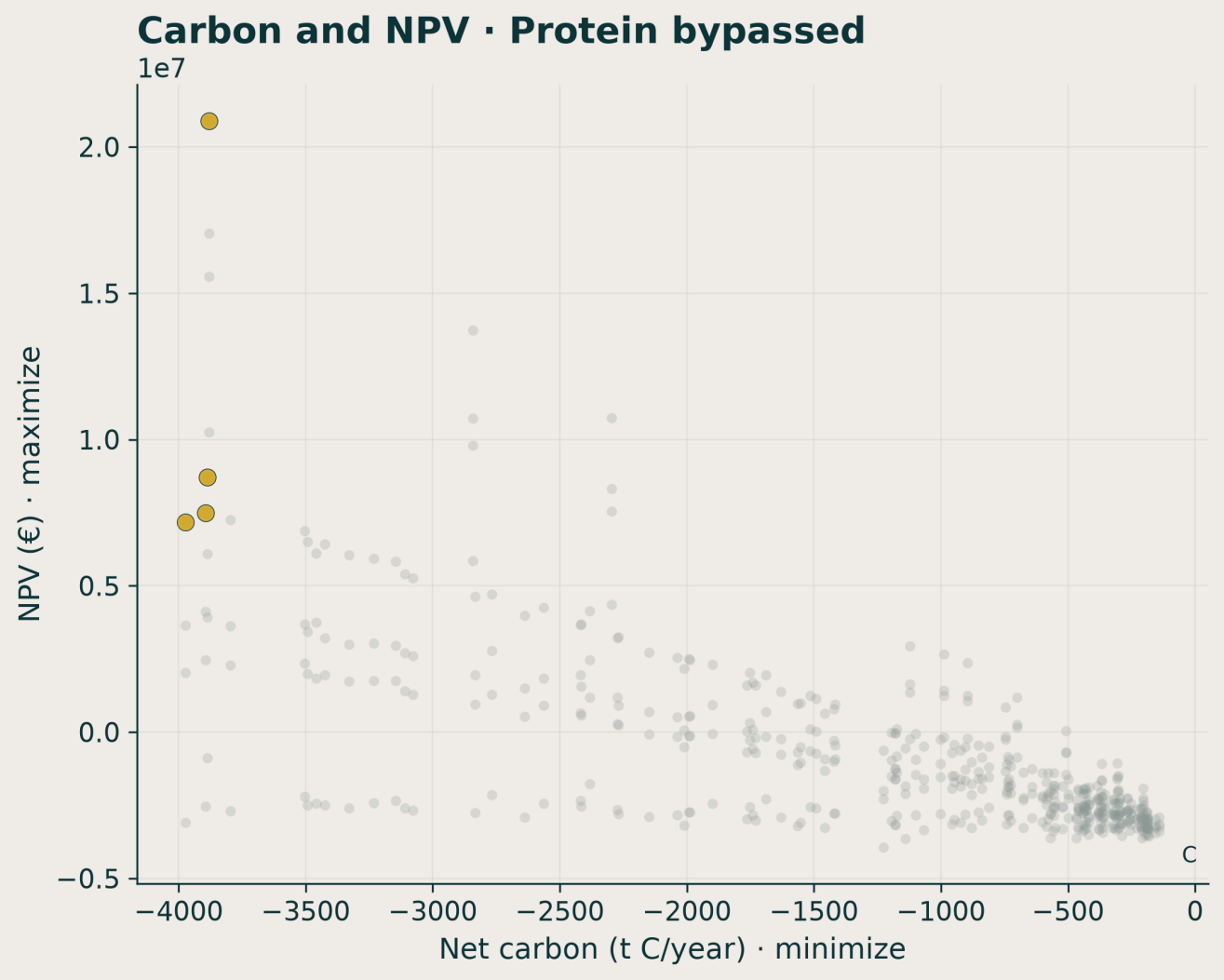


Figure 49. Carbon and NPV · Protein bypassed

protein_bypass / case C | Net carbon: t C/year; NPV: € | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

pareto net carbon vs npv protein bypass case D

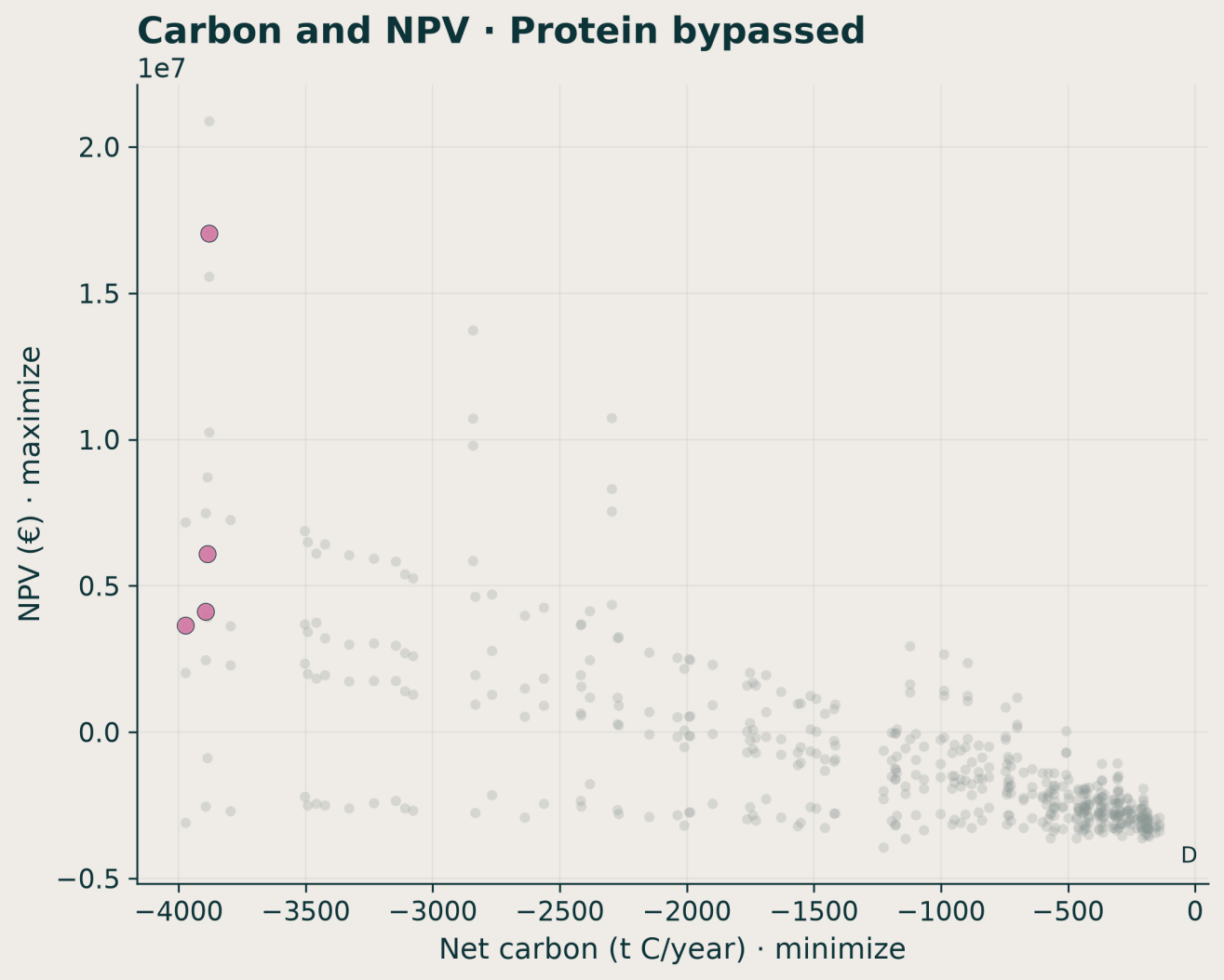


Figure 50. Carbon and NPV · Protein bypassed

protein_bypass / case D | Net carbon: t C/year; NPV: € | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

pareto 3d carbon npv support protein bypass

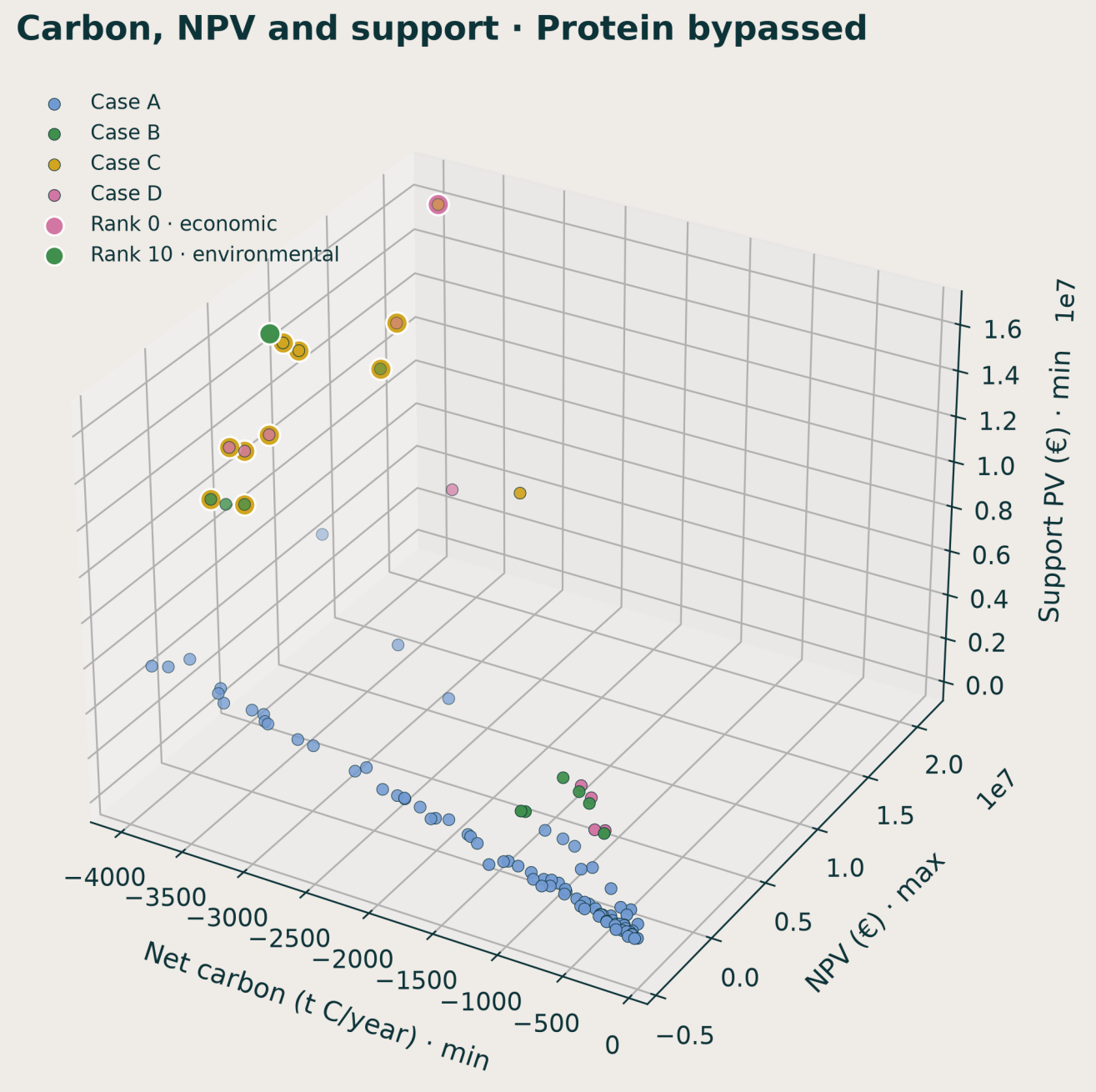


Figure 51. Three-objective Pareto front

Protein bypassed | t C/year; €; € present value | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate kpi heatmap protein bypass

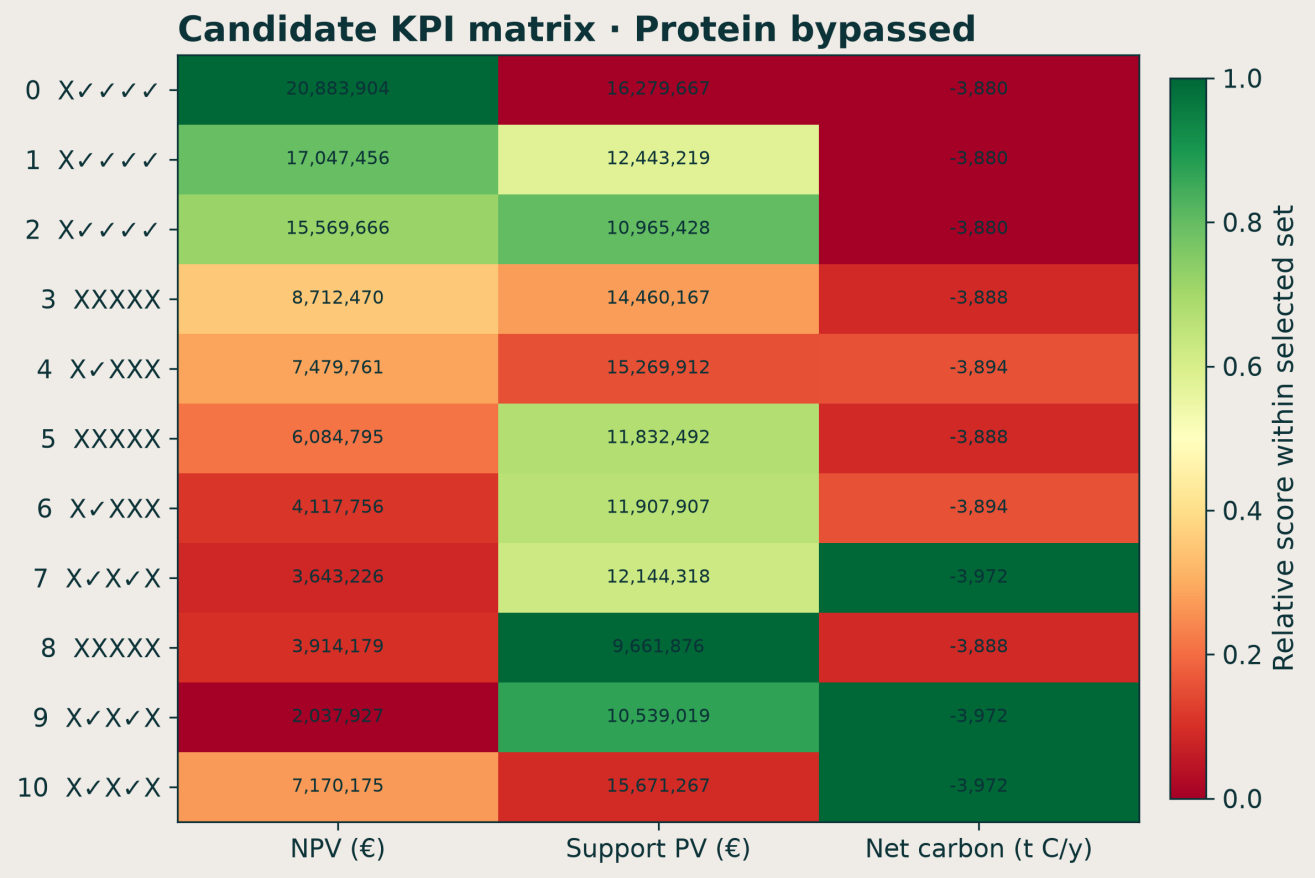


Figure 52. Selected candidate KPI comparison; color normalized by objective direction

Protein bypassed | €, € present value, t C/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate economics protein bypass

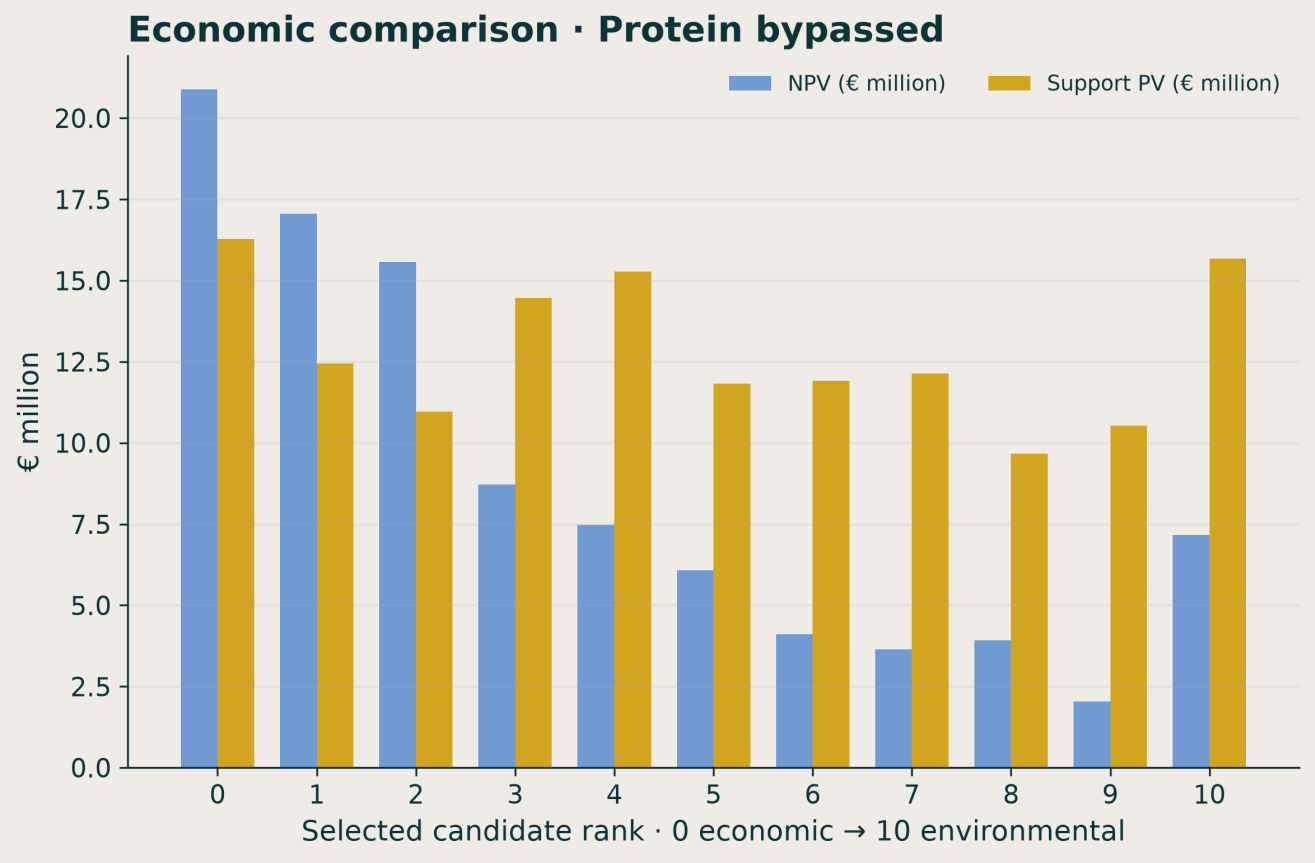


Figure 53. NPV and assumed public support PV for selected Pareto candidates

Protein bypassed | € million | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate environment protein bypass

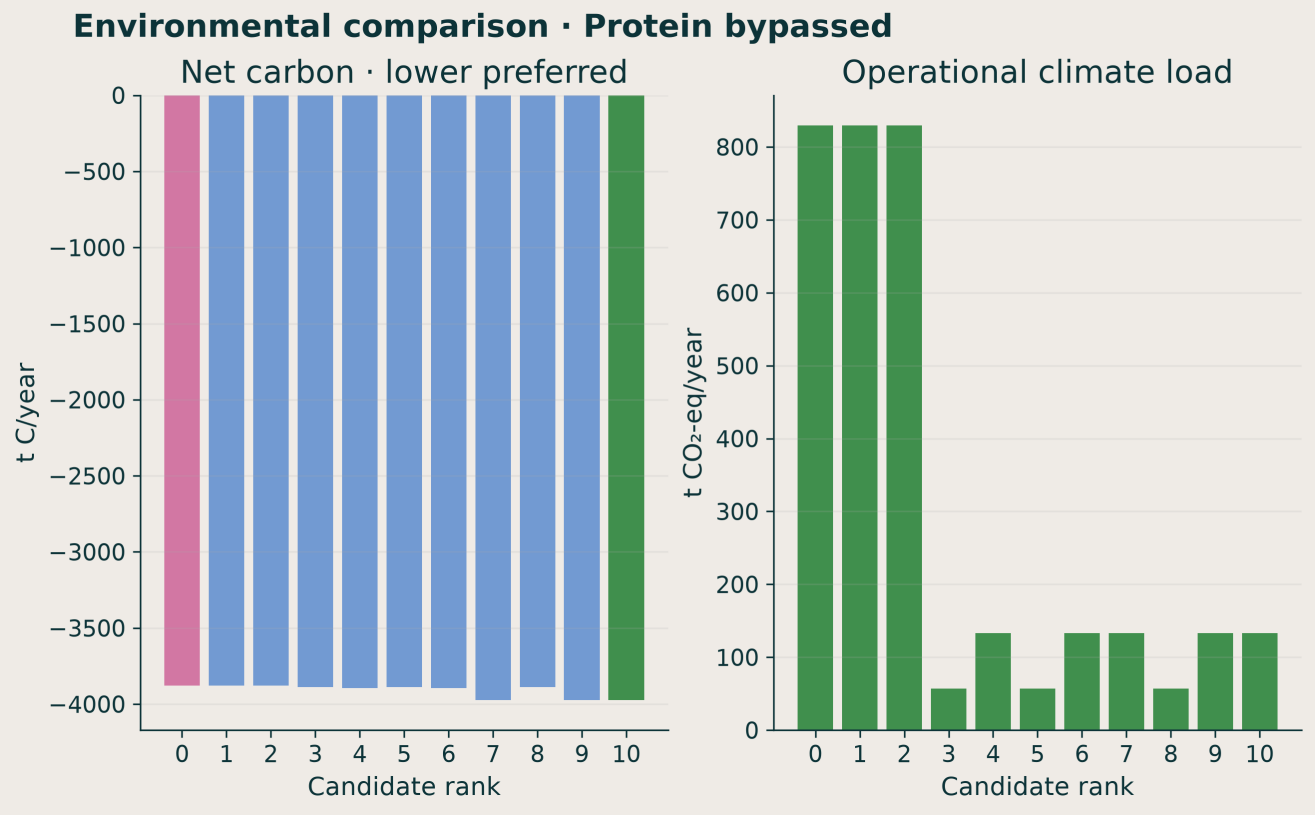


Figure 54. Net stored carbon objective and attributional operational climate load

Protein bypassed | t C/year; t CO□-eq/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

carbon load vs stored carbon protein bypass

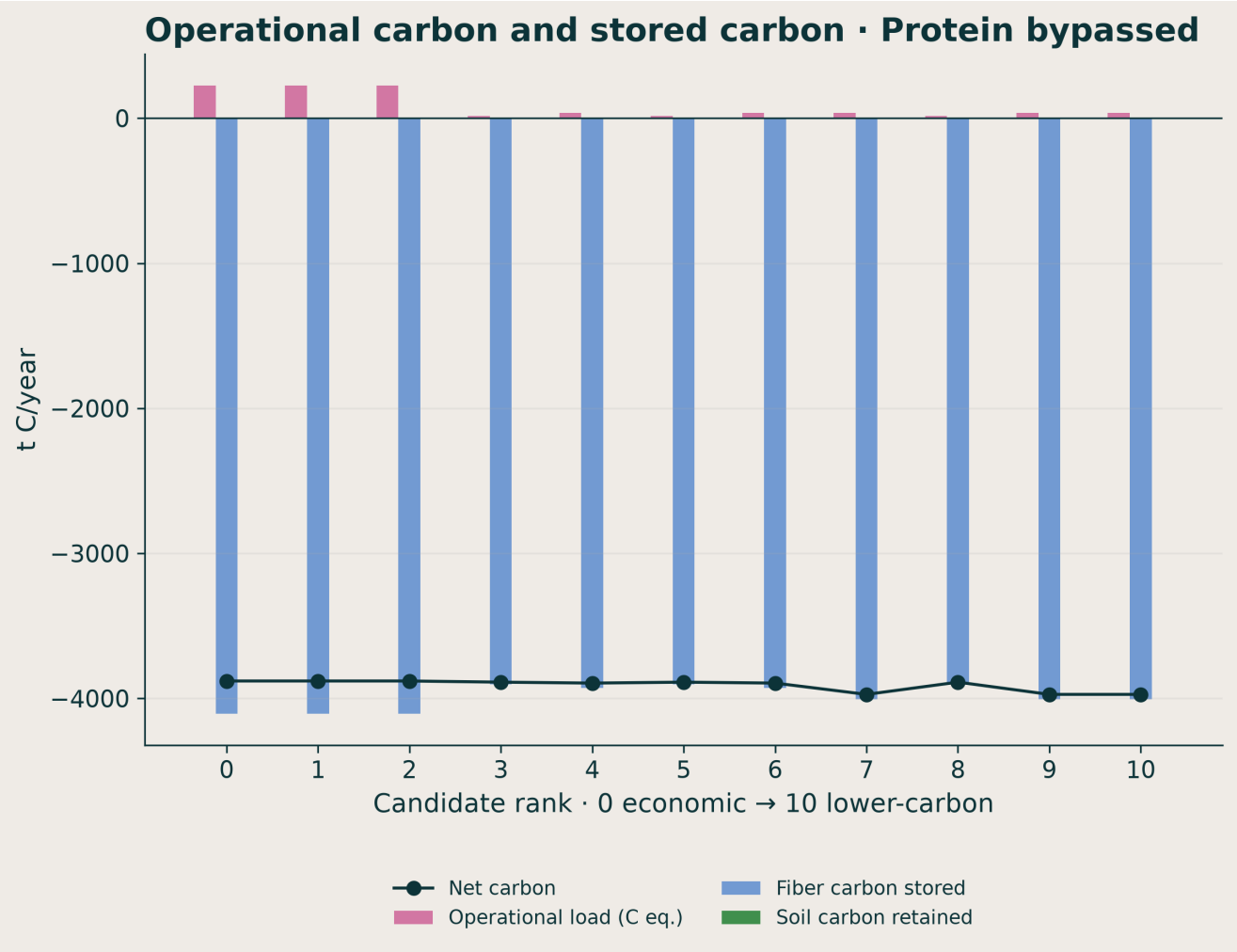


Figure 55. Attributional operational carbon-equivalent load minus fiber and soil retained product carbon

Protein bypassed | t C/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate products

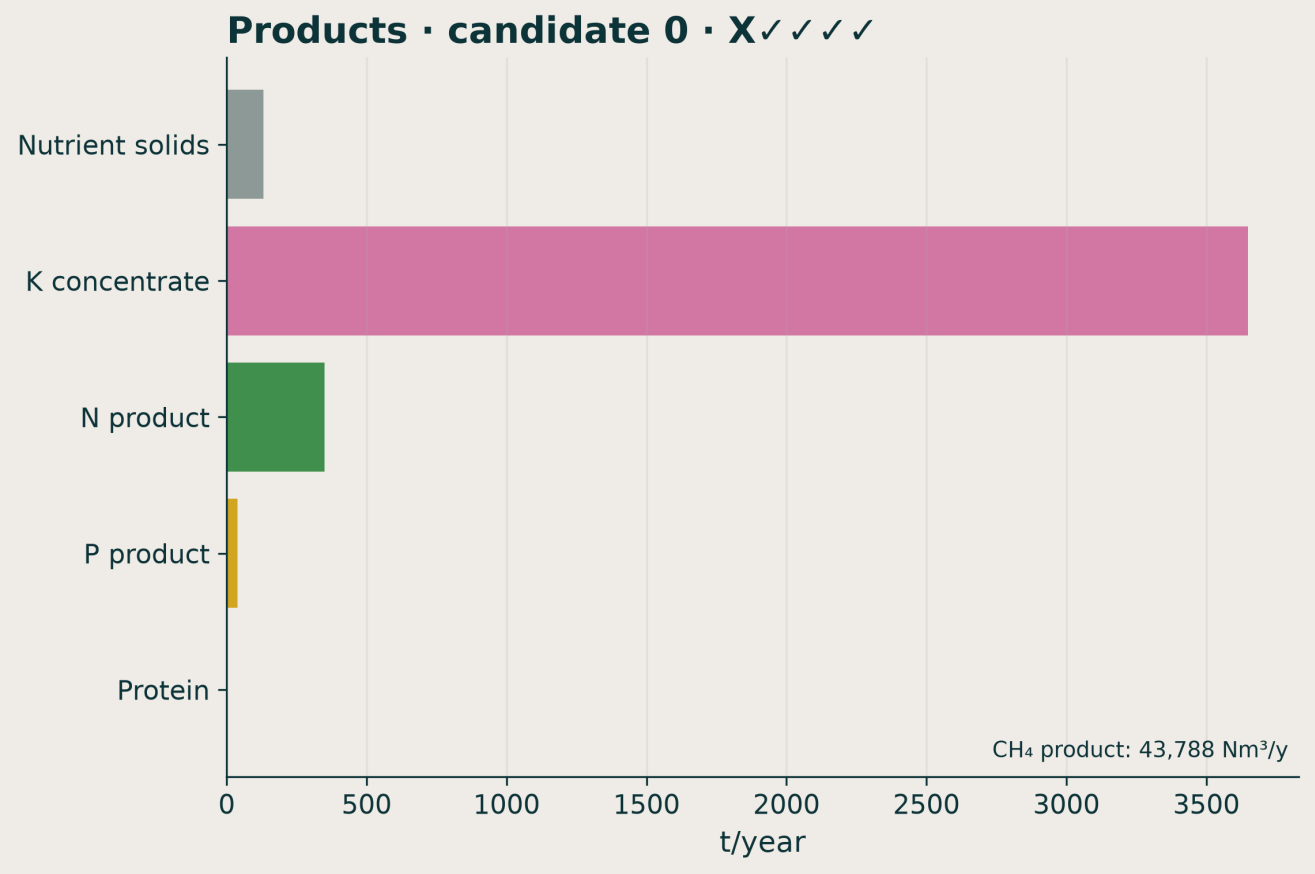


Figure 56. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_bypass; rank 0; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate utilities

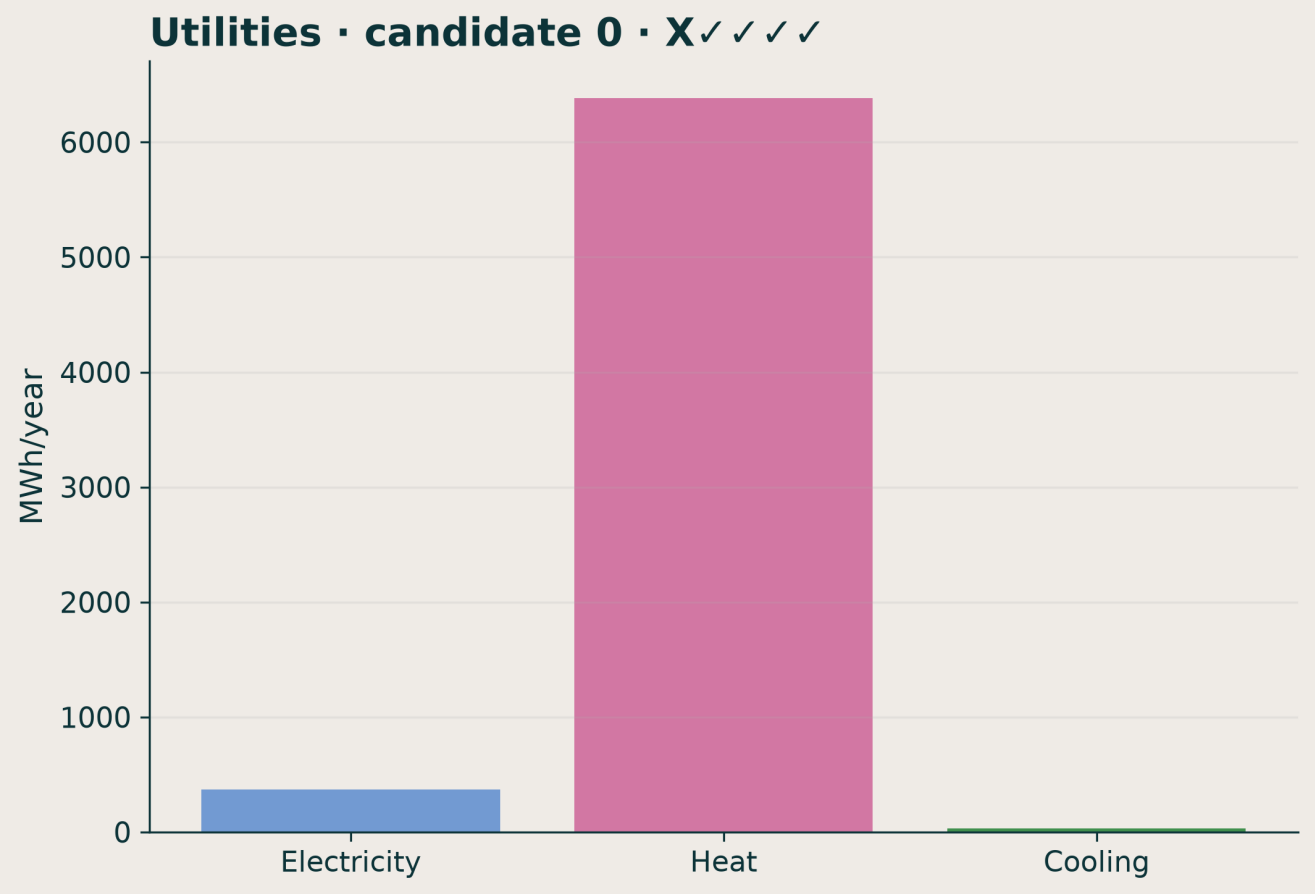


Figure 57. Annual electricity, heating and cooling duty

protein_bypass; rank 0; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate kpis

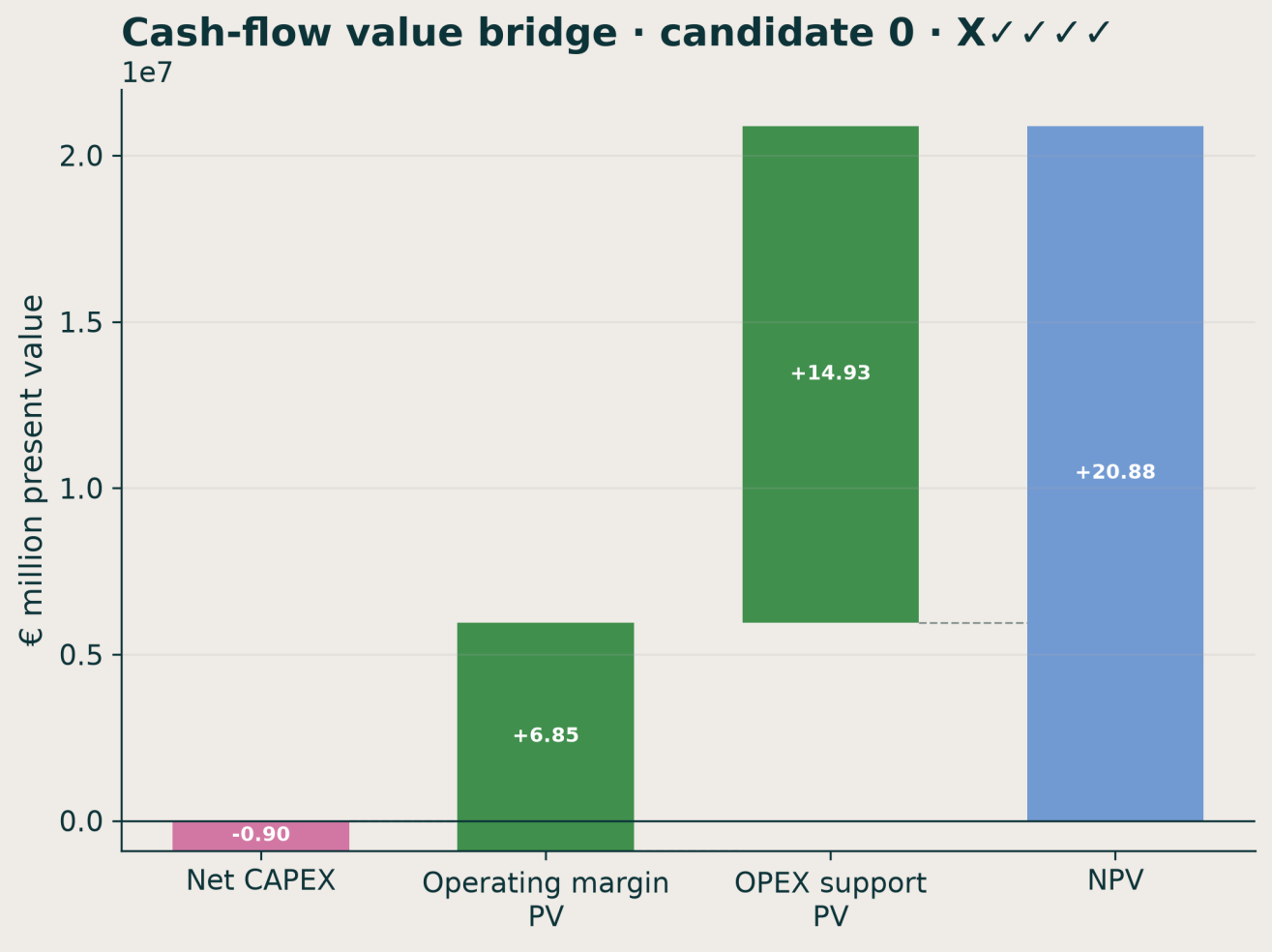


Figure 58. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 0; support C | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate products

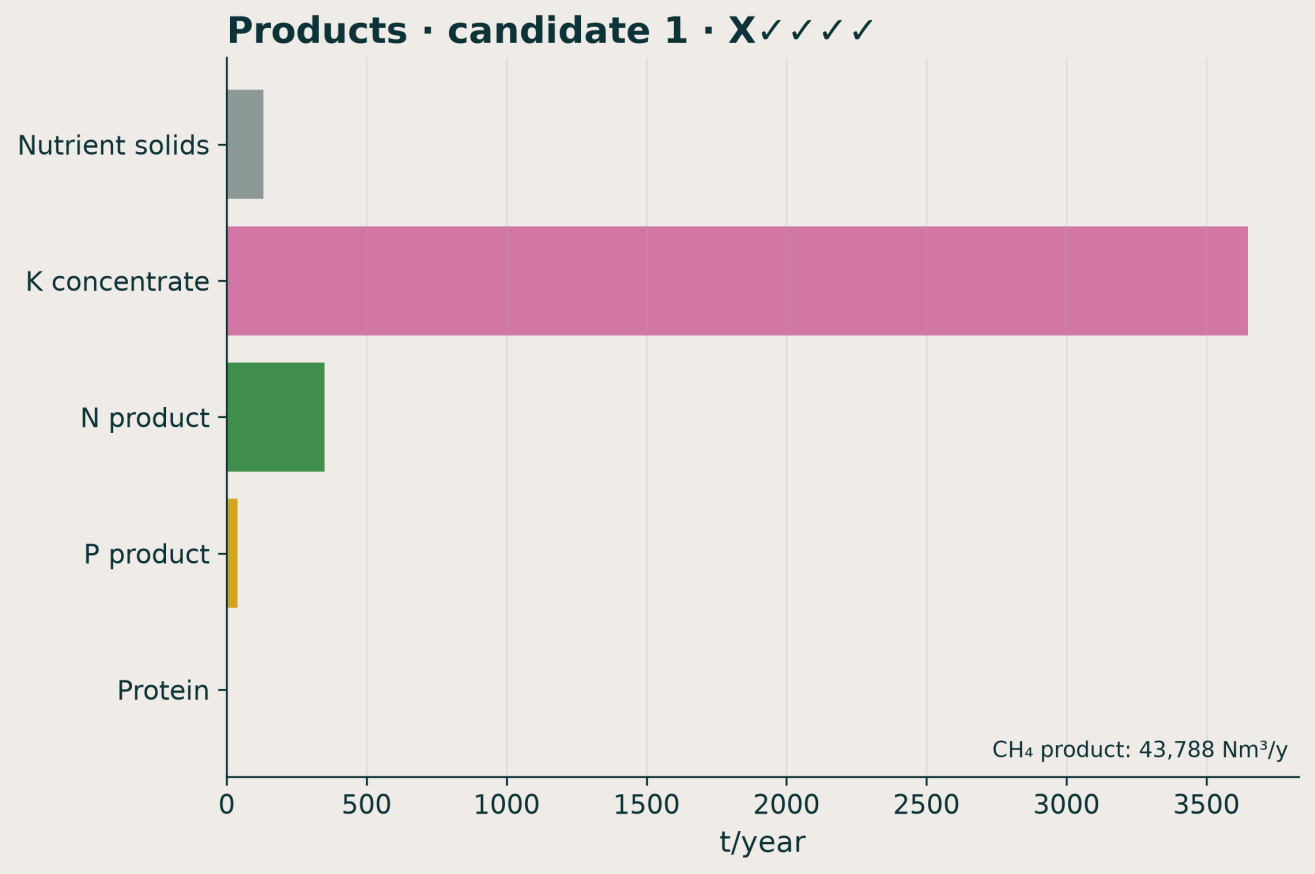


Figure 59. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_bypass; rank 1; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate utilities

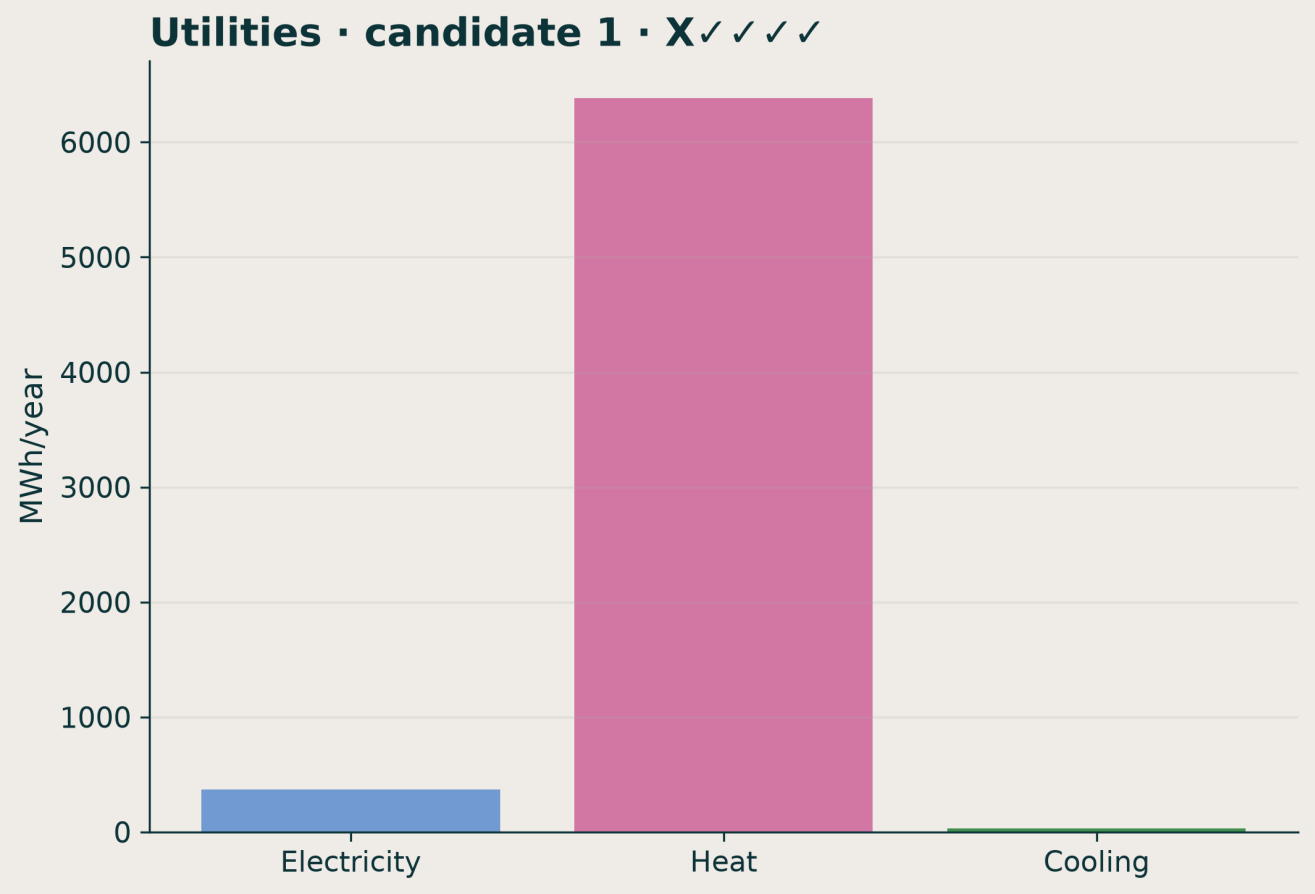


Figure 60. Annual electricity, heating and cooling duty

protein_bypass; rank 1; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate kpis

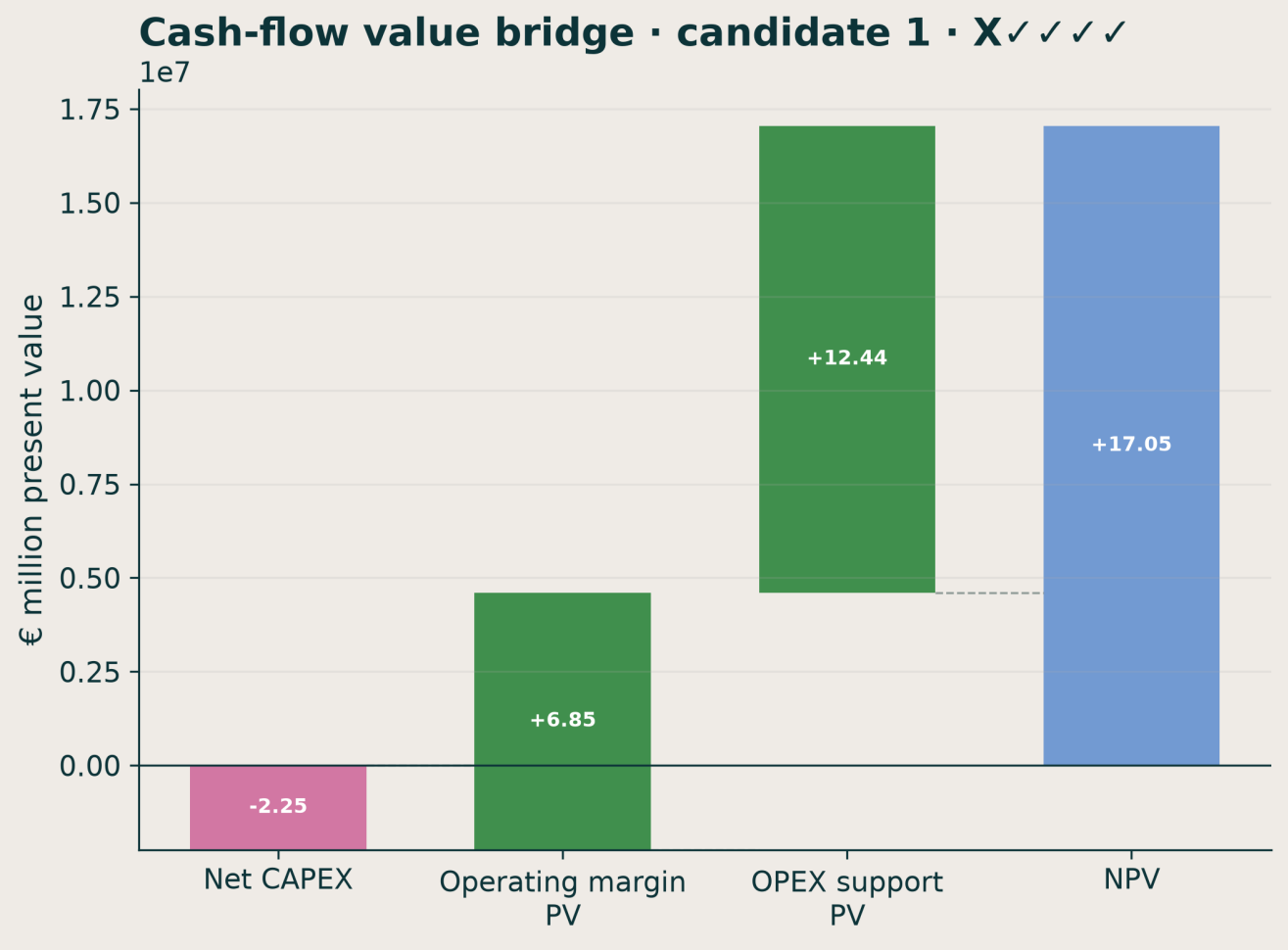


Figure 61. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 1; support D | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate products

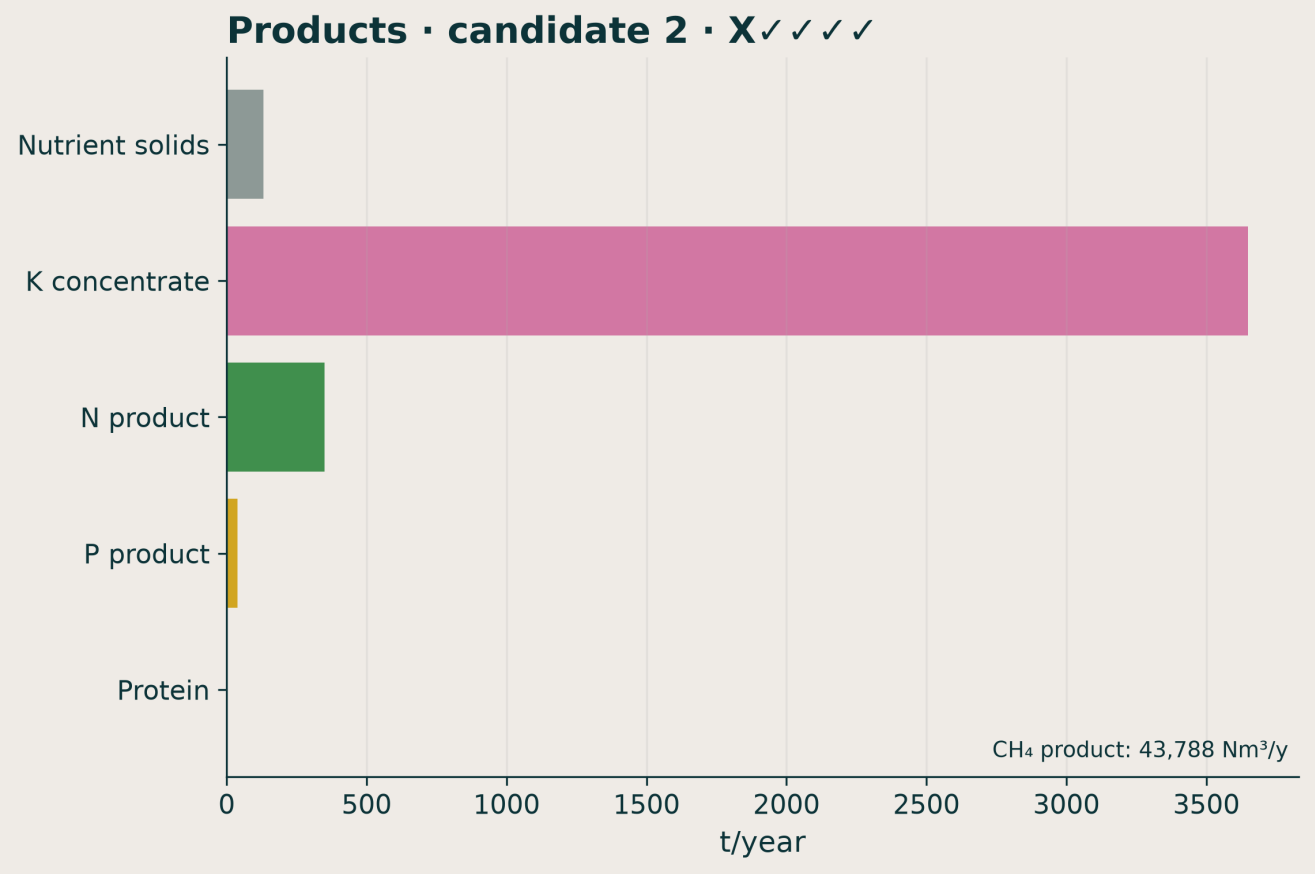


Figure 62. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_bypass; rank 2; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate utilities

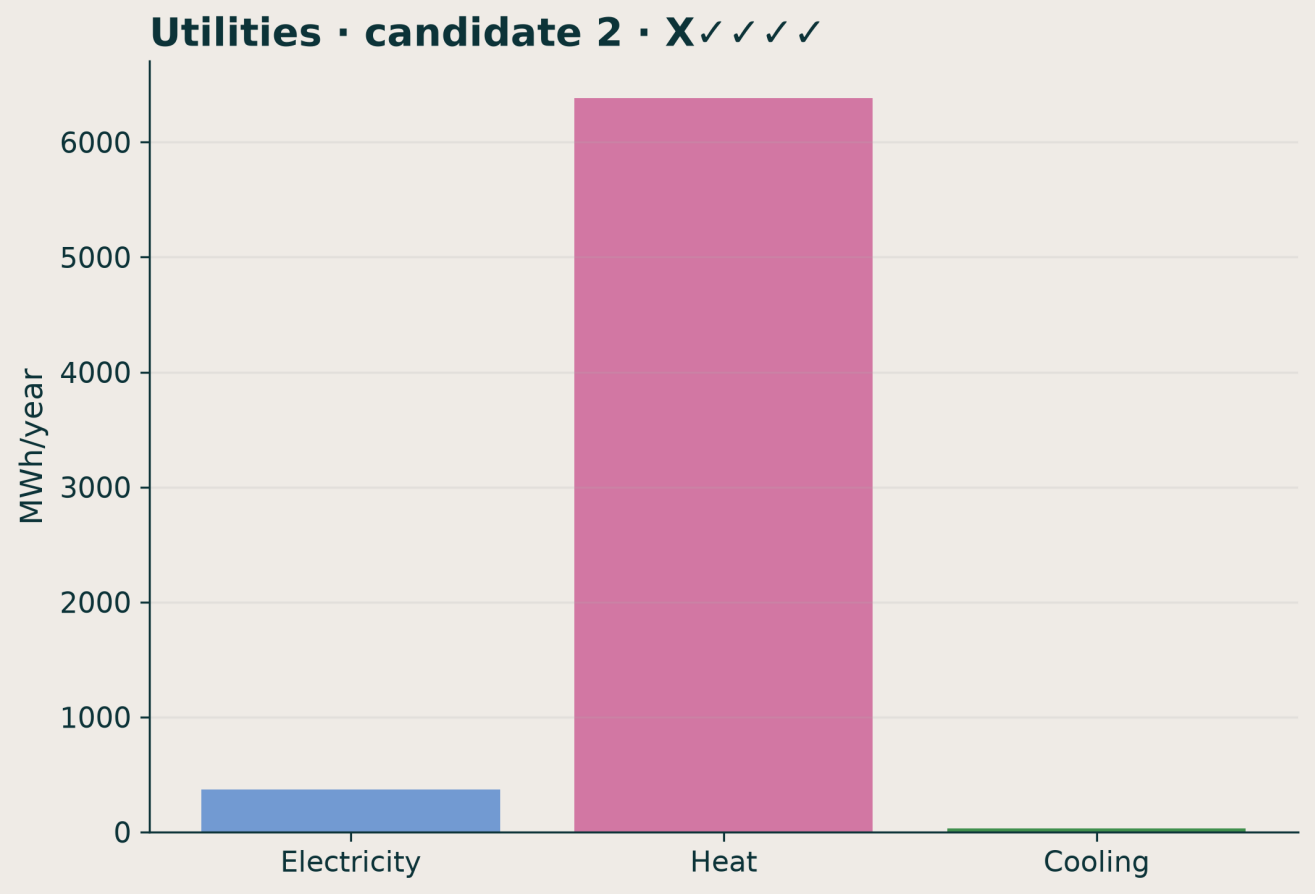


Figure 63. Annual electricity, heating and cooling duty

protein_bypass; rank 2; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate kpis

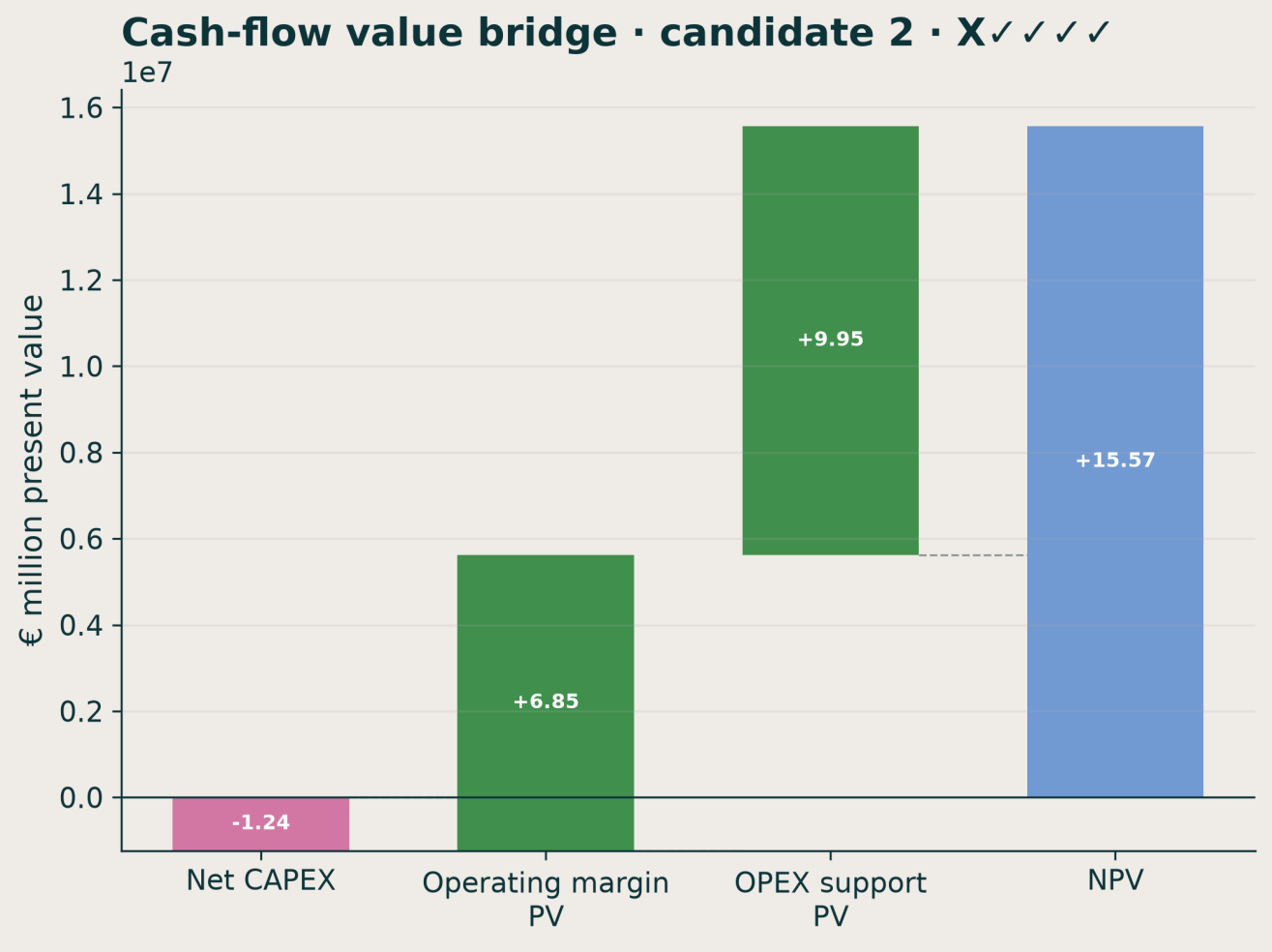


Figure 64. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 2; support B | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate products

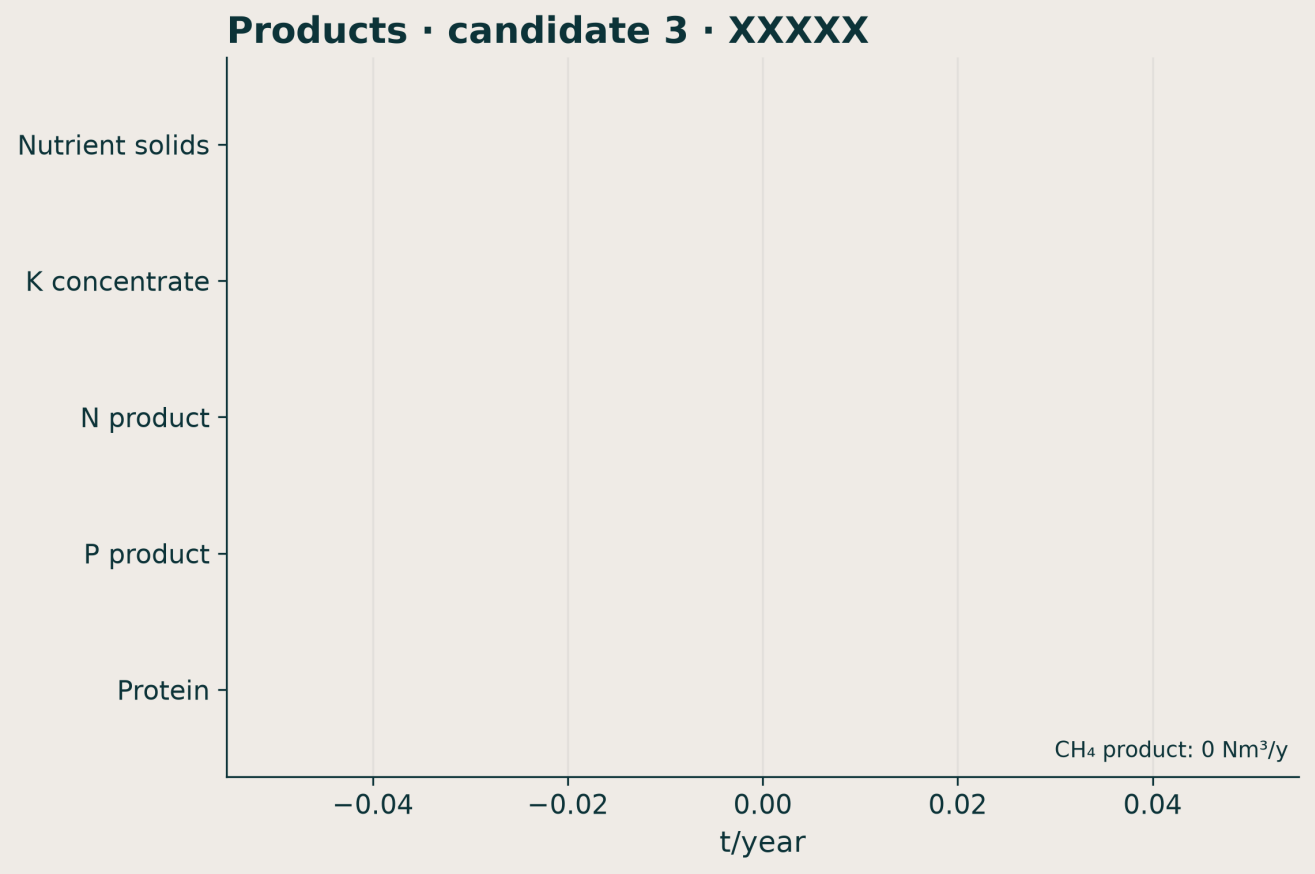


Figure 65. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_bypass; rank 3; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate utilities

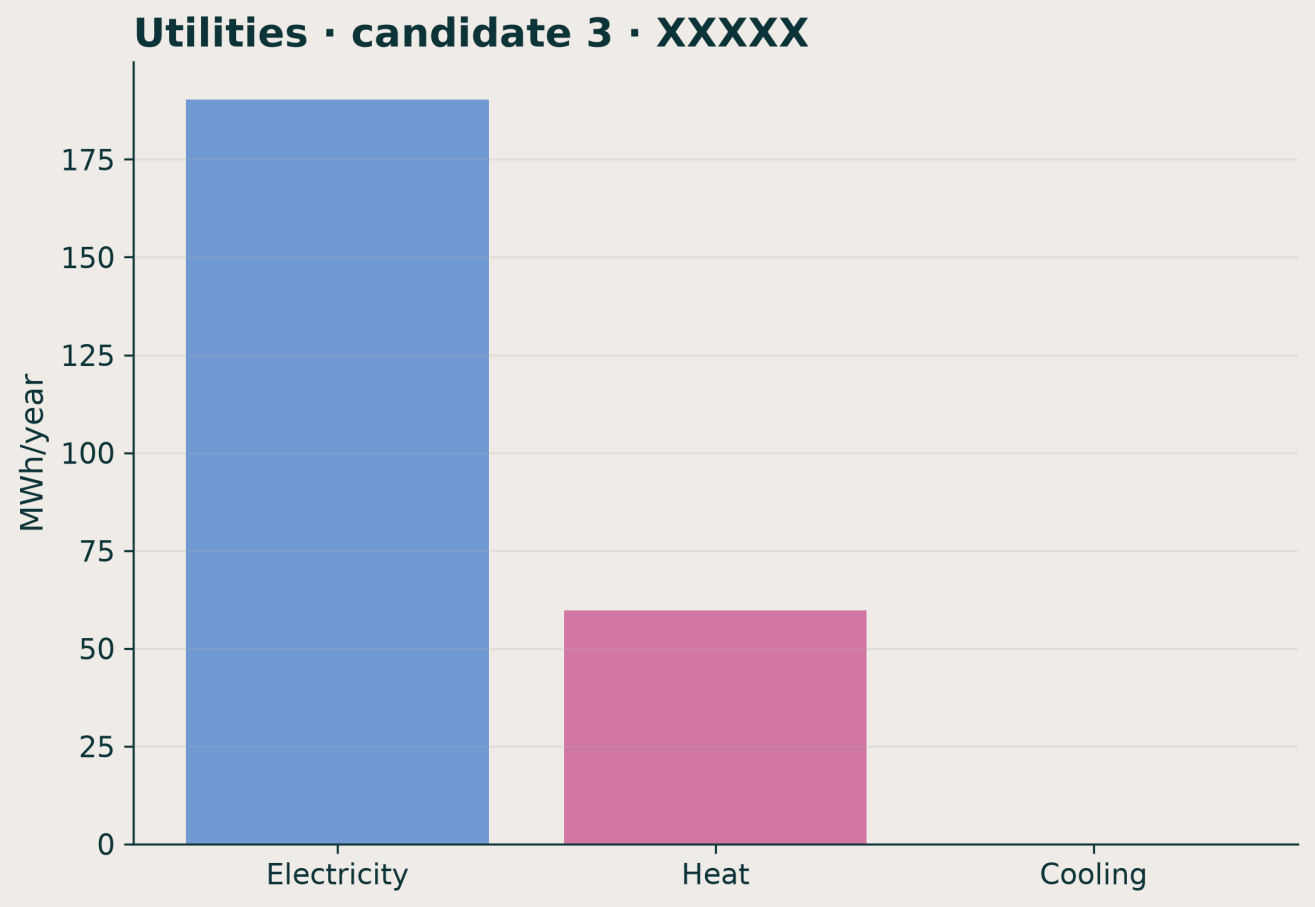


Figure 66. Annual electricity, heating and cooling duty

protein_bypass; rank 3; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate kpis

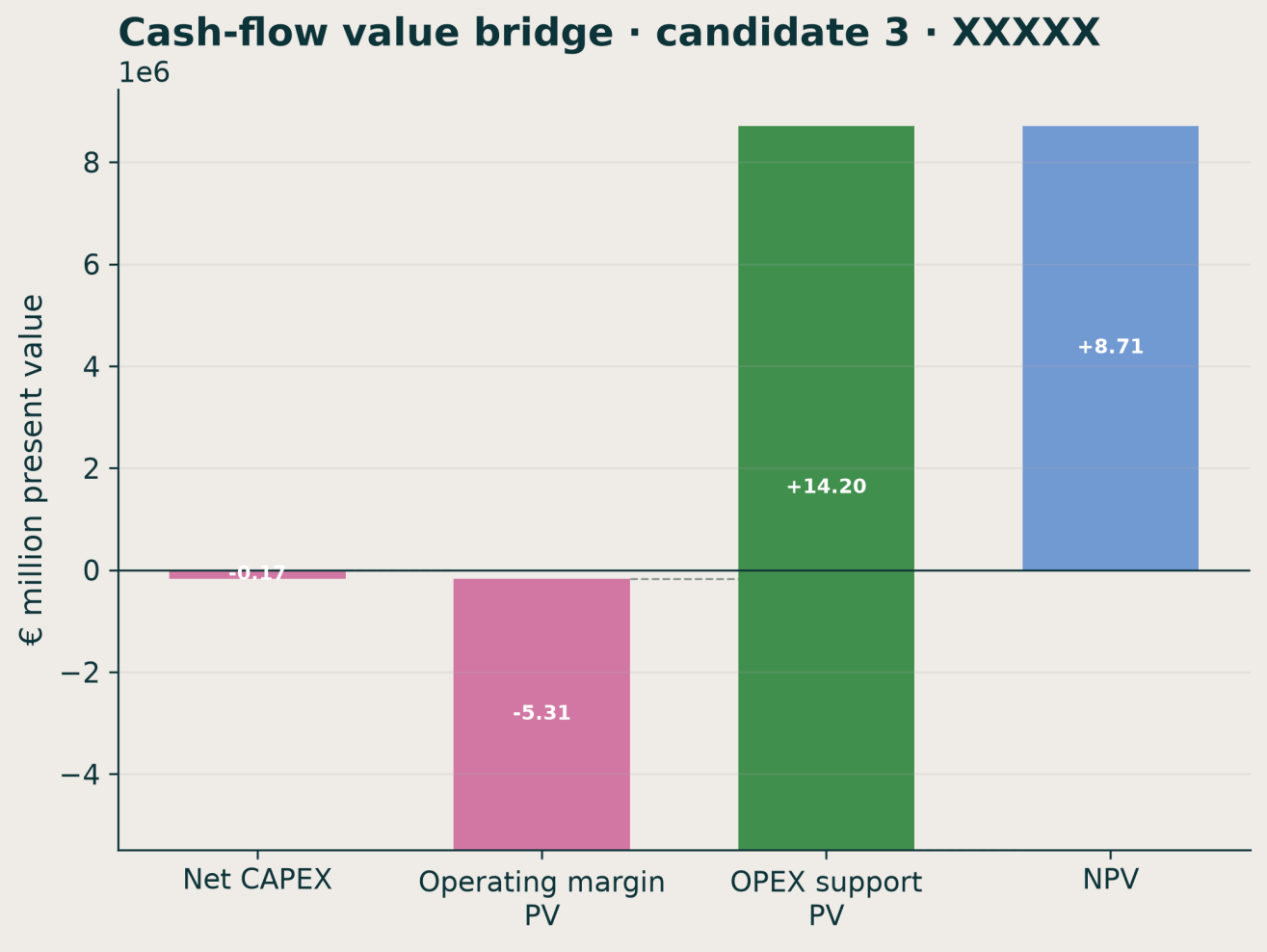


Figure 67. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 3; support C | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate products

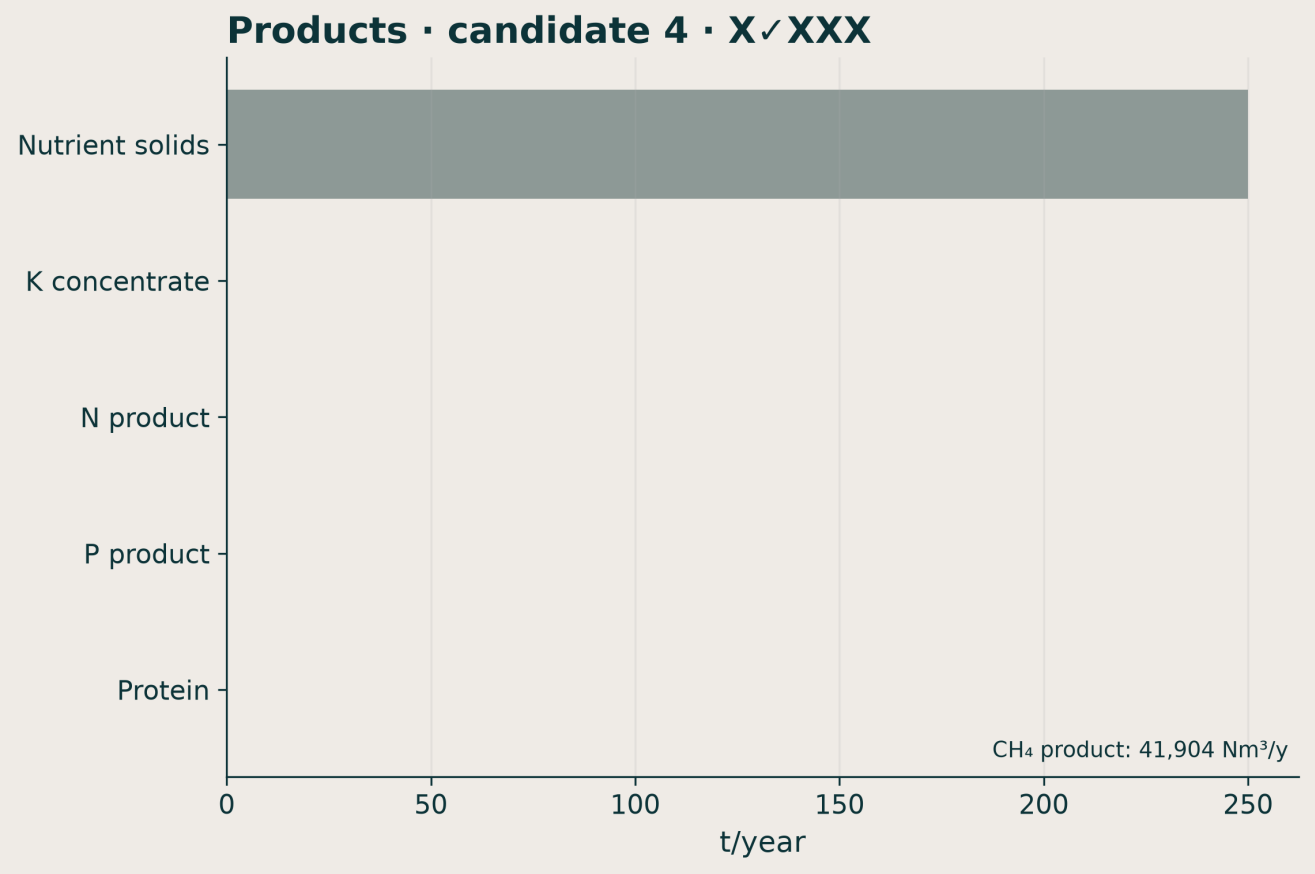


Figure 68. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_bypass; rank 4; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR1', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate utilities

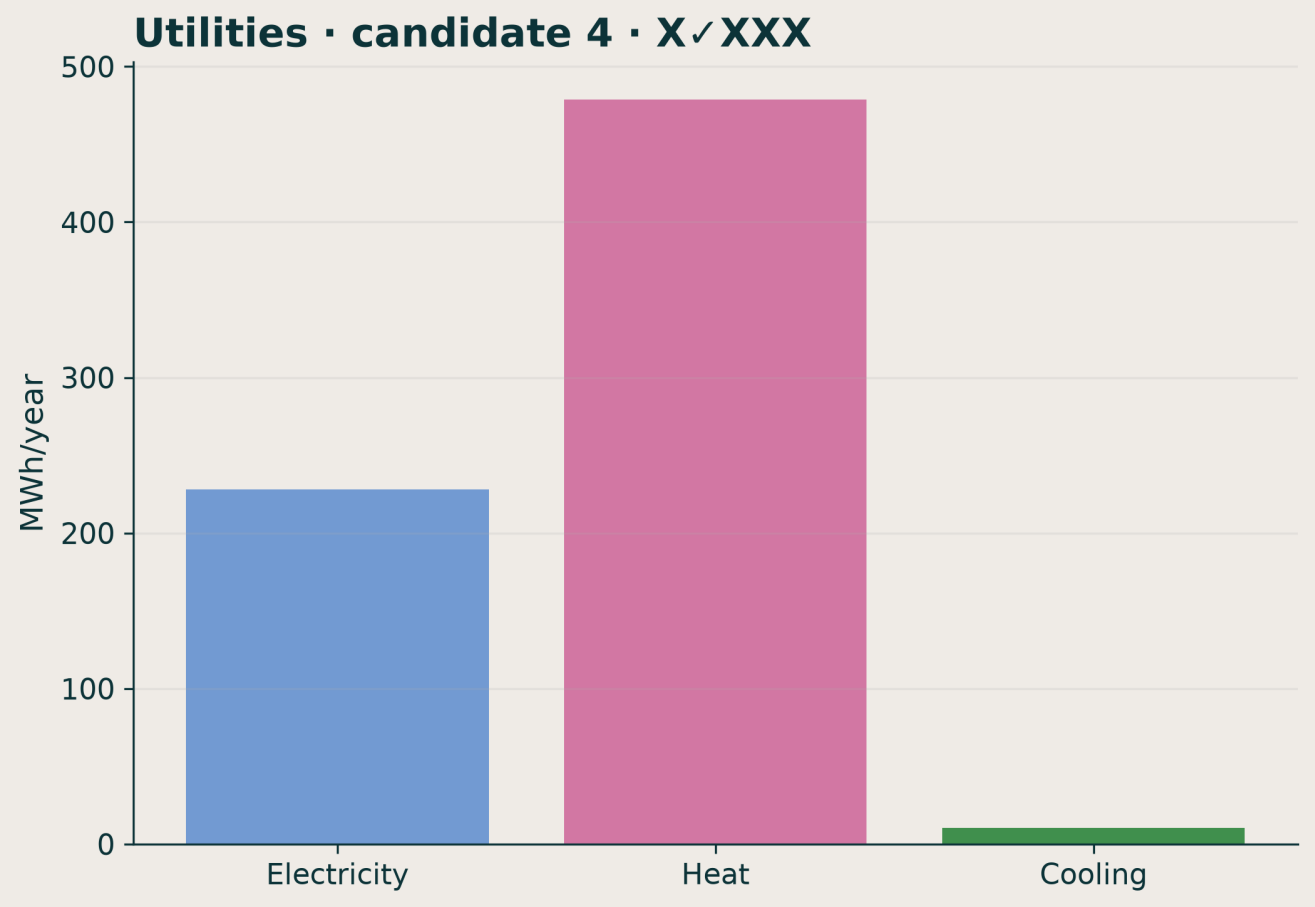


Figure 69. Annual electricity, heating and cooling duty

protein_bypass; rank 4; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR1', 'process_order': 'A'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate kpis

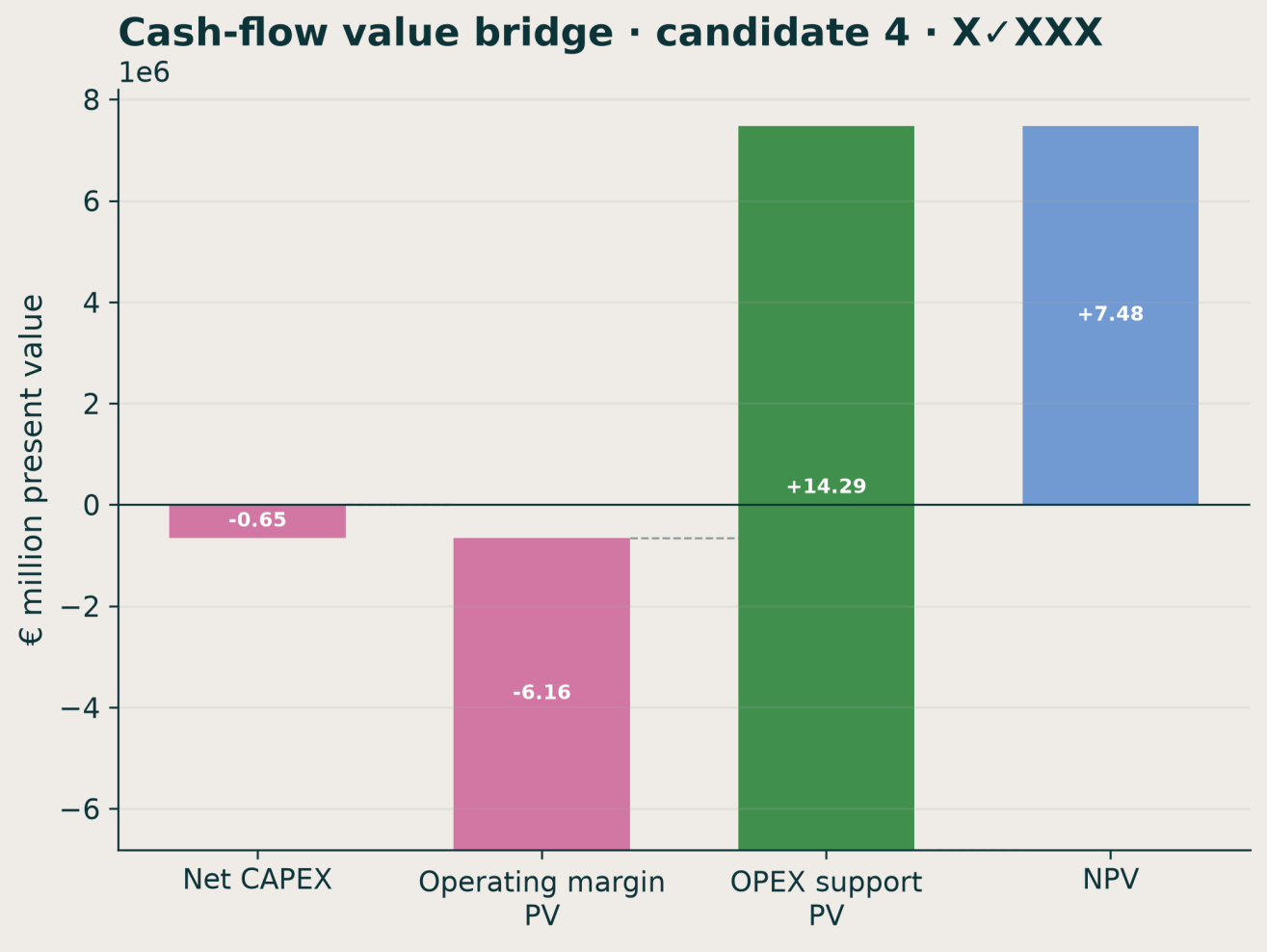


Figure 70. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 4; support C | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate products

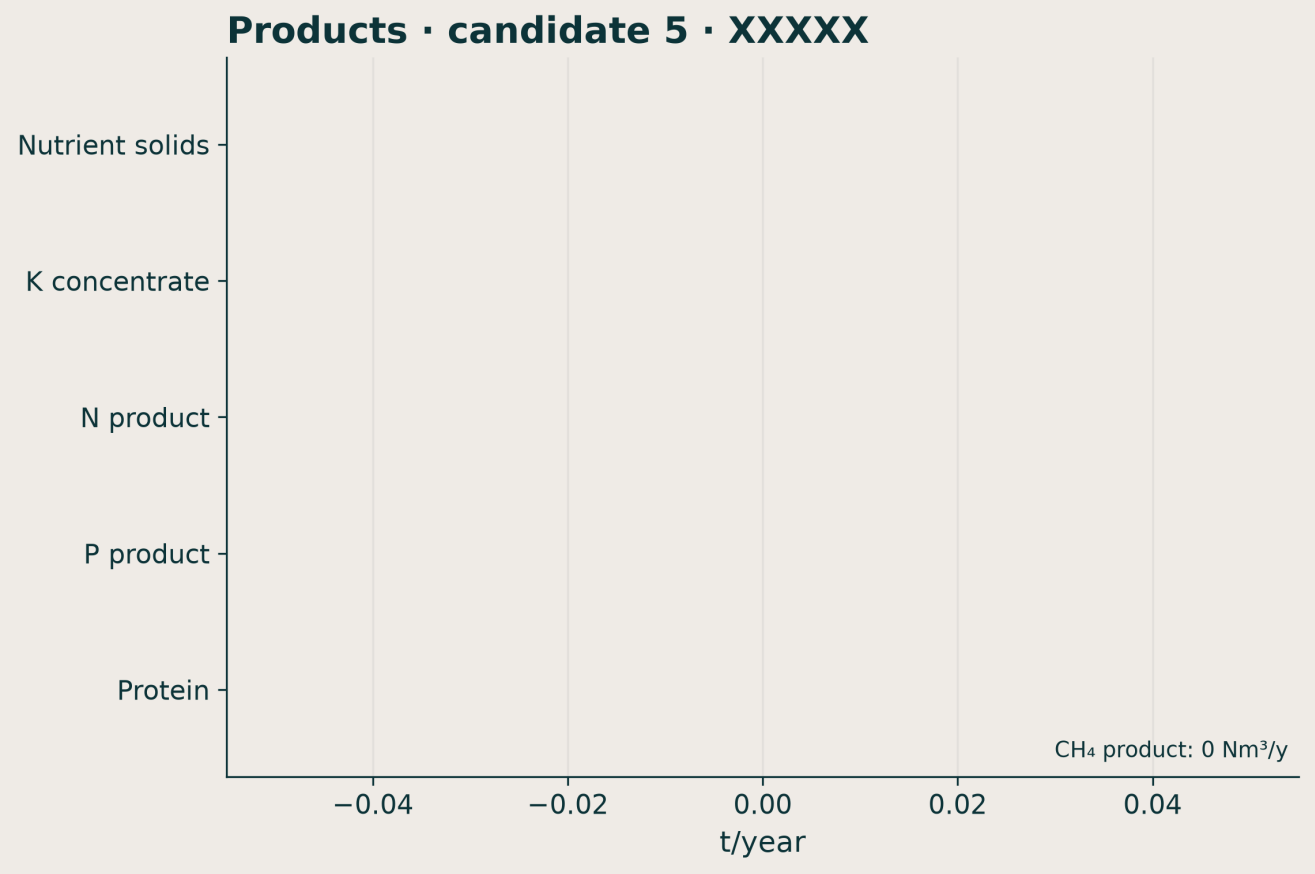


Figure 71. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_bypass; rank 5; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate utilities

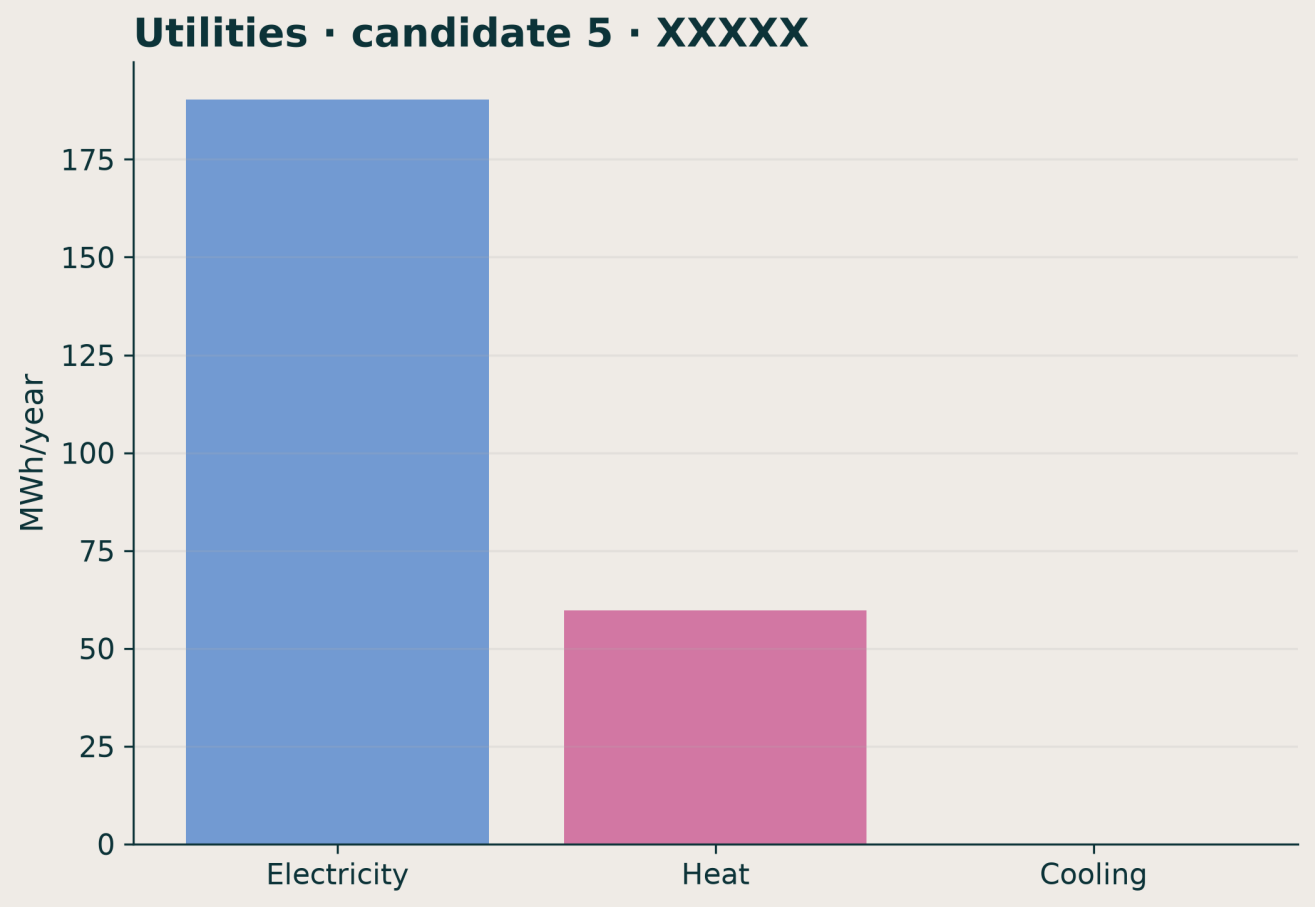


Figure 72. Annual electricity, heating and cooling duty

protein_bypass; rank 5; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate kpis

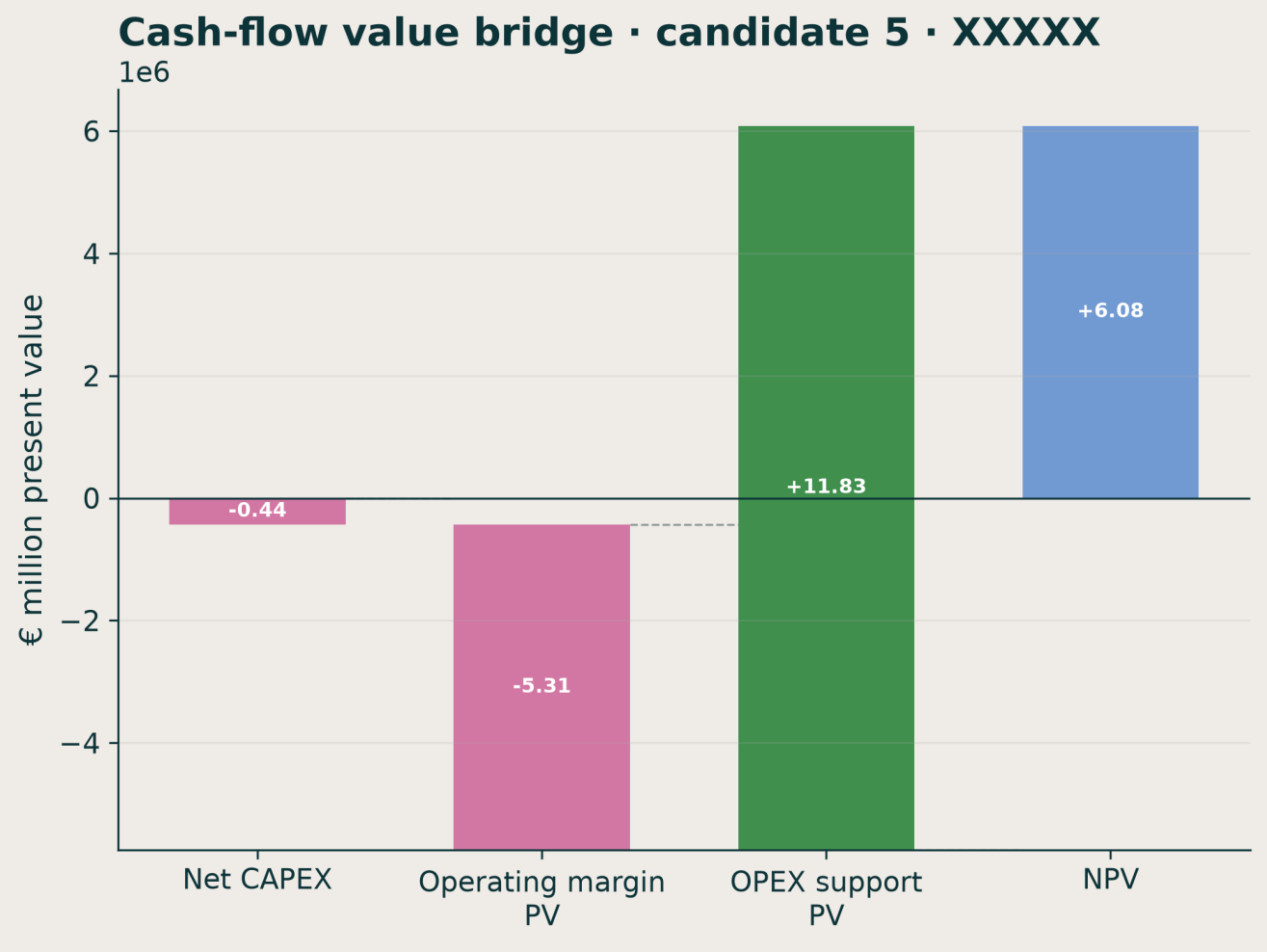


Figure 73. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 5; support D | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate products

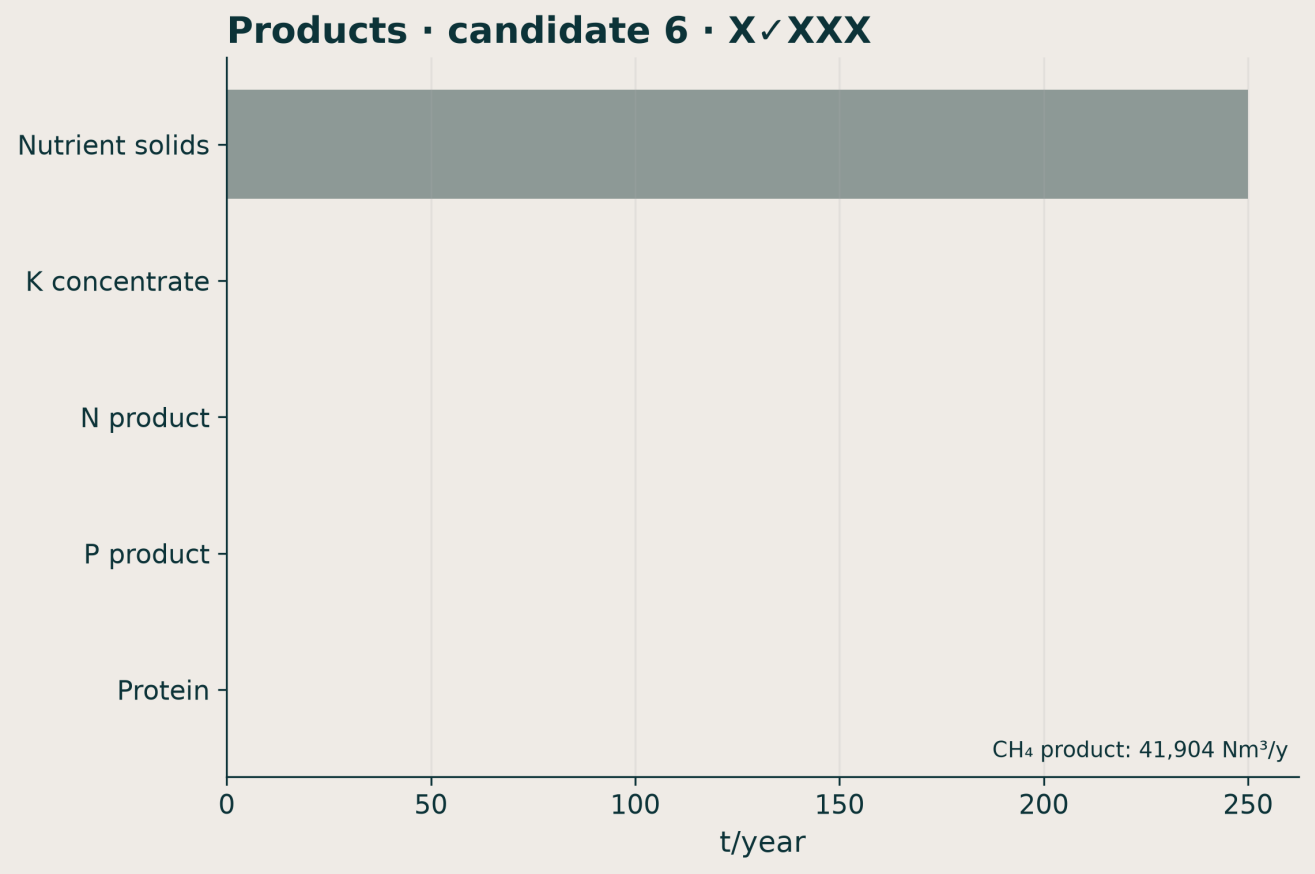


Figure 74. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_bypass; rank 6; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR1', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate utilities

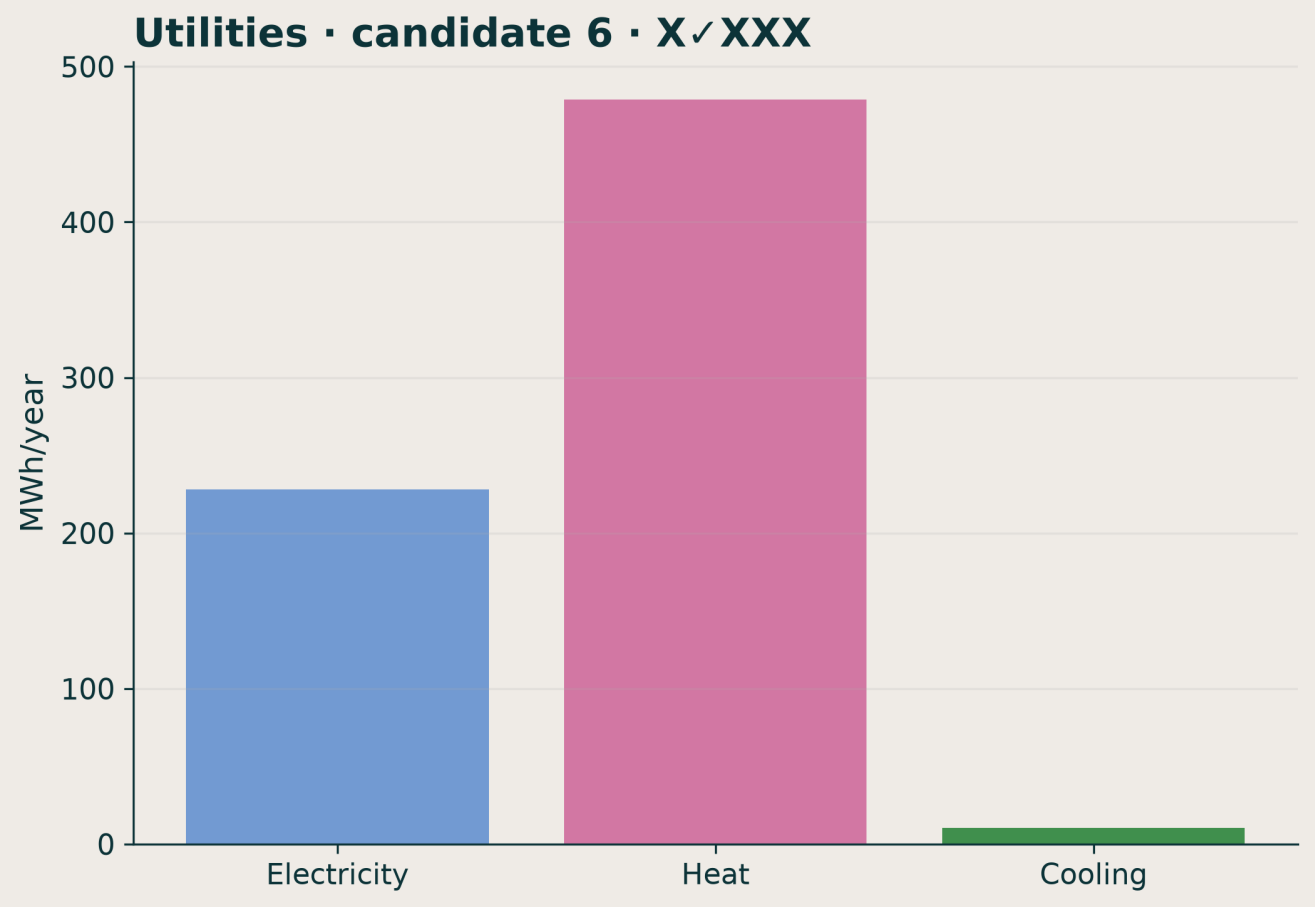


Figure 75. Annual electricity, heating and cooling duty

protein_bypass; rank 6; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR1', 'process_order': 'A'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate kpis

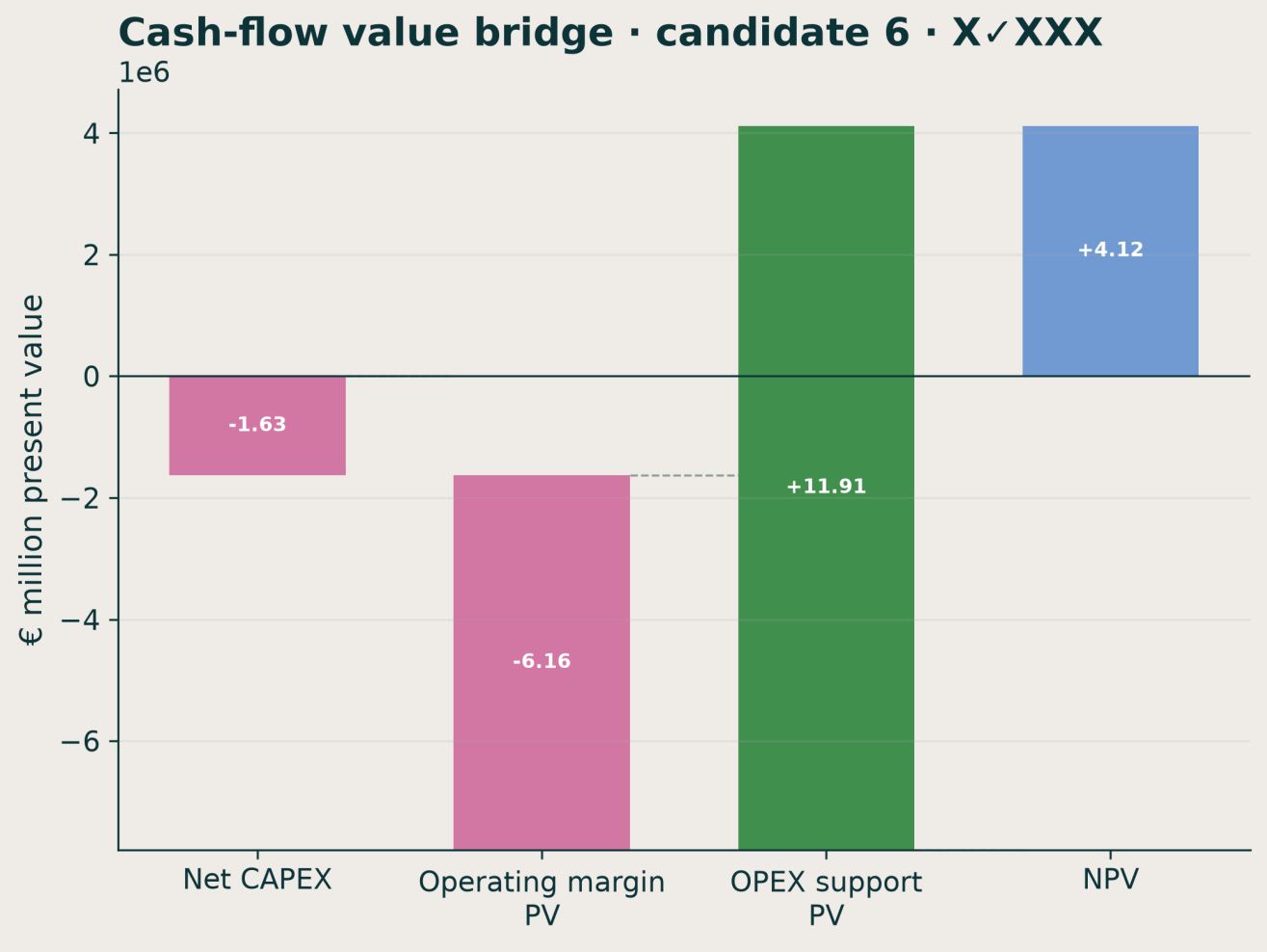


Figure 76. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 6; support D | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate products

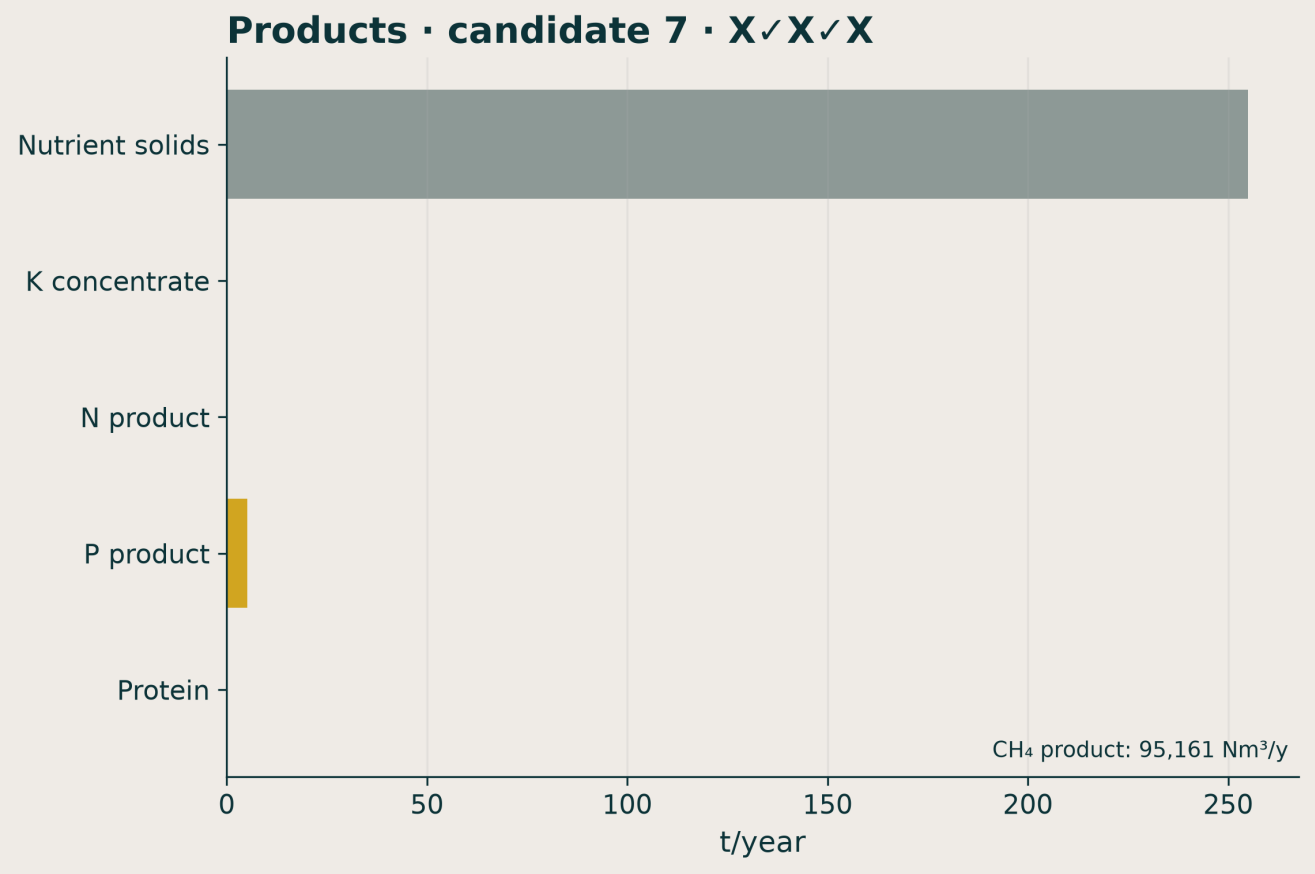


Figure 77. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_bypass; rank 7; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR2', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate utilities

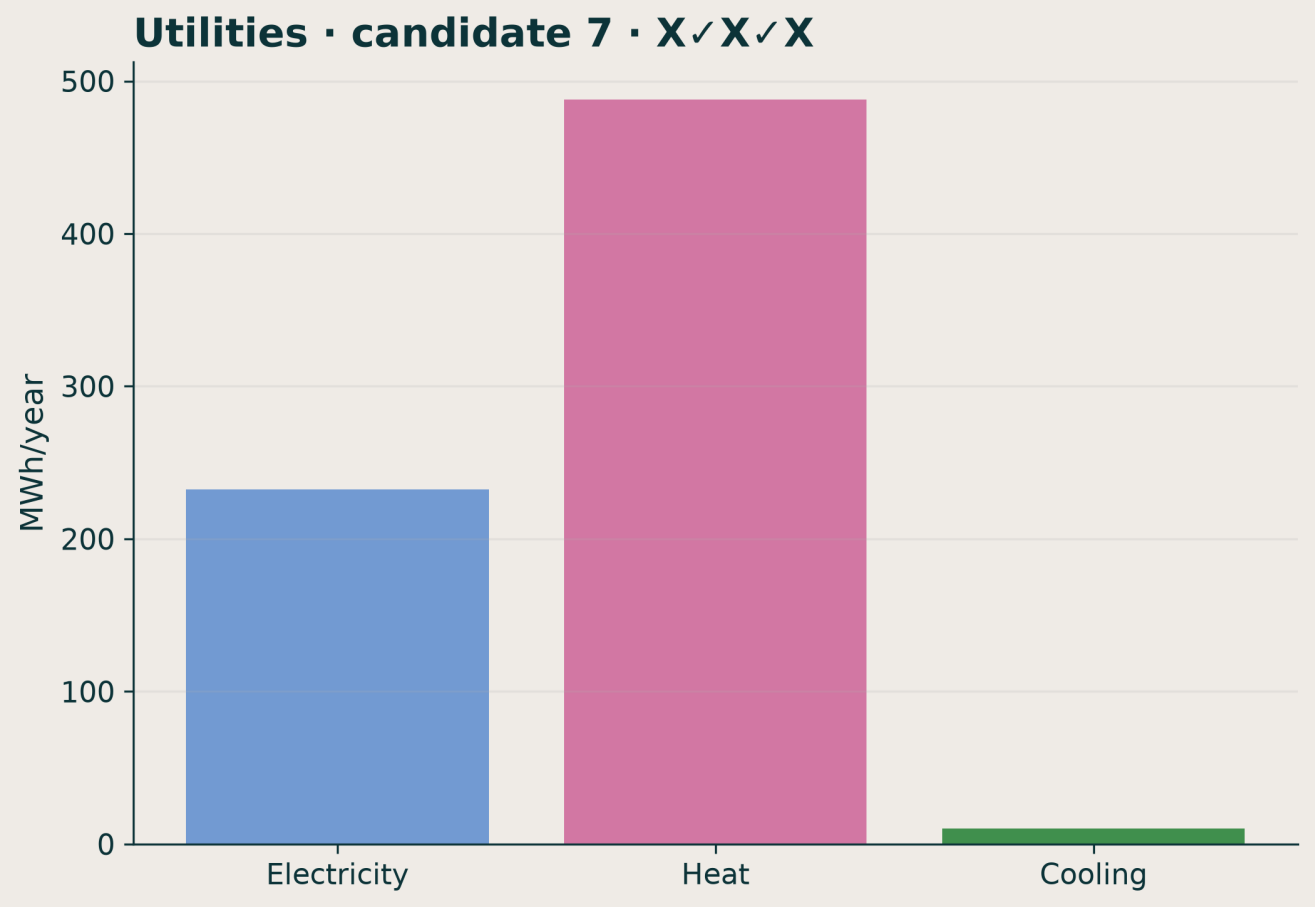


Figure 78. Annual electricity, heating and cooling duty

protein_bypass; rank 7; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR2', 'process_order': 'A'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate kpis

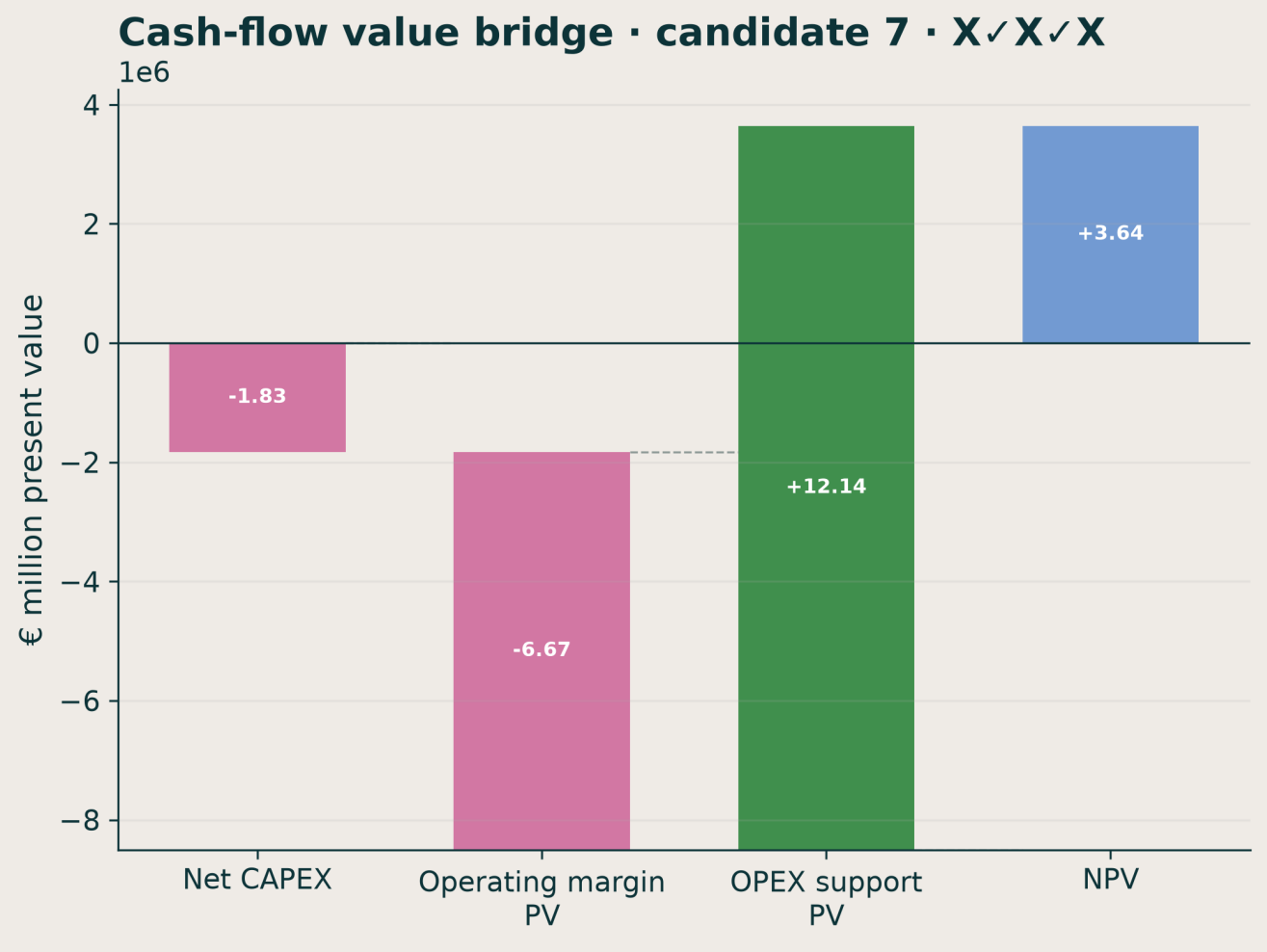


Figure 79. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 7; support D | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate products

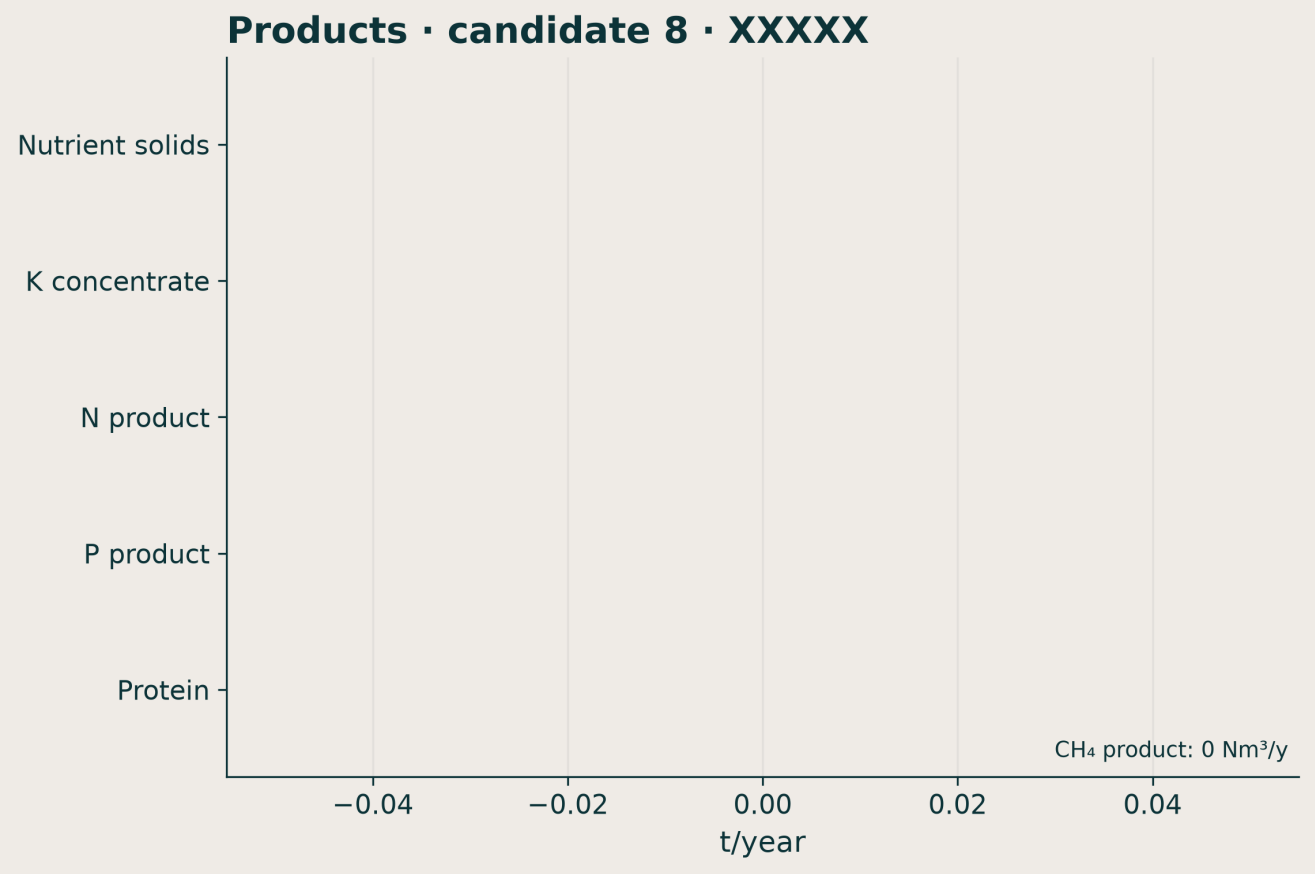


Figure 80. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_bypass; rank 8; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate utilities

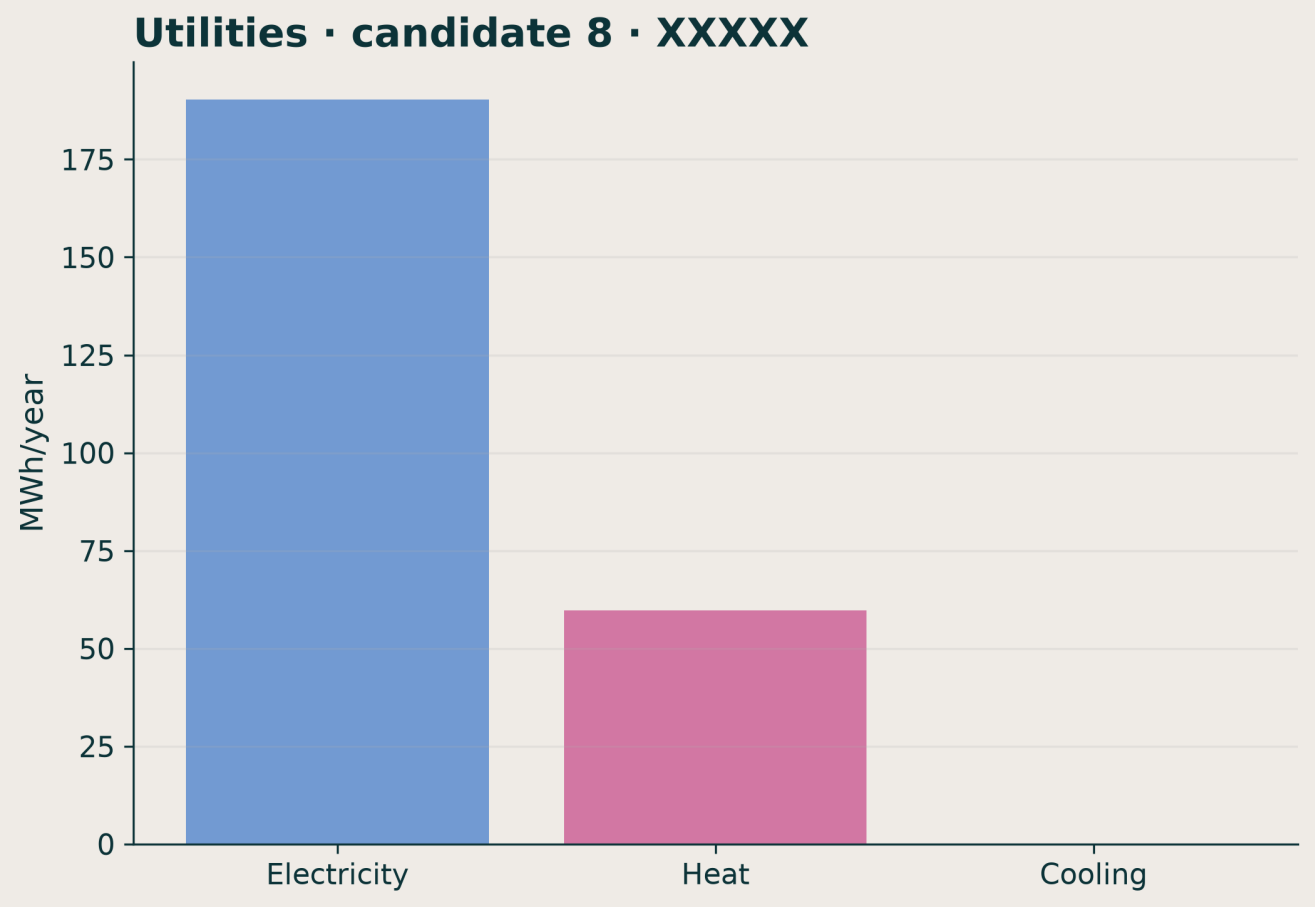


Figure 81. Annual electricity, heating and cooling duty

protein_bypass; rank 8; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate kpis

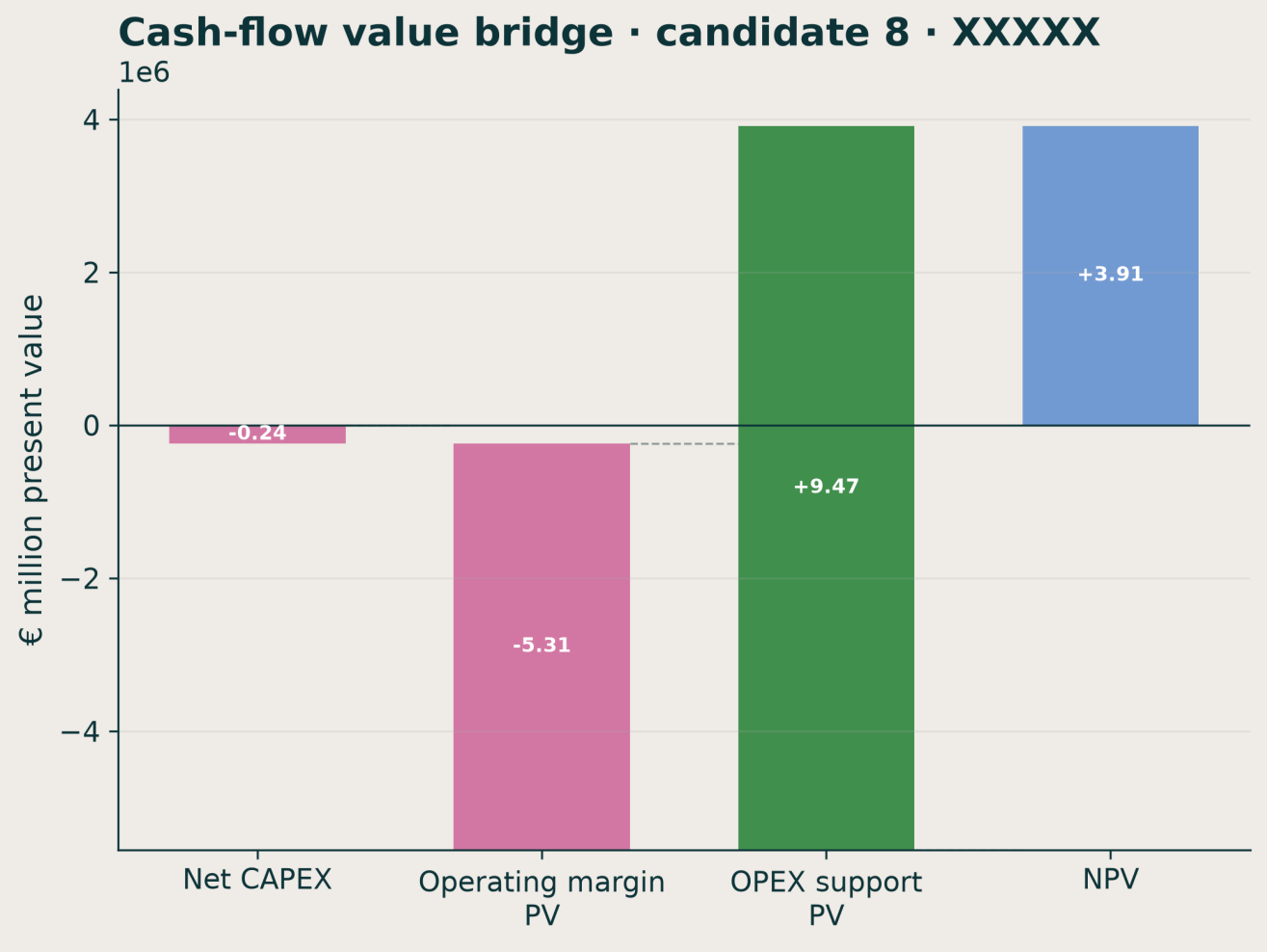


Figure 82. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 8; support B | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate products

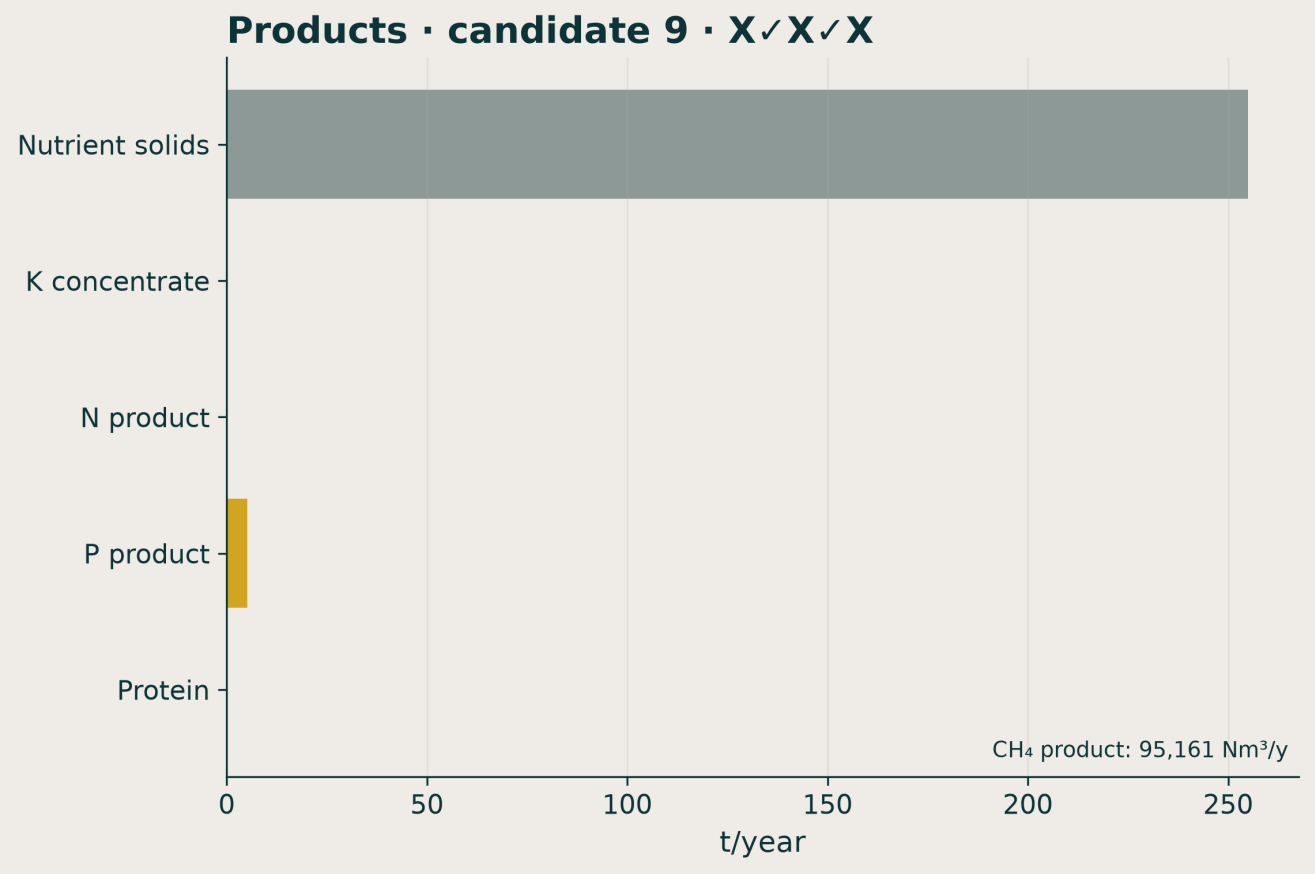


Figure 83. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_bypass; rank 9; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR2', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate utilities

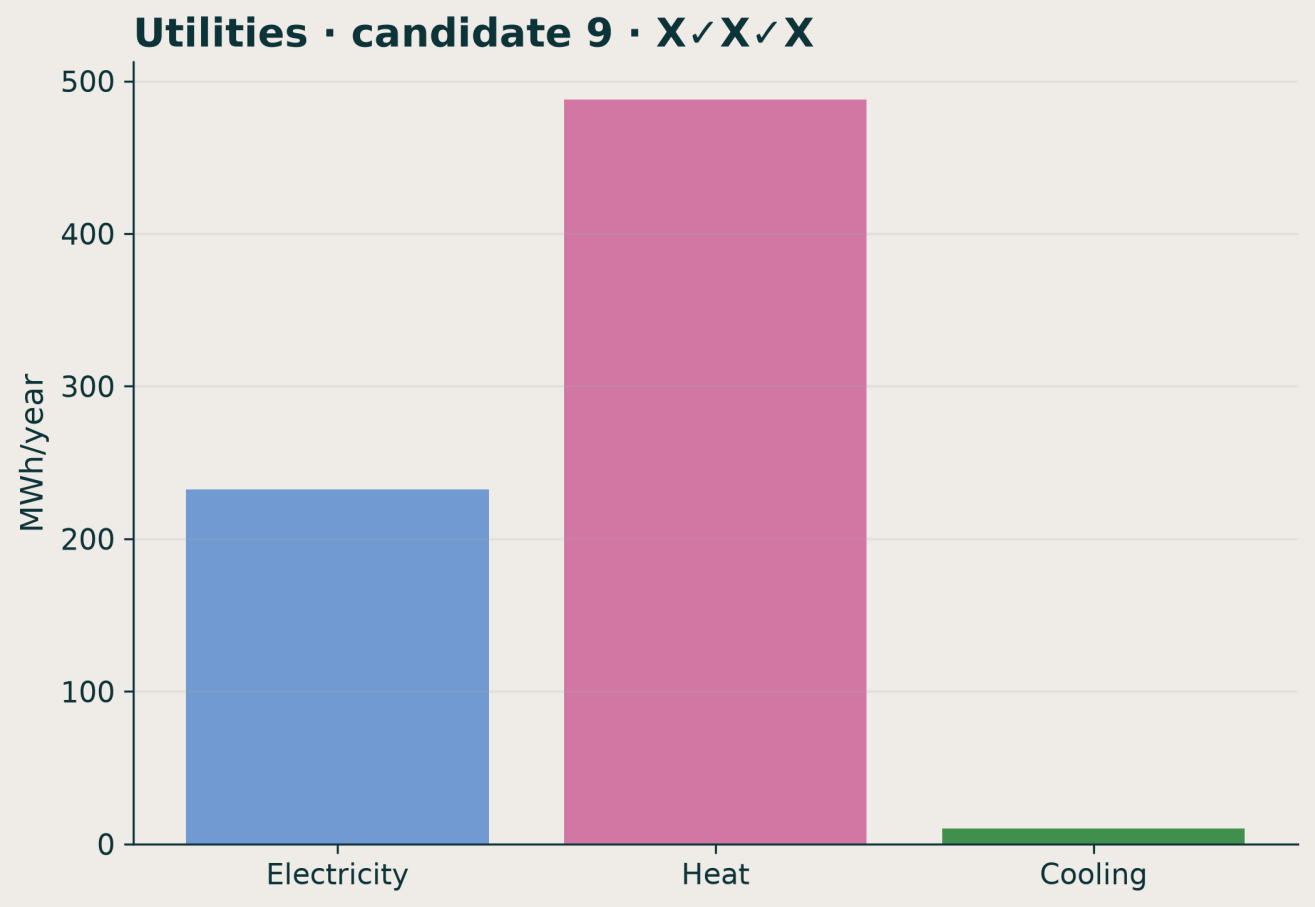


Figure 84. Annual electricity, heating and cooling duty

protein_bypass; rank 9; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR2', 'process_order': 'A'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate kpis

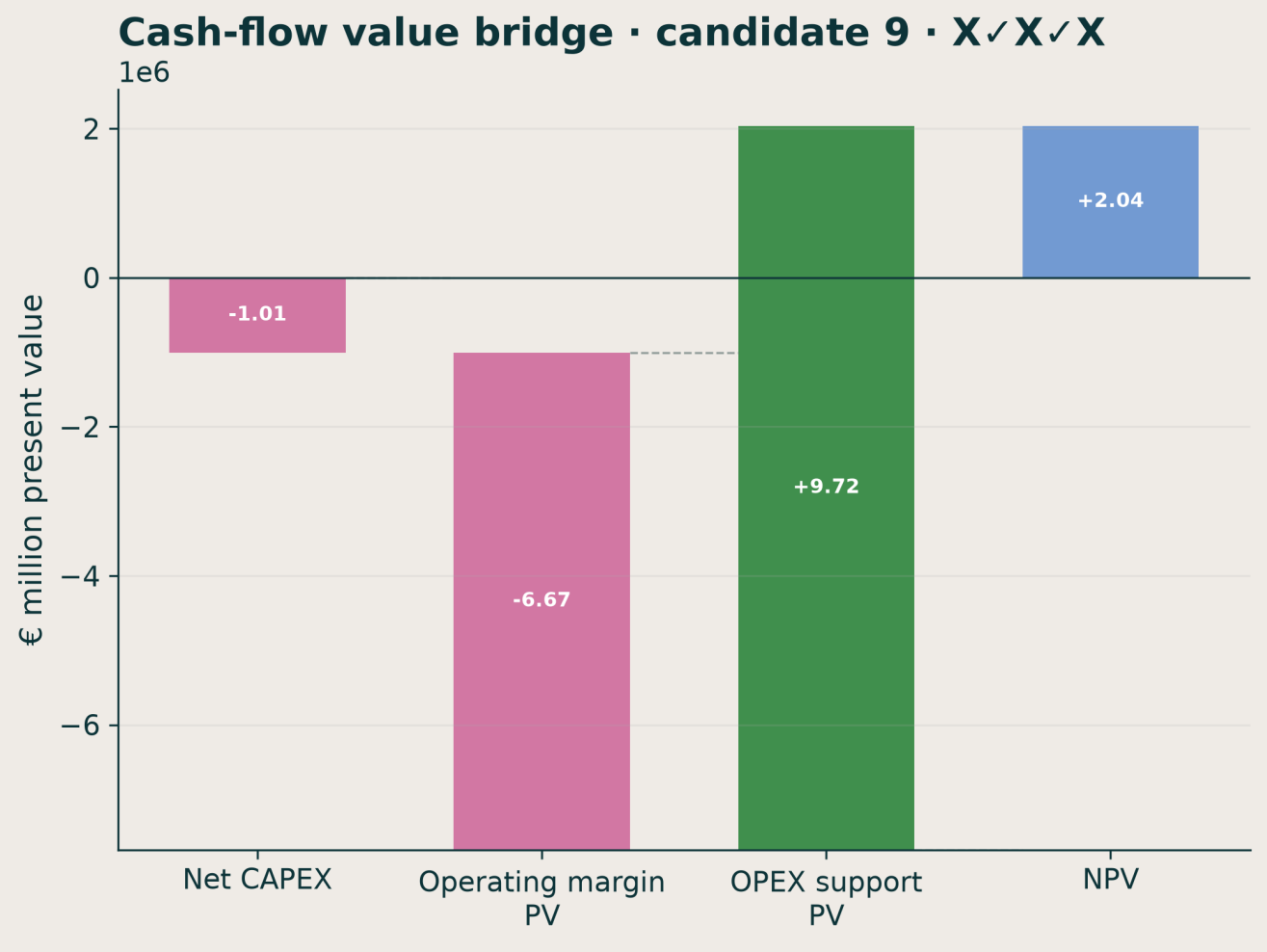


Figure 85. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 9; support B | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate products

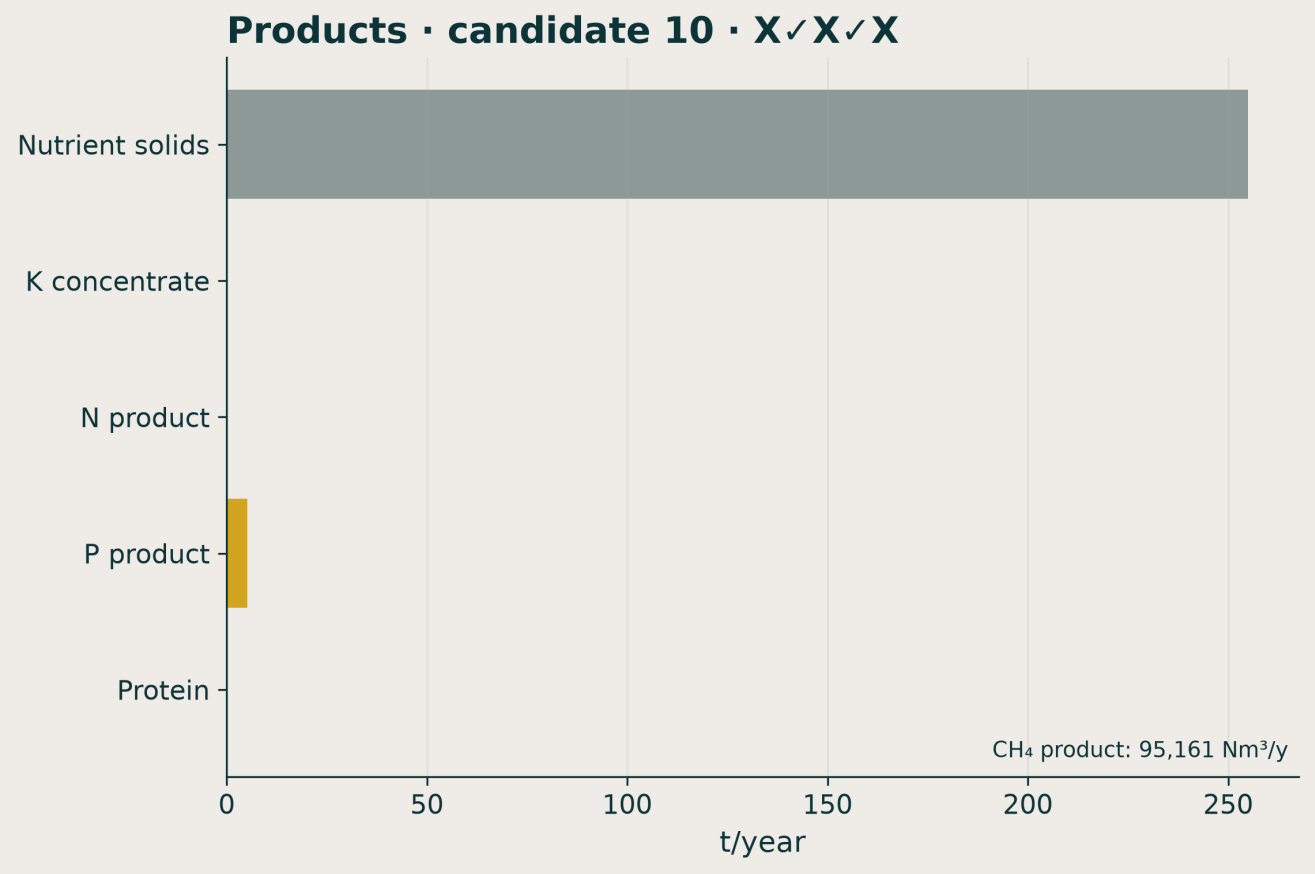


Figure 86. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_bypass; rank 10; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR2', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate utilities

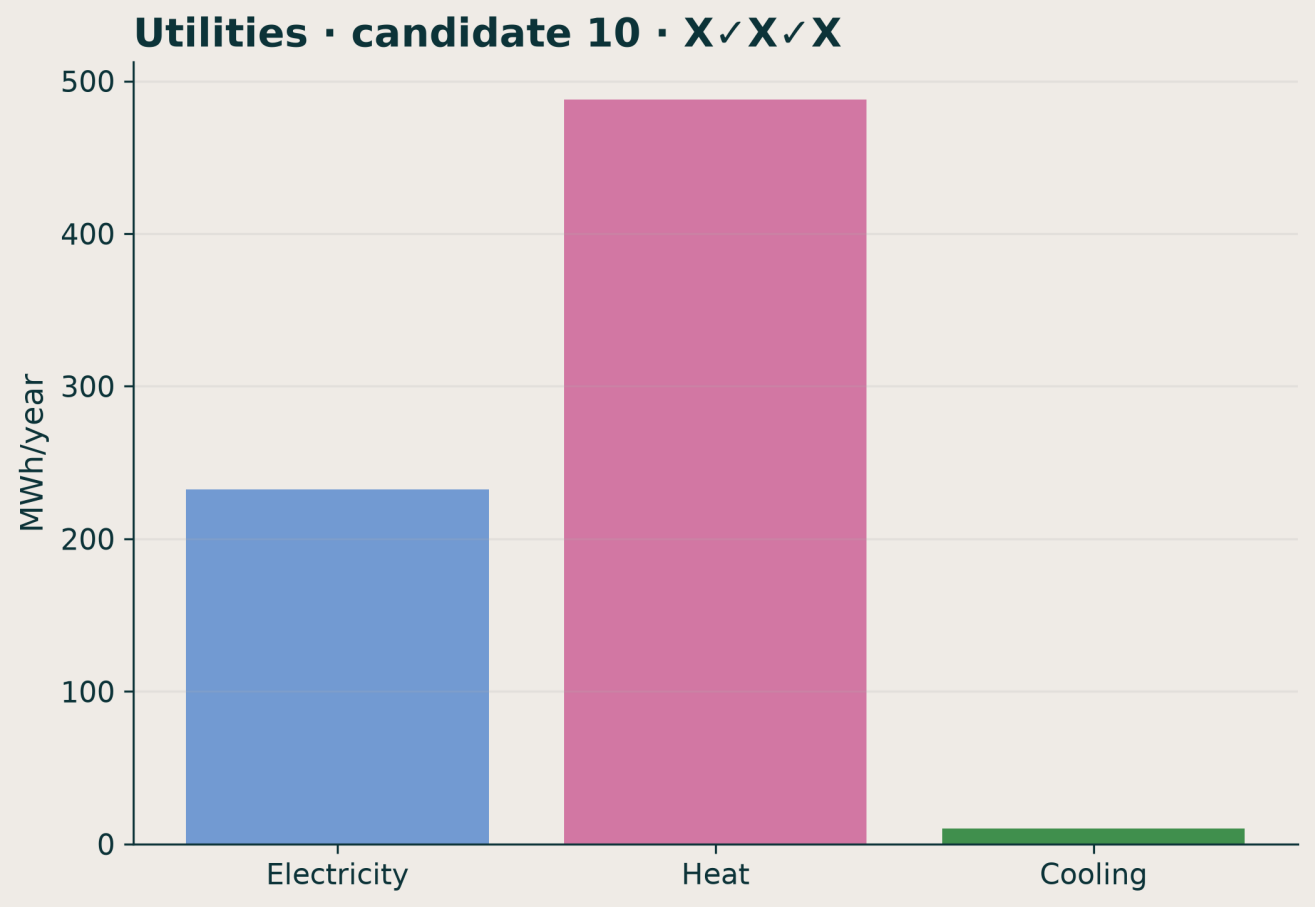


Figure 87. Annual electricity, heating and cooling duty

protein_bypass; rank 10; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR2', 'process_order': 'A'} | MWh/year | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

candidate kpis

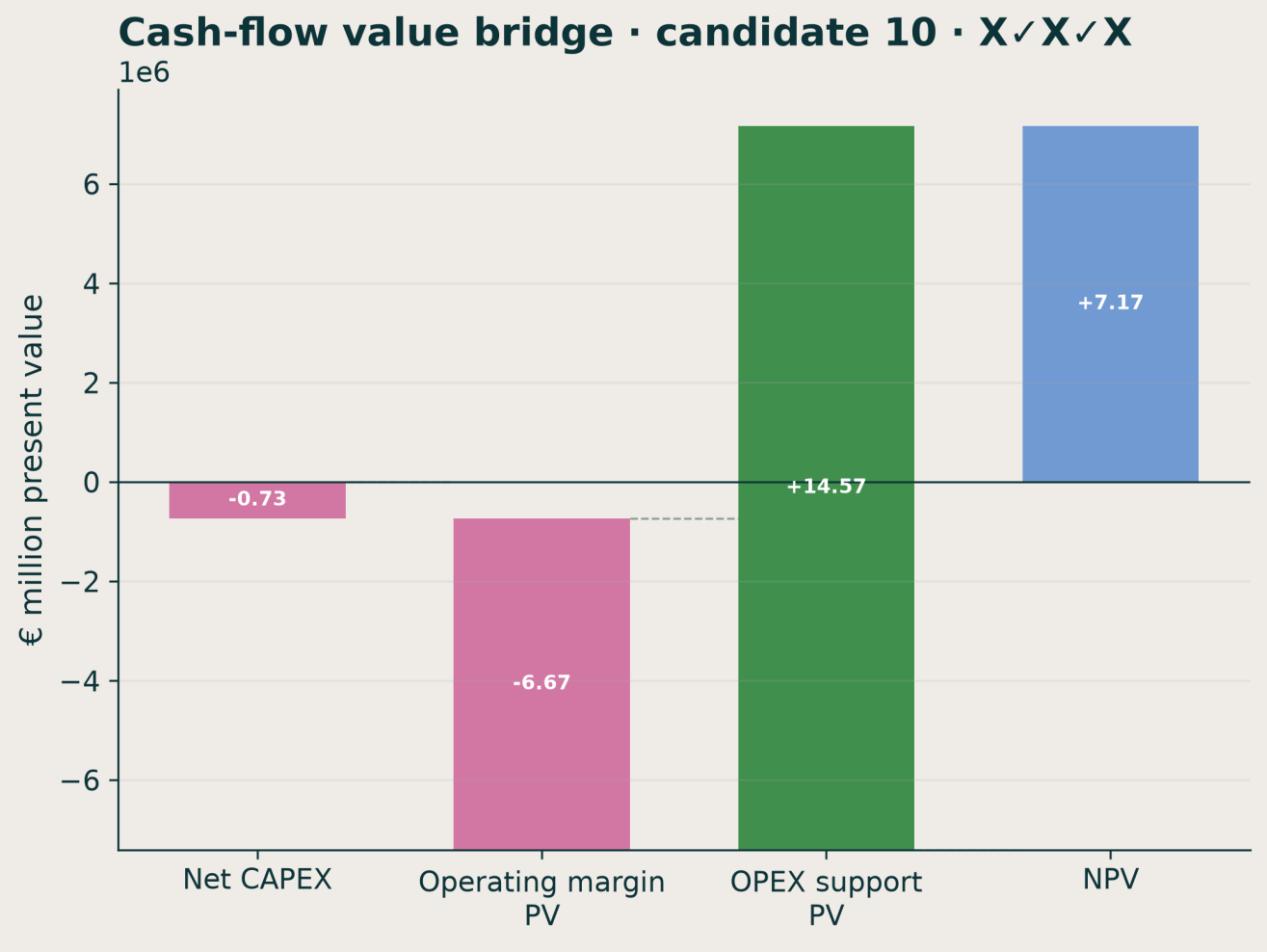


Figure 88. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 10; support C | € million present value | source: Greenhub\results\pareto_rvo_2026\protein_bypass_batch.json

baseline rvo support kpis

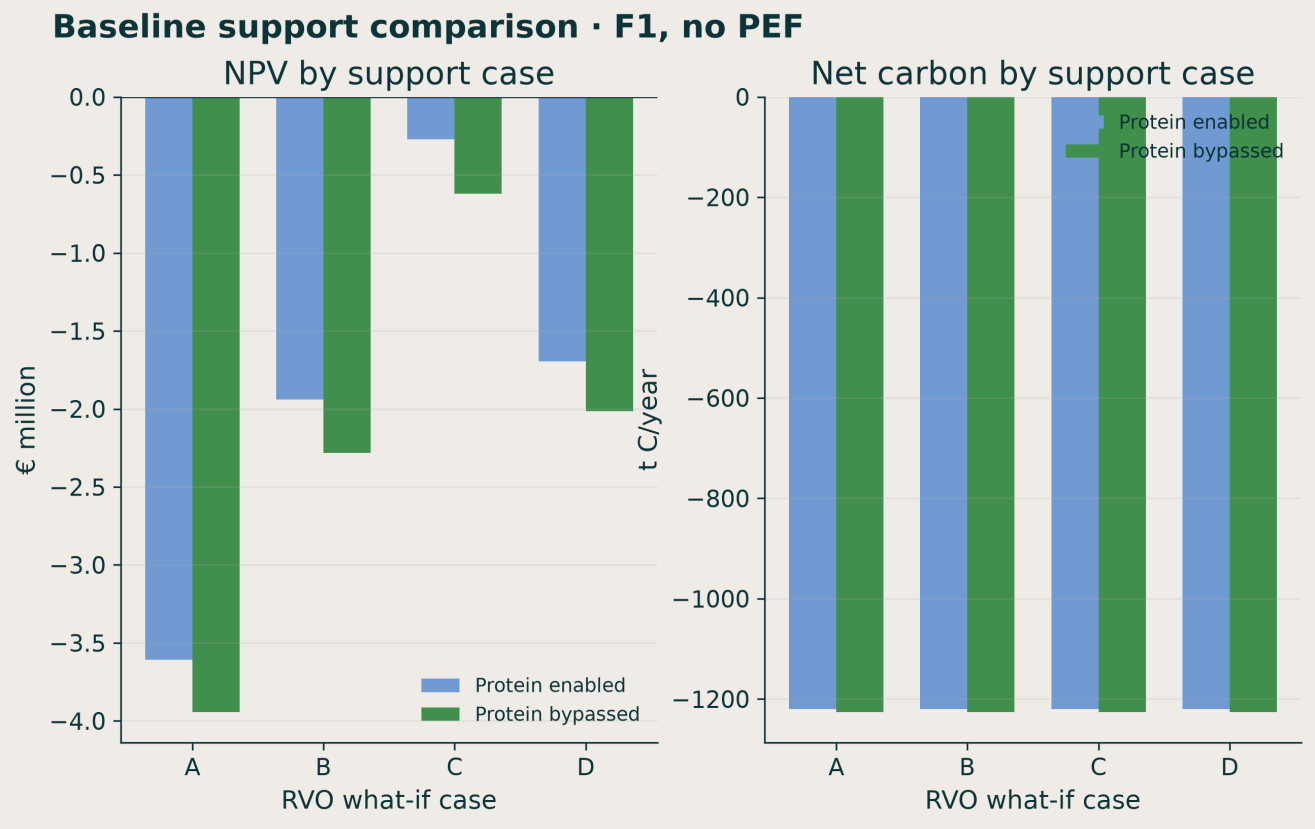


Figure 89. Baseline NPV and net carbon across paired RVO screening cases

No-PEF baseline; protein enabled and bypassed | € million; t C/year | source: Greenhub\results\pareto_rvo_2026

baseline utilities protein comparison

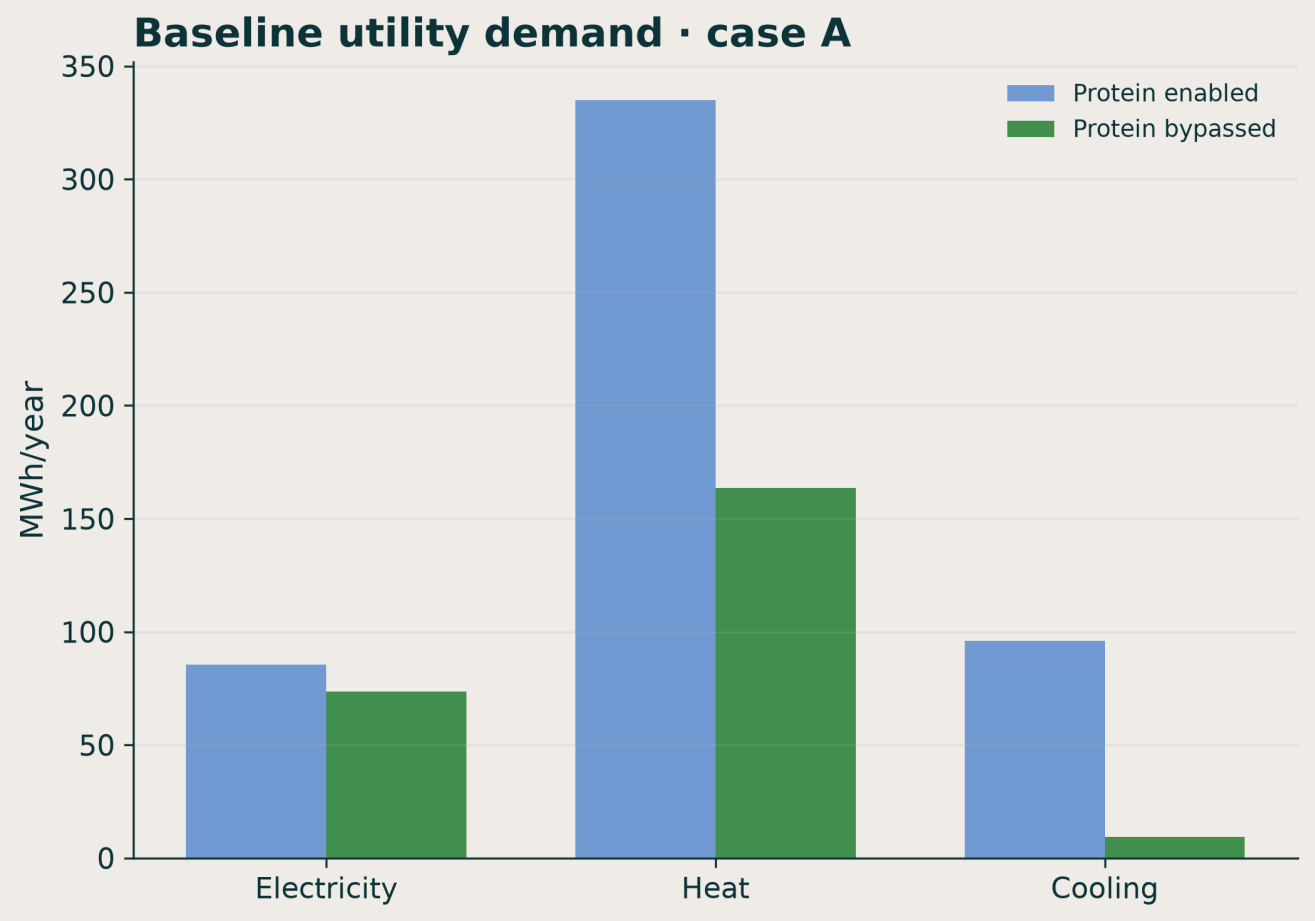


Figure 90. Baseline utility comparison for protein route and bypass

No-PEF baseline; case A | MWh/year | source: Greenhub\results\pareto_rvo_2026

rvo support case npv comparison

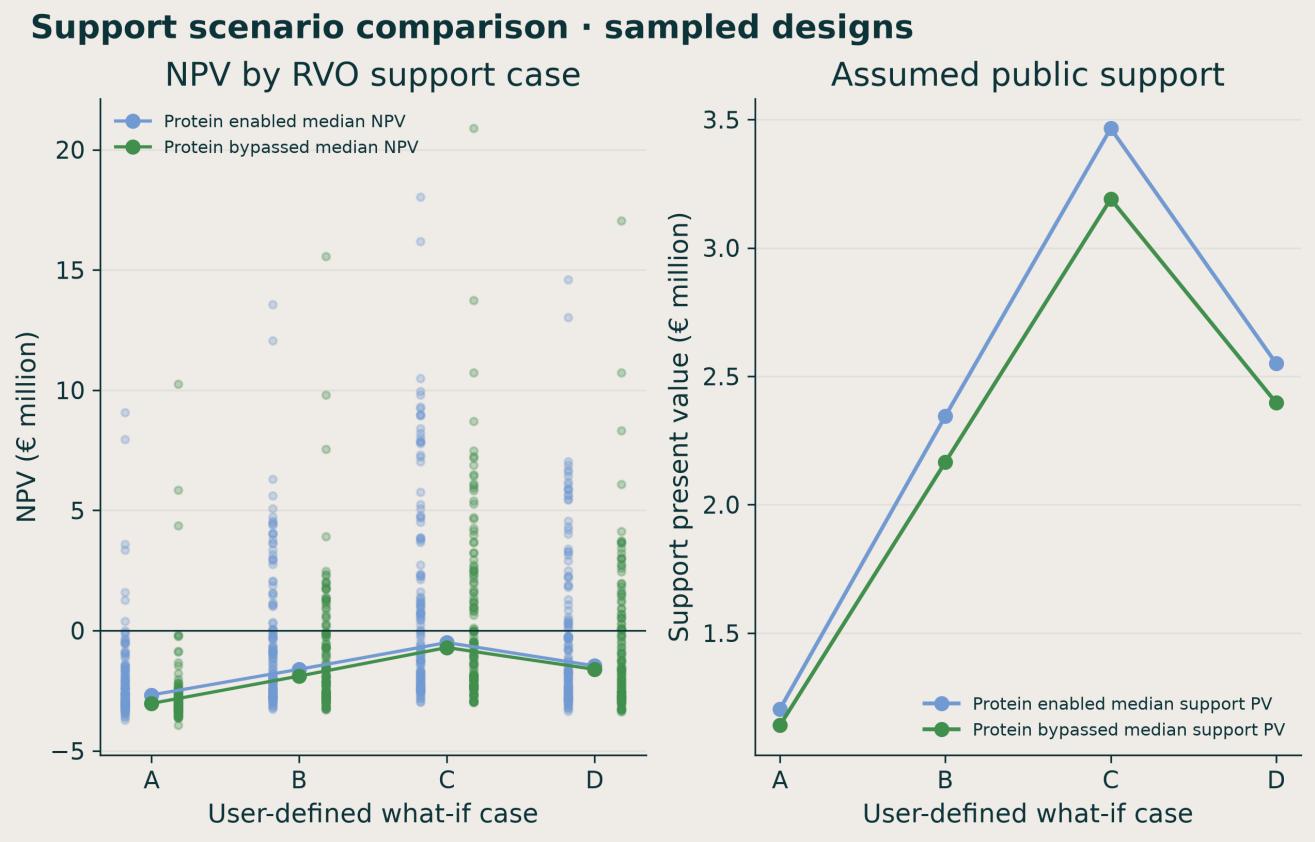


Figure 91. NPV samples and median support present value by support case

Protein enabled and bypassed; paired support cases | € million | source: Greenhub\results\pareto_rvo_2026

endpoint product yields

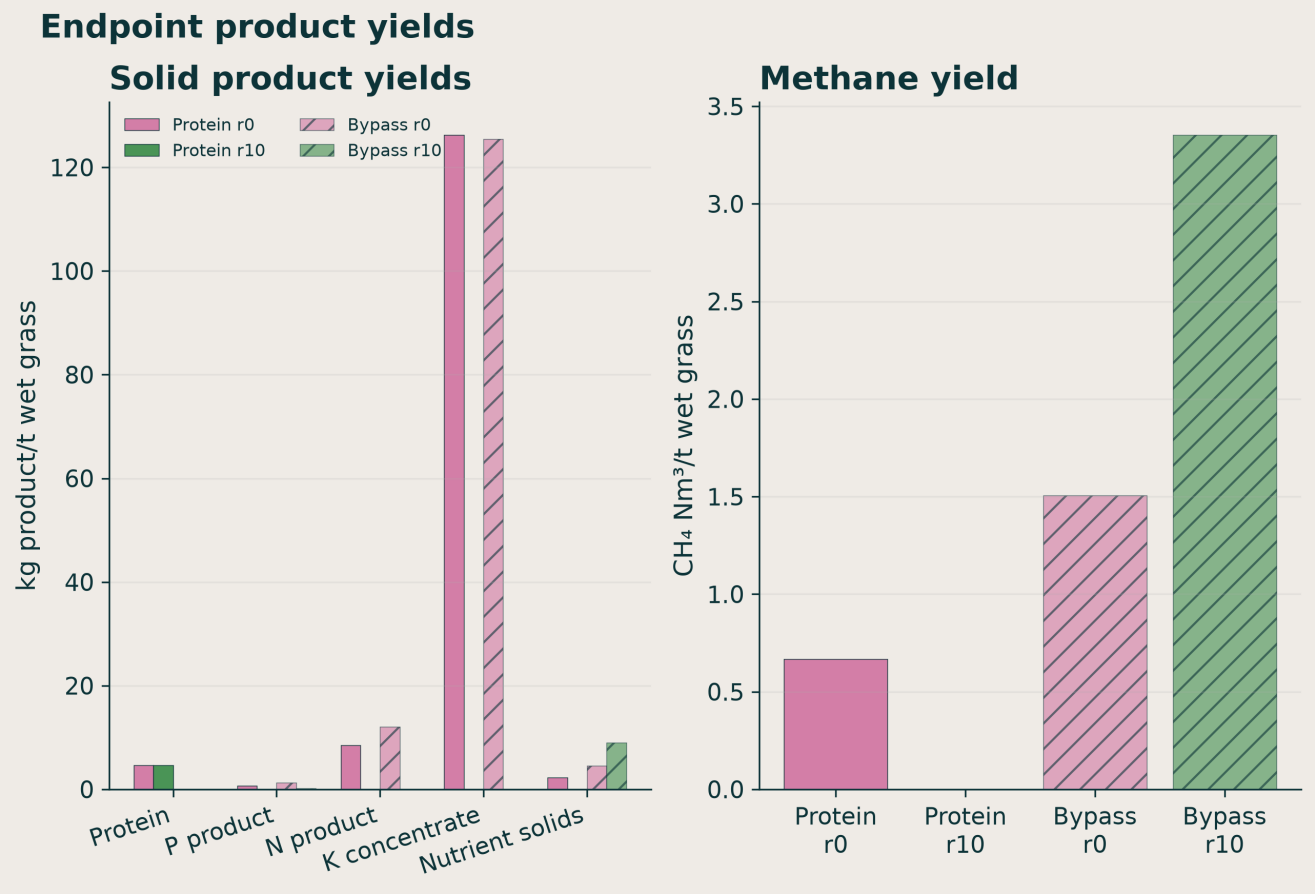


Figure 92. Solid product and methane yields at selected economic and carbon endpoints

Rank 0 and 10 from the three-objective front; no PEF | kg product/t wet grass; CH4 Nm3/t wet grass | source: Greenhub\results\pareto_rvo_2026

energy intensity by candidate

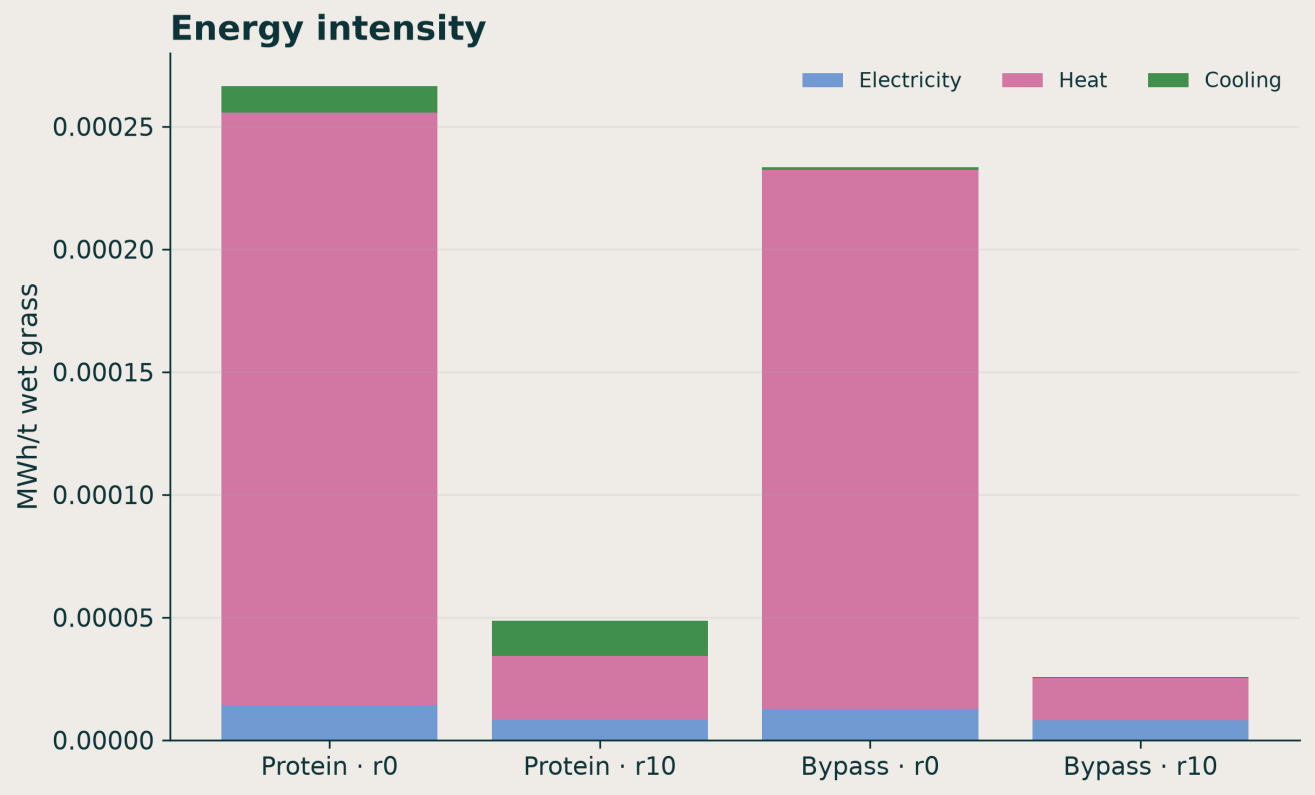


Figure 93. Electricity, heat and cooling intensity at selected endpoint candidates

Rank 0 and 10 from the three-objective front; no PEF | MWh/t wet grass | source: Greenhub\results\pareto_rvo_2026

npv profit by feed scale

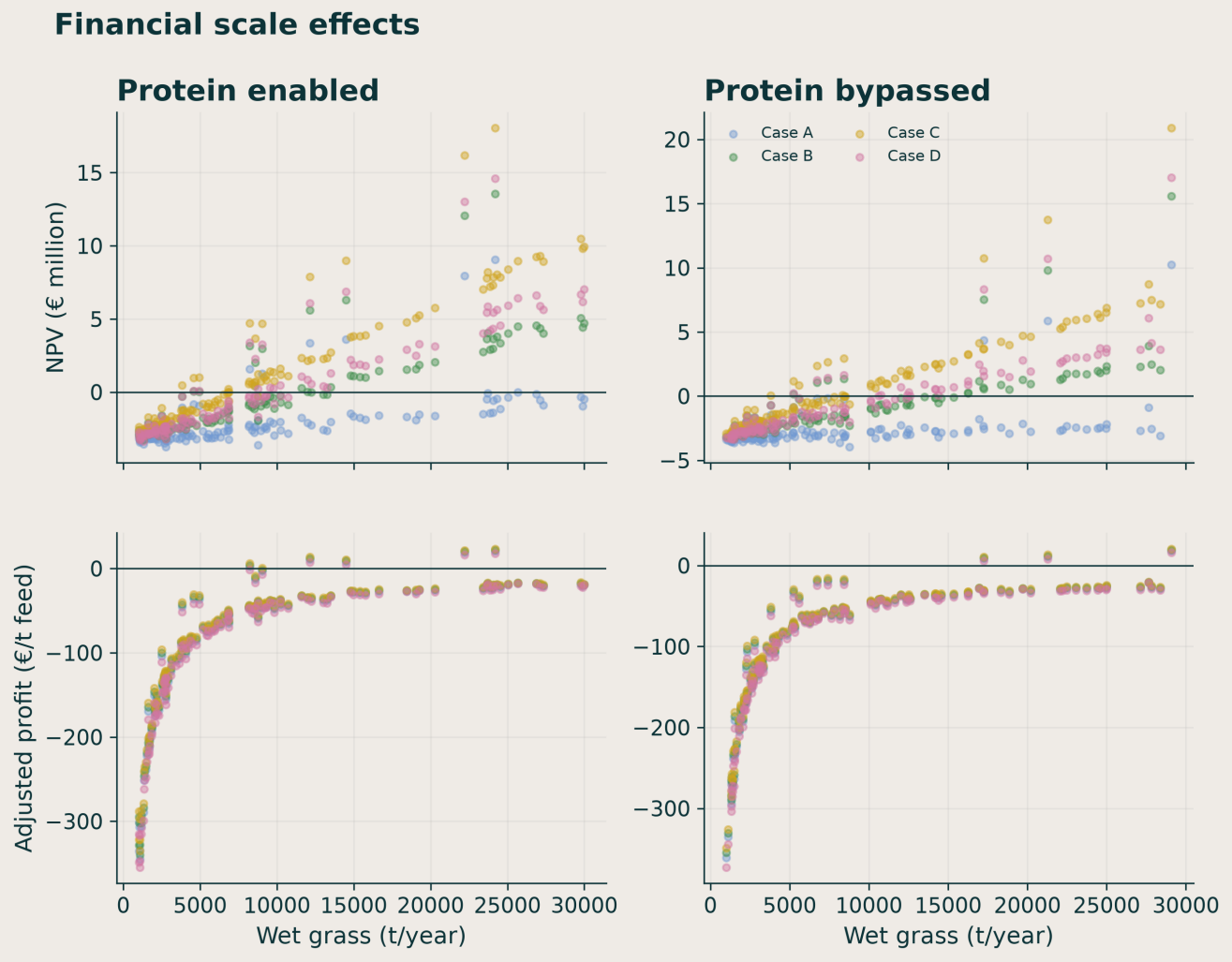


Figure 94. Screening NPV and adjusted annual profit across evaluated feed scales

All evaluated designs; cases A-D; design-space scatter, not probabilistic uncertainty | NPV: € million; profit intensity: €/t wet grass | source: Greenhub\results\pareto_rvo_2026

rvo support composition

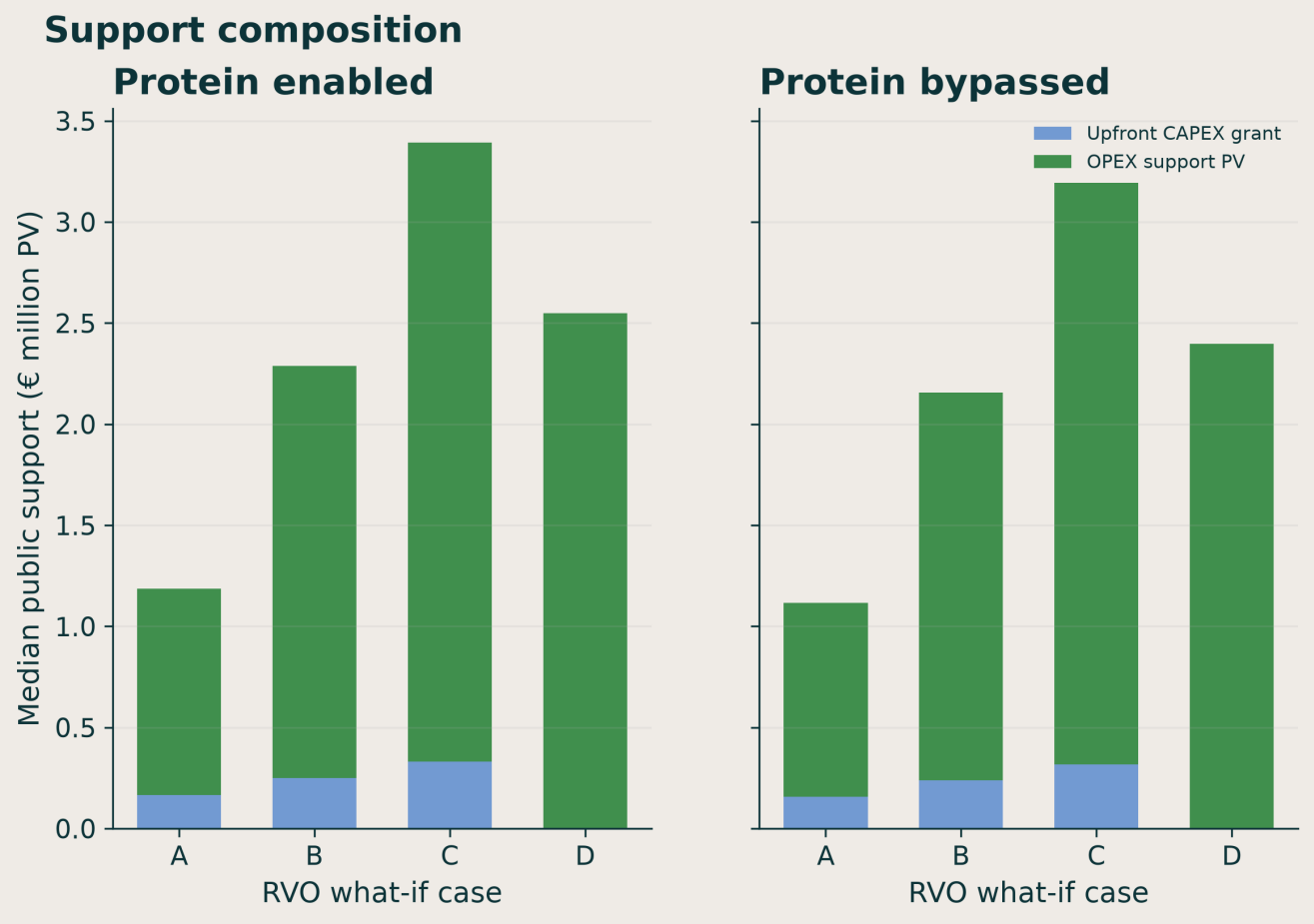


Figure 95. Median upfront grant and discounted operating-support components across paired support cases

Median across evaluated designs by protein group | € million present value | source: Greenhub\results\pareto_rvo_2026

operational climate by stage

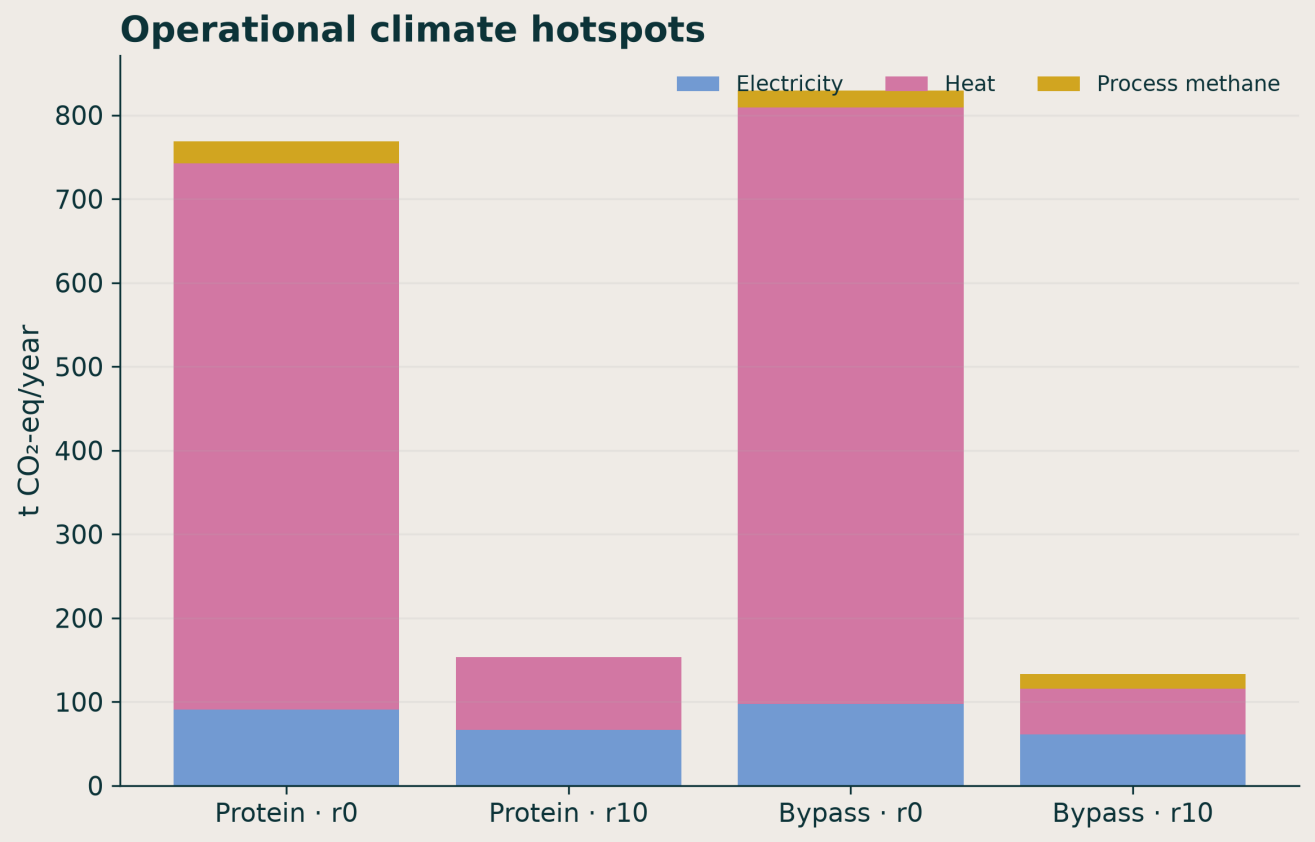


Figure 96. Electricity, heat and characterized process-methane burdens for representative Pareto endpoints

Rank 0 and 10; protein enabled and bypassed; attributional gate-to-gate loads only | t CO2-eq/year | source: Greenhub\results\pareto_rvo_2026\stored decision_support.json

opex categories by endpoint

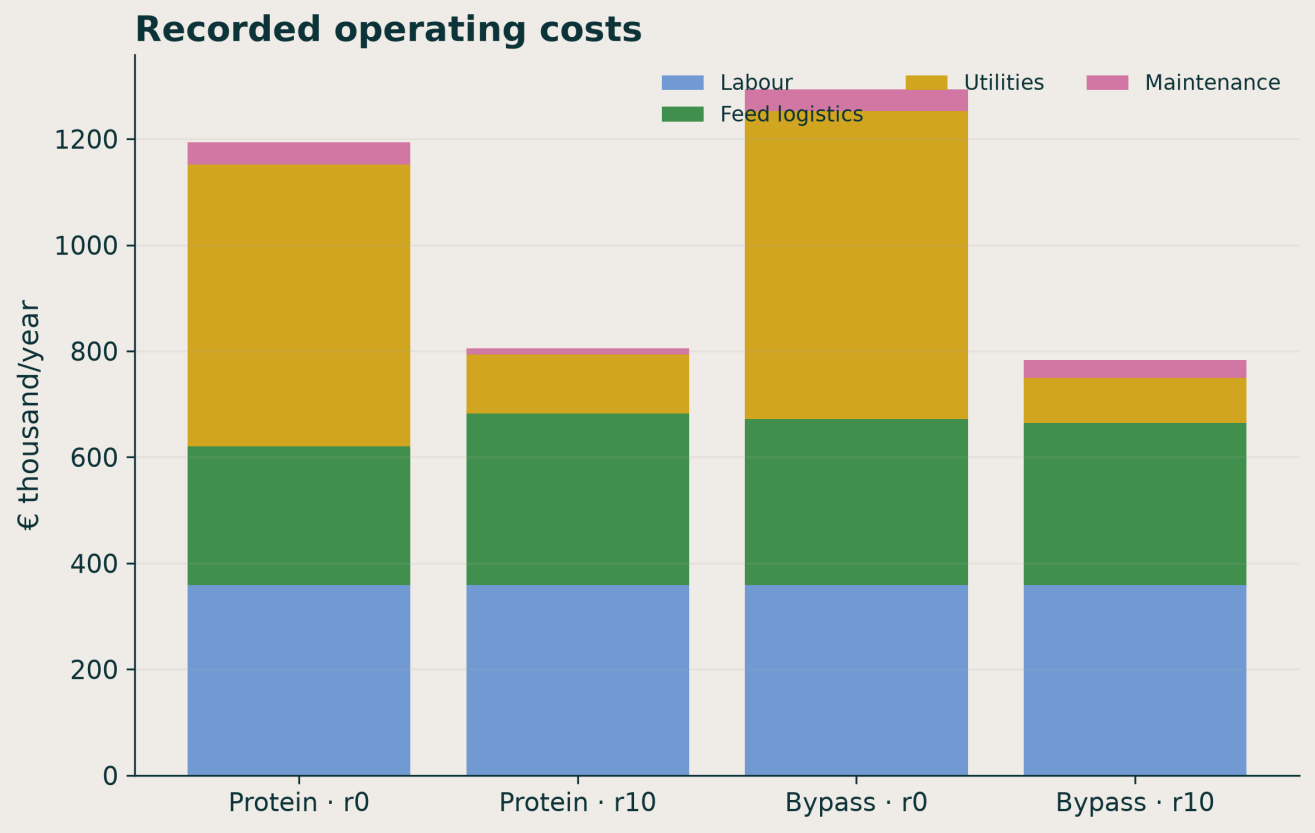


Figure 97. Recorded system-level OPEX categories at representative Pareto endpoints; shared costs are not product-allocated

Rank 0 and 10; protein enabled and bypassed; recorded cost lines only | € thousand/year | source: Greenhub\results\pareto_rvo_2026\stored decision_support.json

equipment energy by endpoint

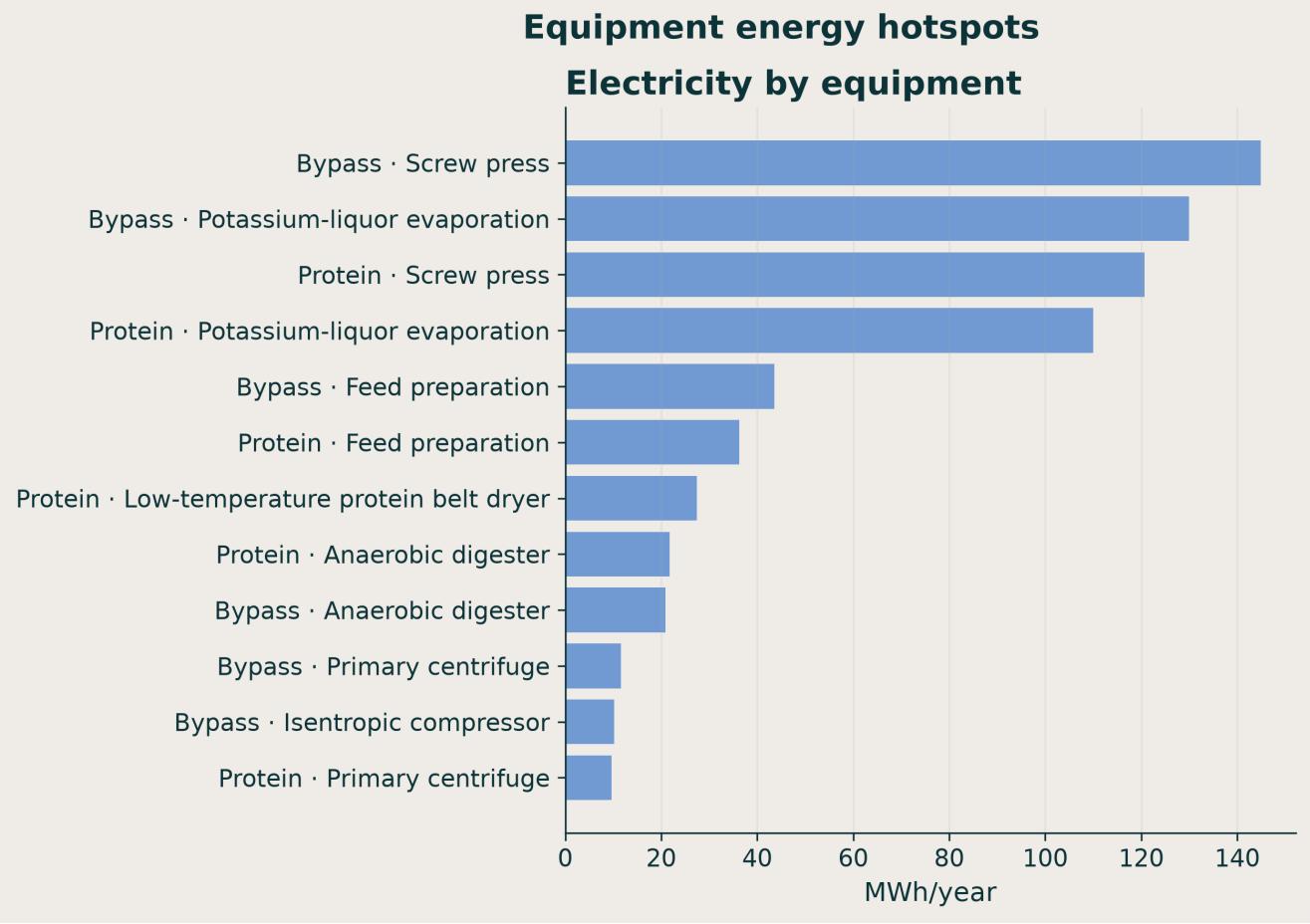


Figure 98. Largest saved equipment electricity duties for economic-endpoint process runs

Rank 0; protein-enabled and protein-bypassed cases; annualized at each run schedule | MWh/year | source: Greenhub\results\pareto_rvo_2026\stored utilities.json